

ODSL Block Course

Lecture 10: Unsupervised Learning / clustering

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ODSL Block Course

25 Mar 2026



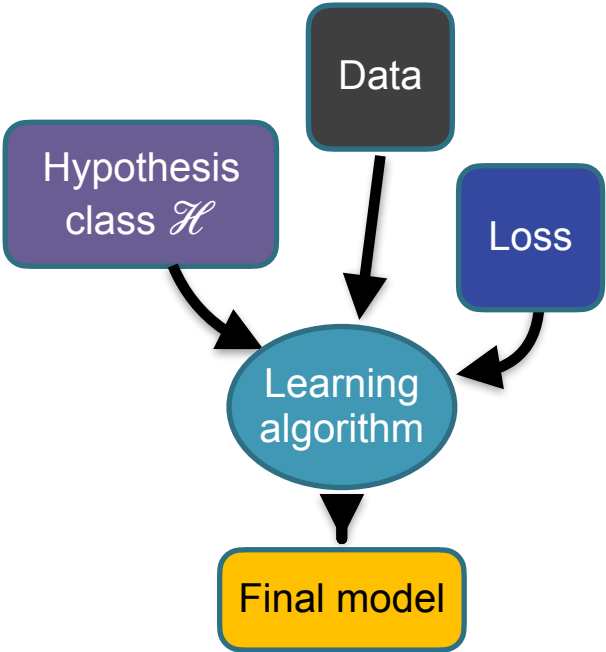
ZOOMING OUT

Neural network

$$f(x; W, b) = NN(x; W, b)$$

Gradient descent

$$\frac{\partial \mathcal{L}}{\partial w}$$



Automatic differentiation

Reverse Mode

$$\vec{z} = f \left(g \left(h \left(x \right) \right) \right)$$

$$\frac{dz_i}{dx} = \frac{df_i}{dg_j} \cdot \frac{dg_j}{dh_k} \cdot \frac{dh_k}{dx}$$

$$\delta \mathcal{L} = 1 \quad J_{f(g)} \quad J_{g(h)} \quad J_{h(x)}$$

“back propagation”



Outline

1

K-means

2

Principle Component Analysis

3

Learning theory

Unsupervised Learning

Supervised Learning
(up till now this week)

Learn $p_{\theta}(y | X)$

targets data

Possible Data Sources

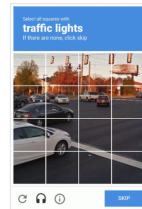
Manual Collection of Data from the real world



$s_1 = (\text{cat image}, \text{"cat"})$

$s_2 = (\text{dog image}, \text{"dog"})$

.....
*Realistic Data but assumptions
about $p(s)$ very hard to control*



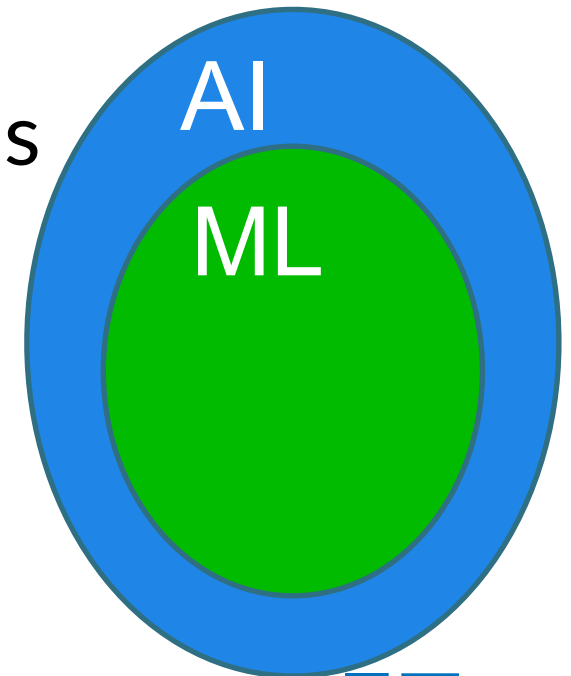
Unsupervised Learning

(Rest of this lecture + Day 3 next week
with NNs)

Learn $p_{\theta}(X)$

□ K-means

□ PCA



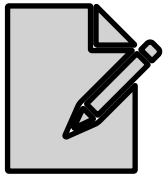
What if you don't have targets?

This lecture: Unsupervised learning

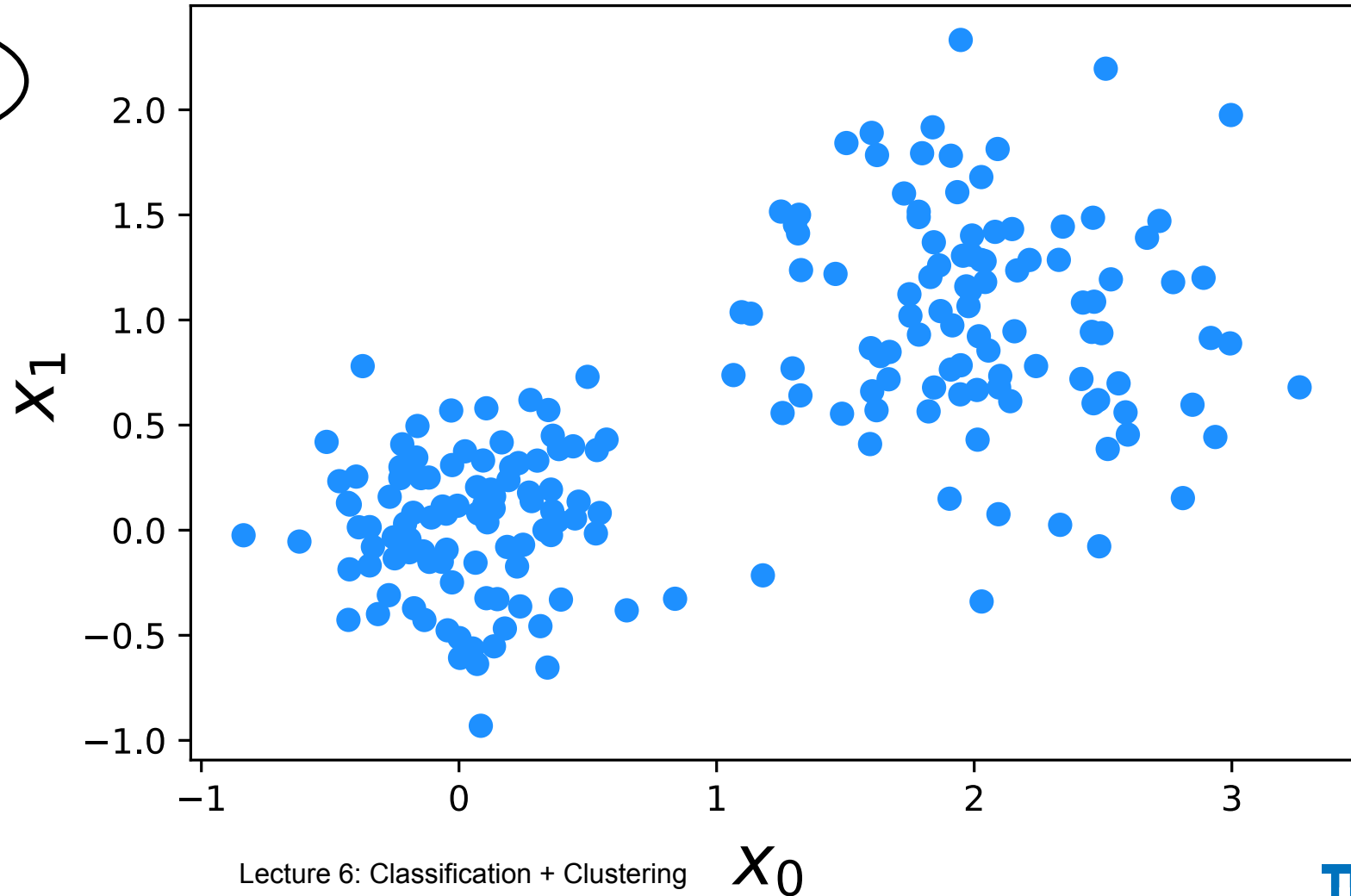
What can you say about this?



Type your As
in the chat!



Write your answers
on your paper.

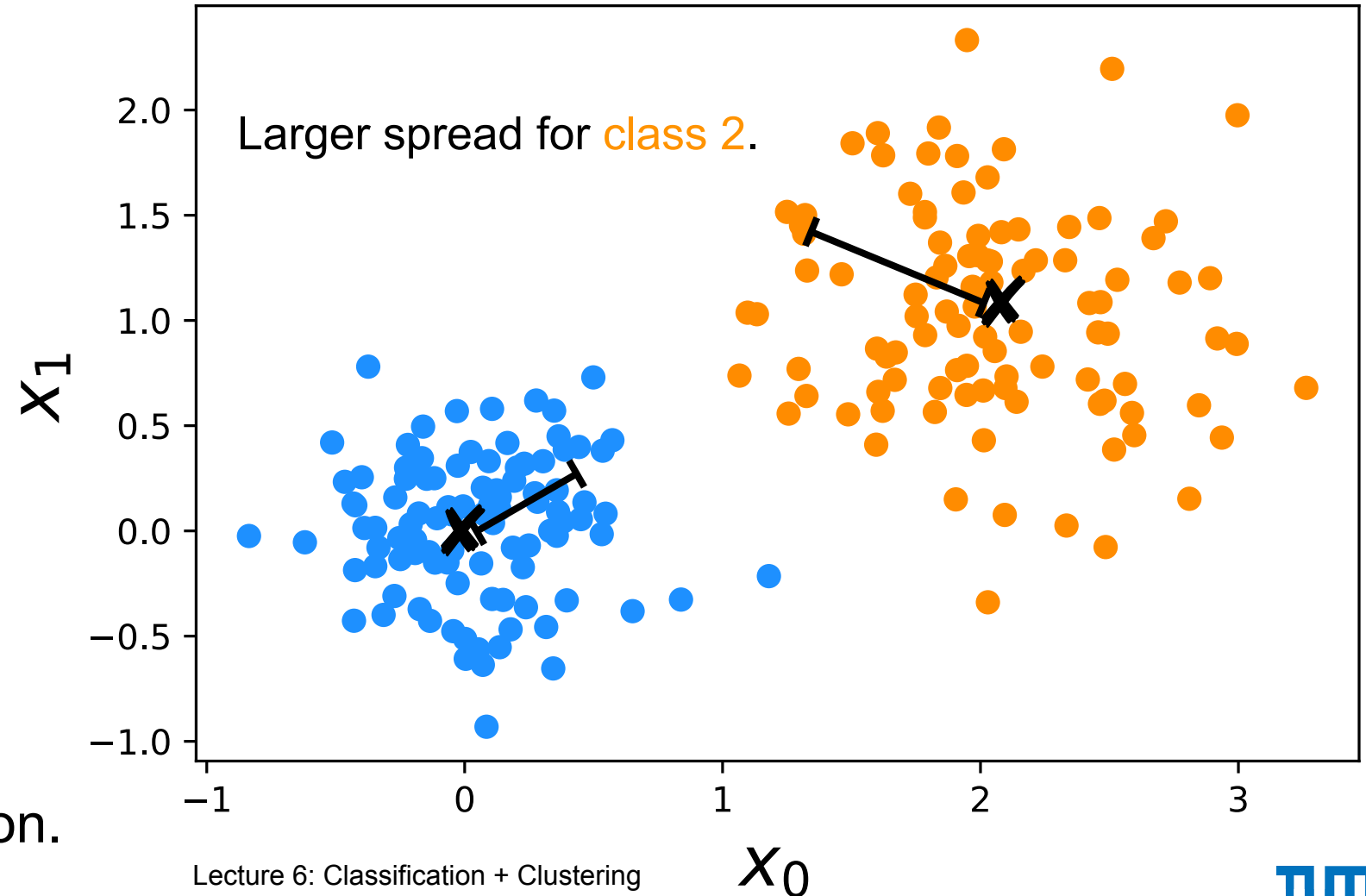


This lecture: Unsupervised learning

What can you say about this?

$$\text{class 1} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, 0.33^2 \cdot \mathbb{I}\right)$$

$$\text{class 2} \sim \mathcal{N}\left(\begin{pmatrix} 2 \\ 1 \end{pmatrix}, 0.45^2 \cdot \mathbb{I}\right)$$



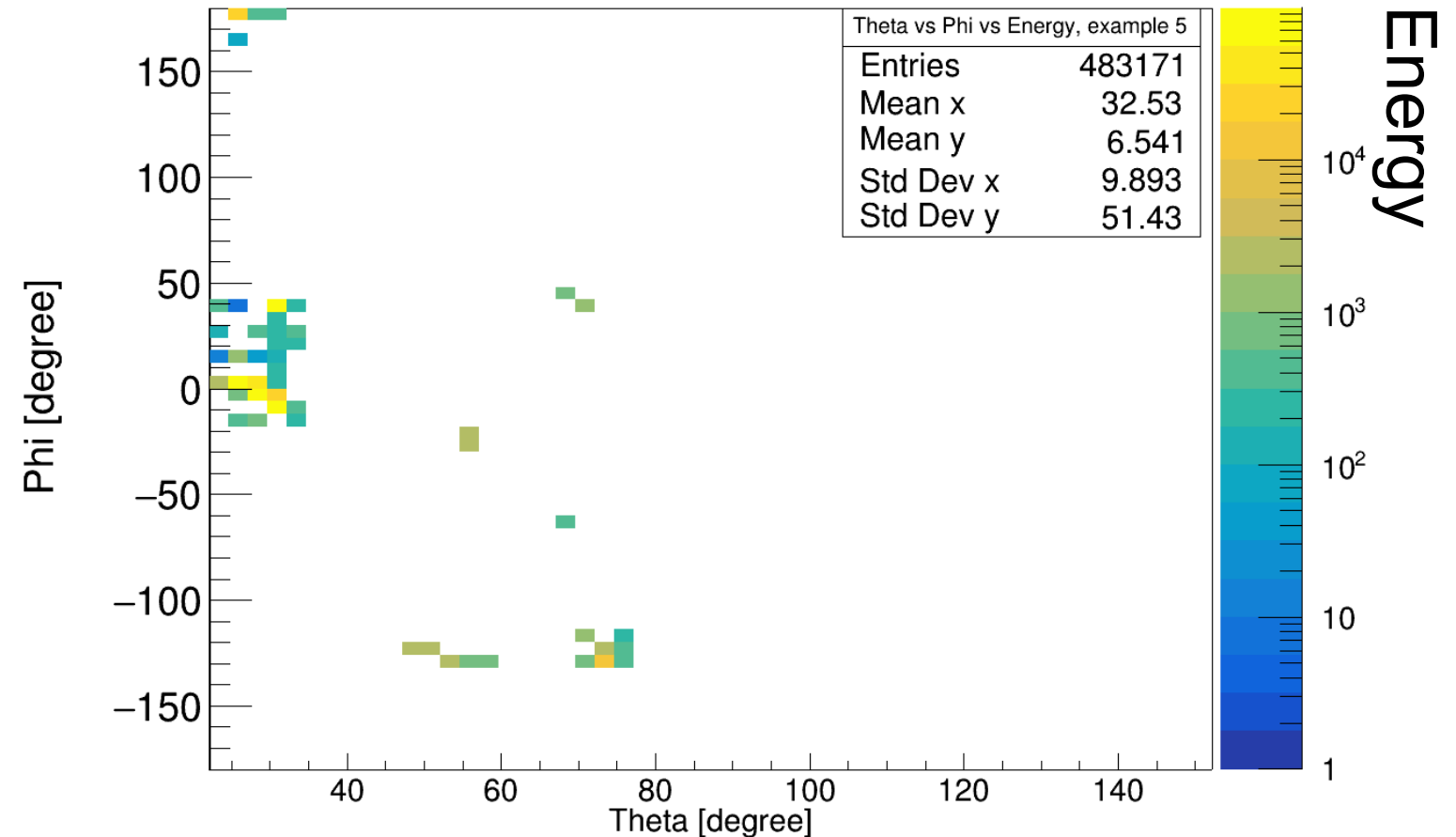
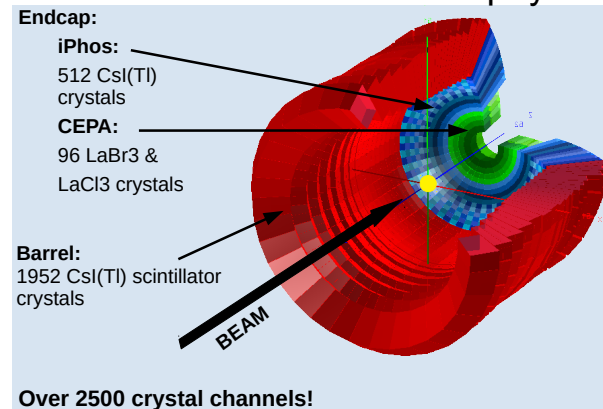
This lecture:
Formalizing this intuition.

Not just an artificial question...

What can you say about this?

CALIFA

Calorimeter data for nuclear physics

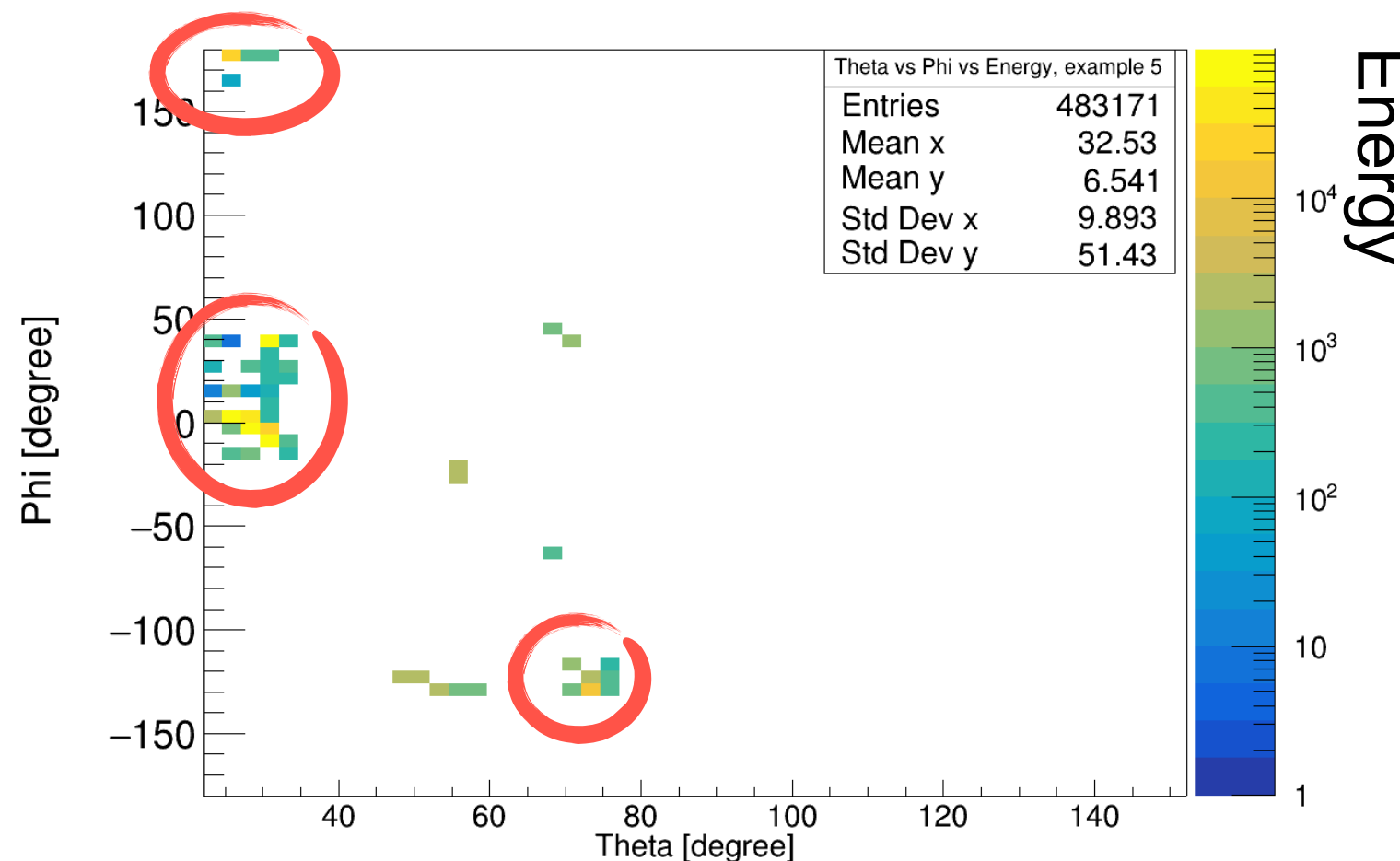
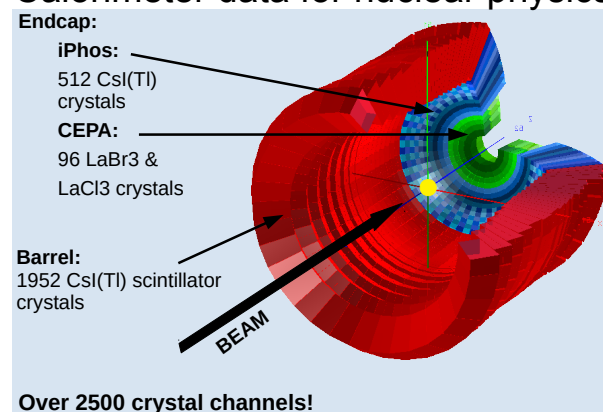


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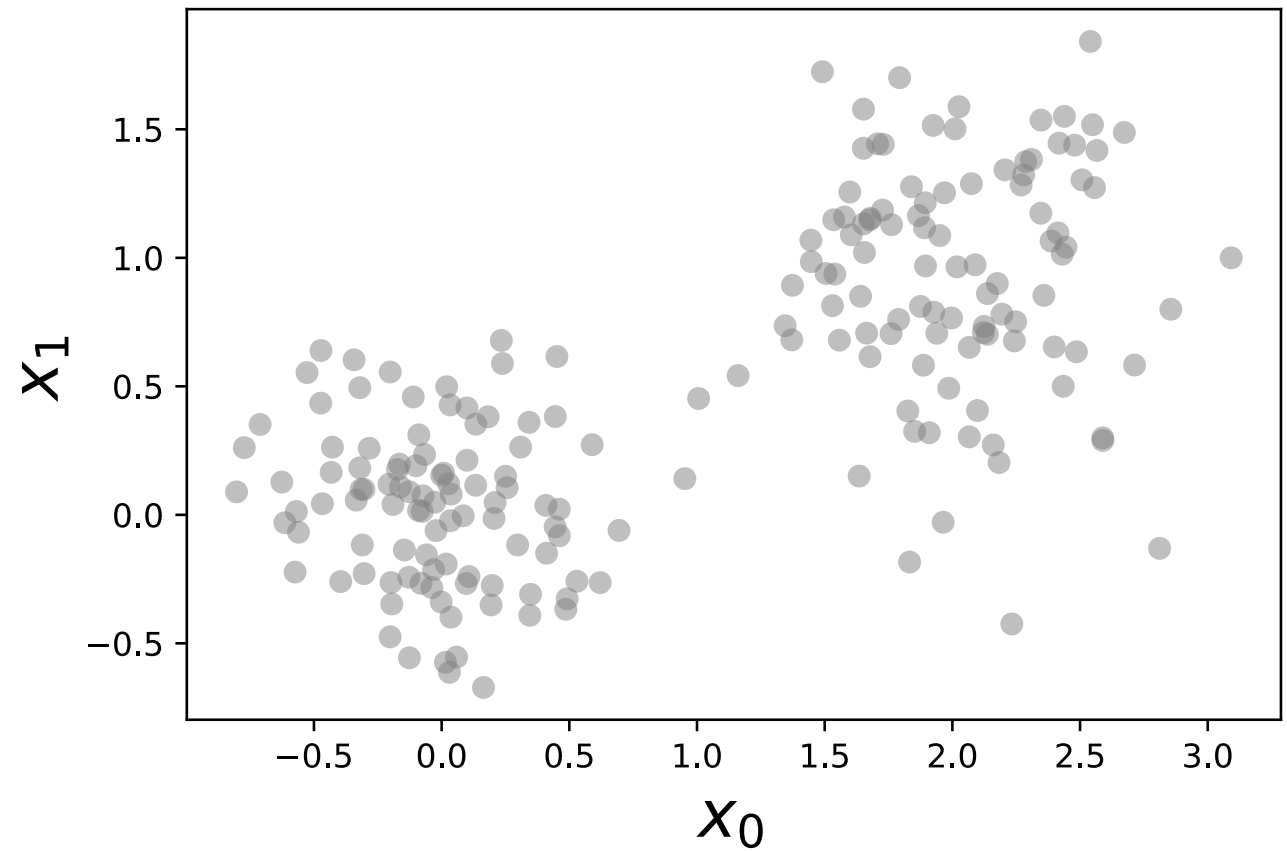
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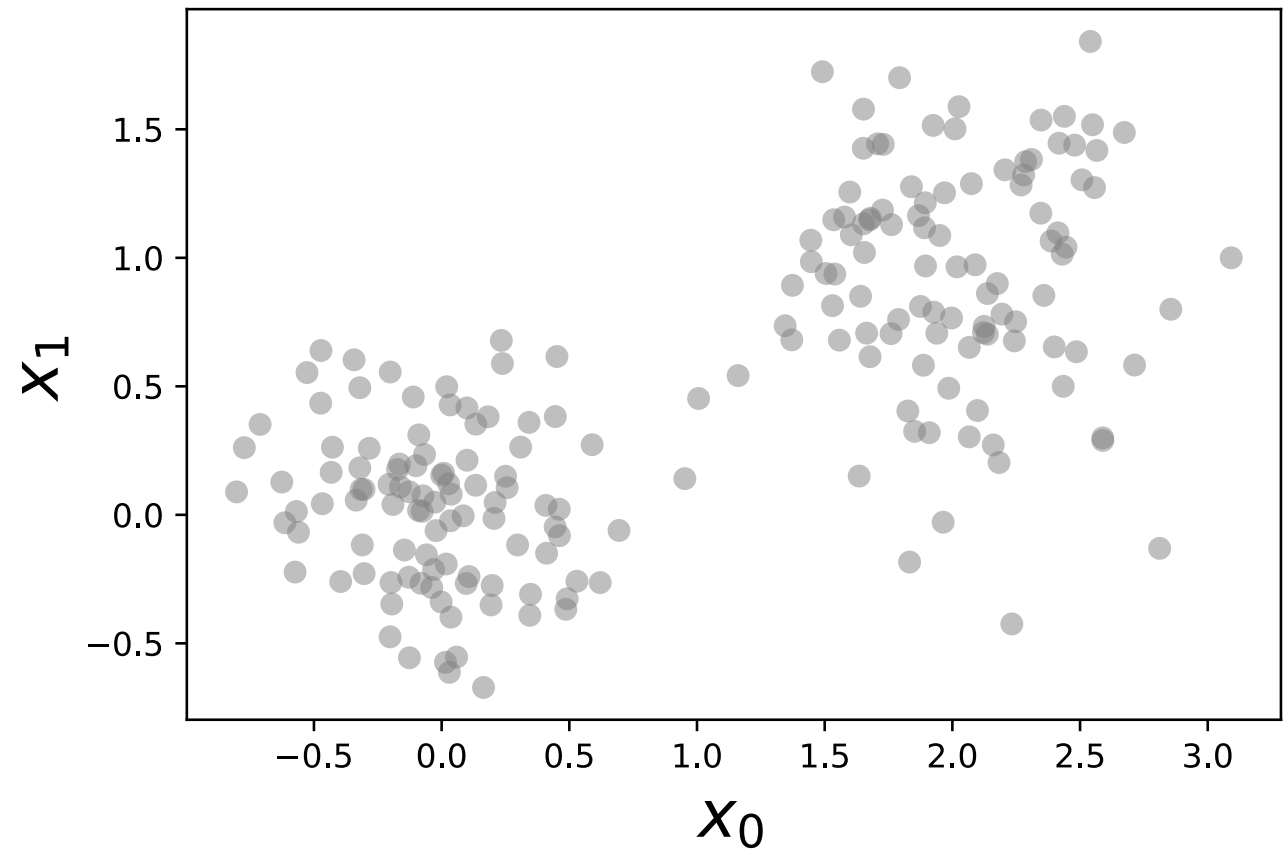
Research question:
Formalizing this intuition.

Example: K-means



Example: K-means

INPUT: # of clusters, K
This example, use $K = 2$.



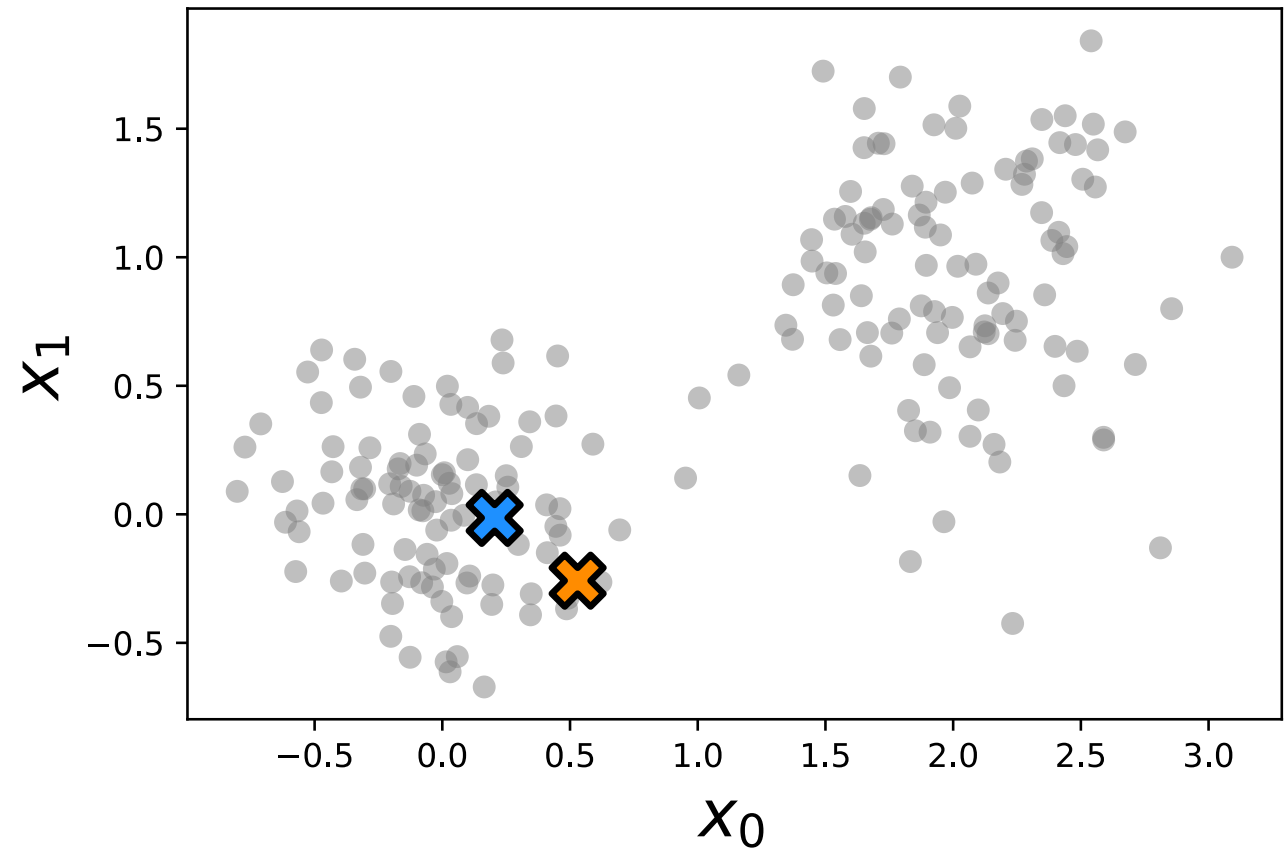
Example: K-means

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1. Initialize cluster centers

Randomly choose K input points for the centroids $\mu_1, \dots, \mu_K$



Example: K-means

iter 0

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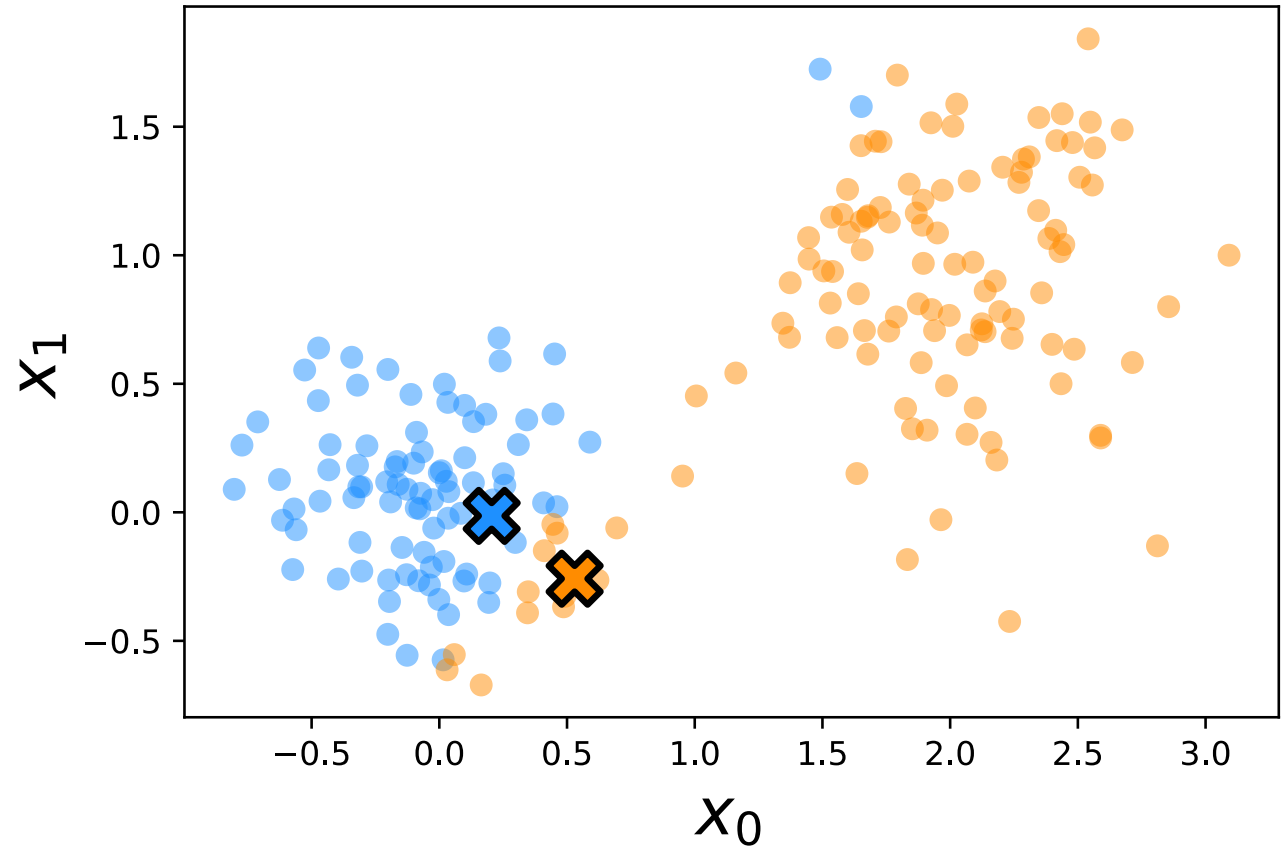
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2. Repeat until convergence:

Assign data points to closest centroids

$$c^{(i)} := \arg \min_j ||x^{(i)} - \mu_j||^2$$



Example: K-means

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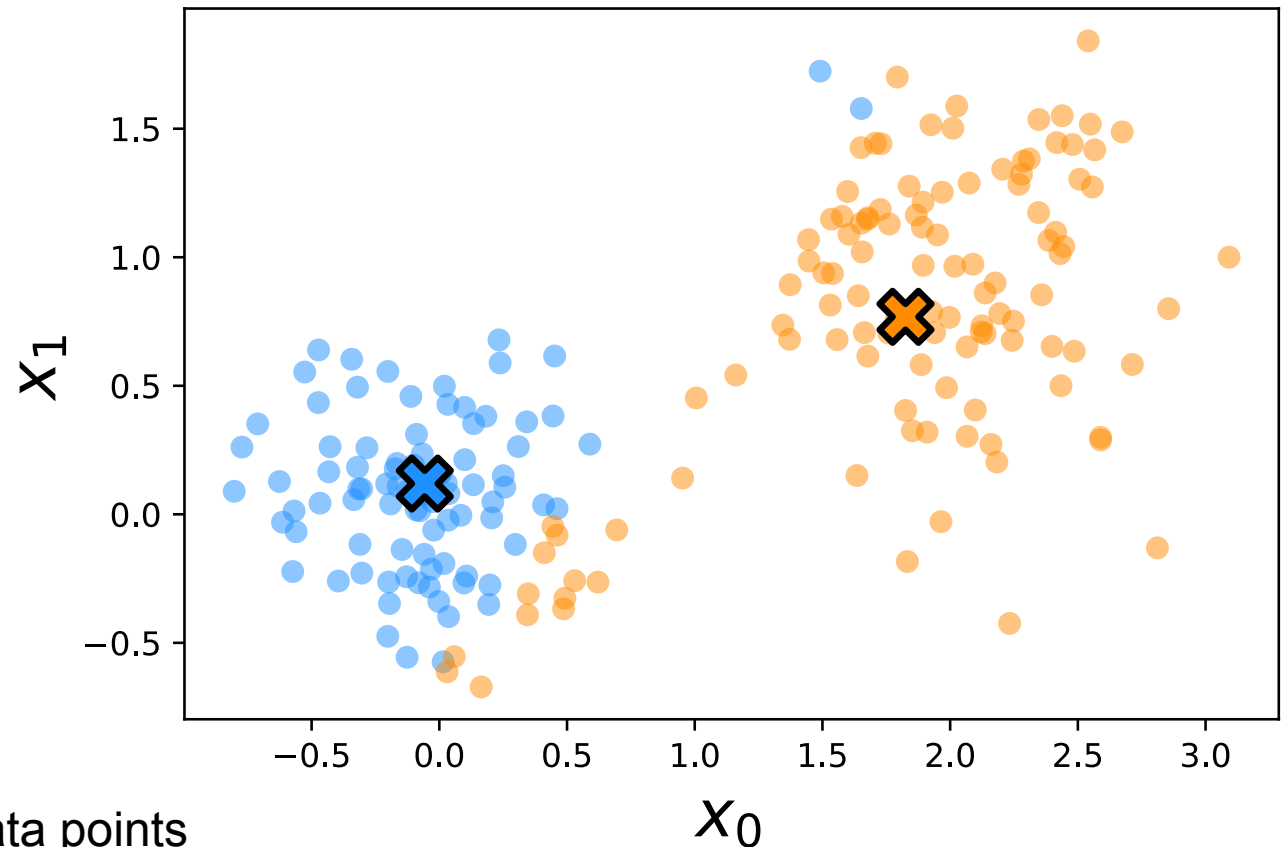
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Reset centroids to mean of associated data points

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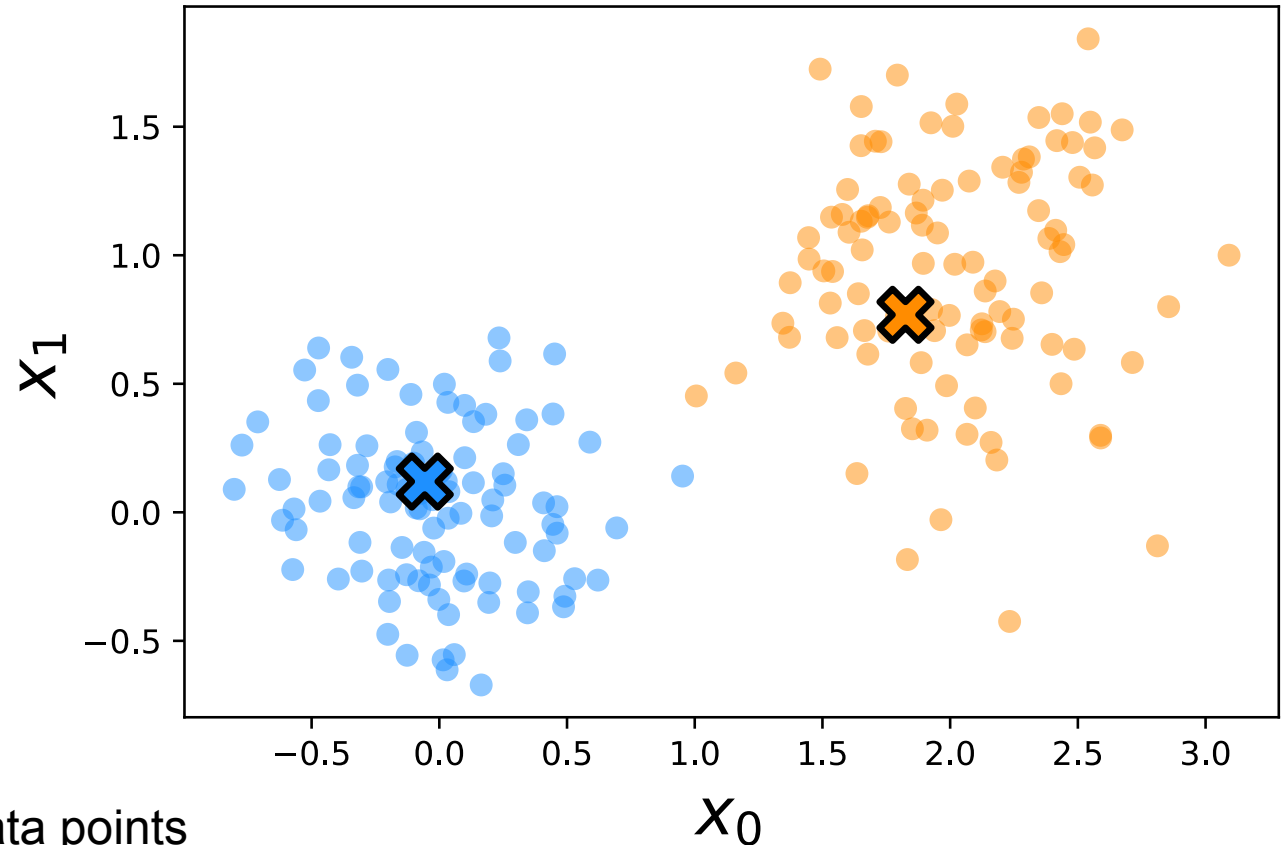
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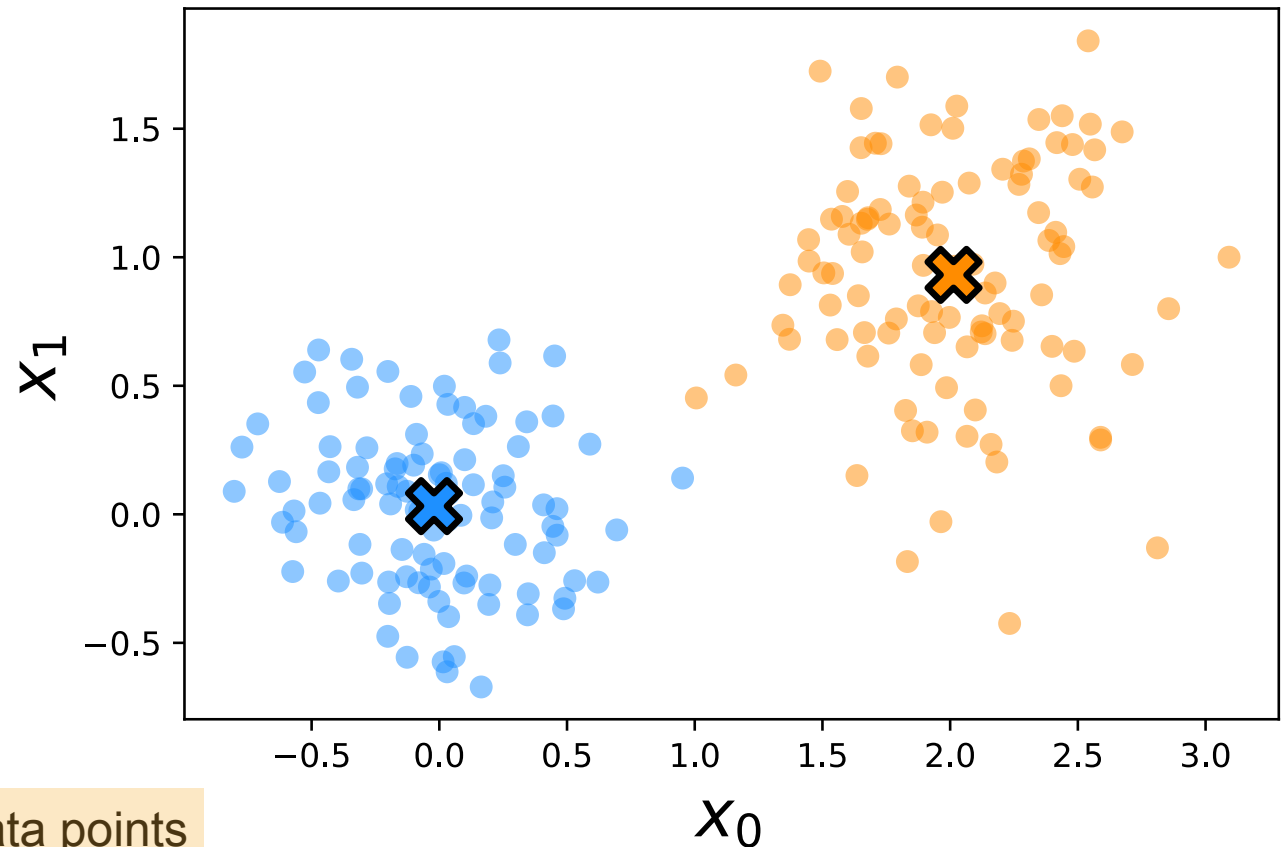
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iter 2

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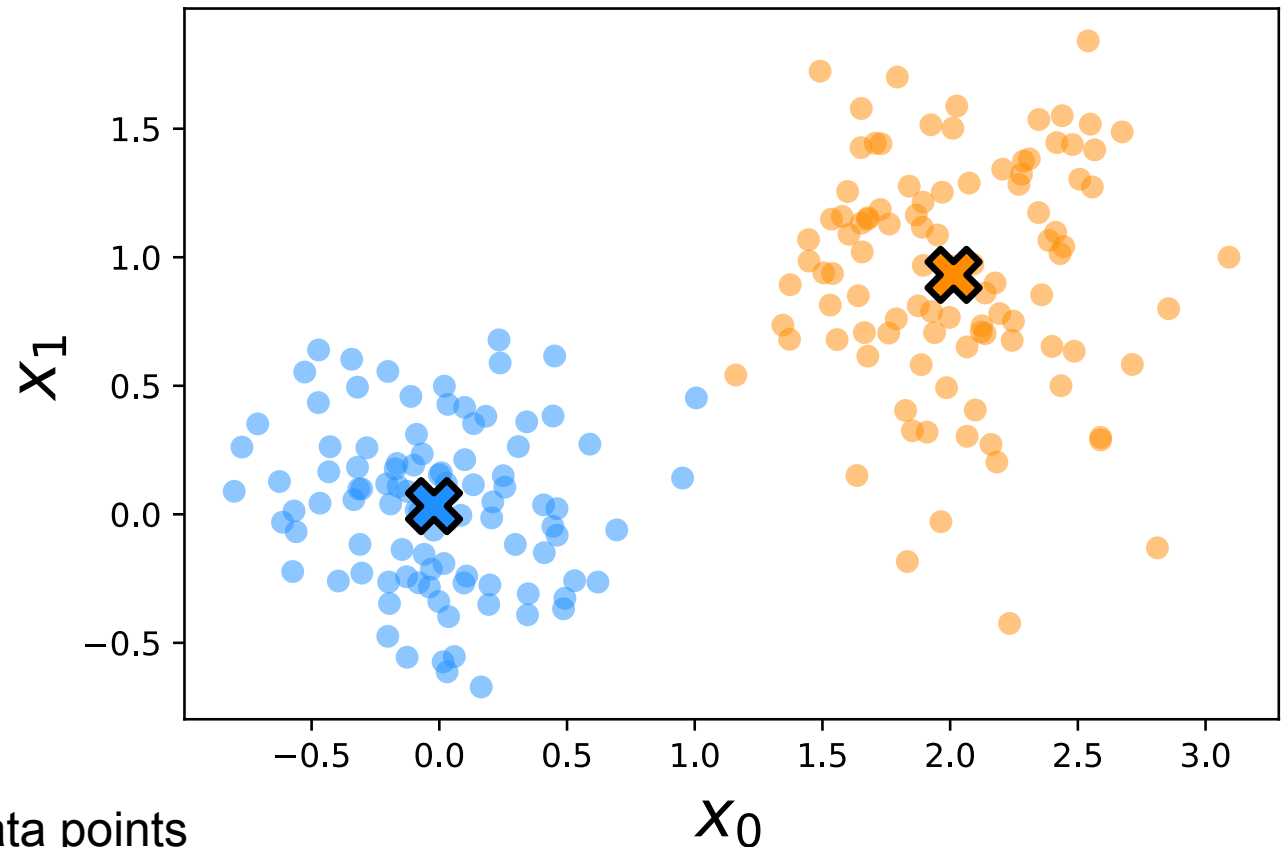
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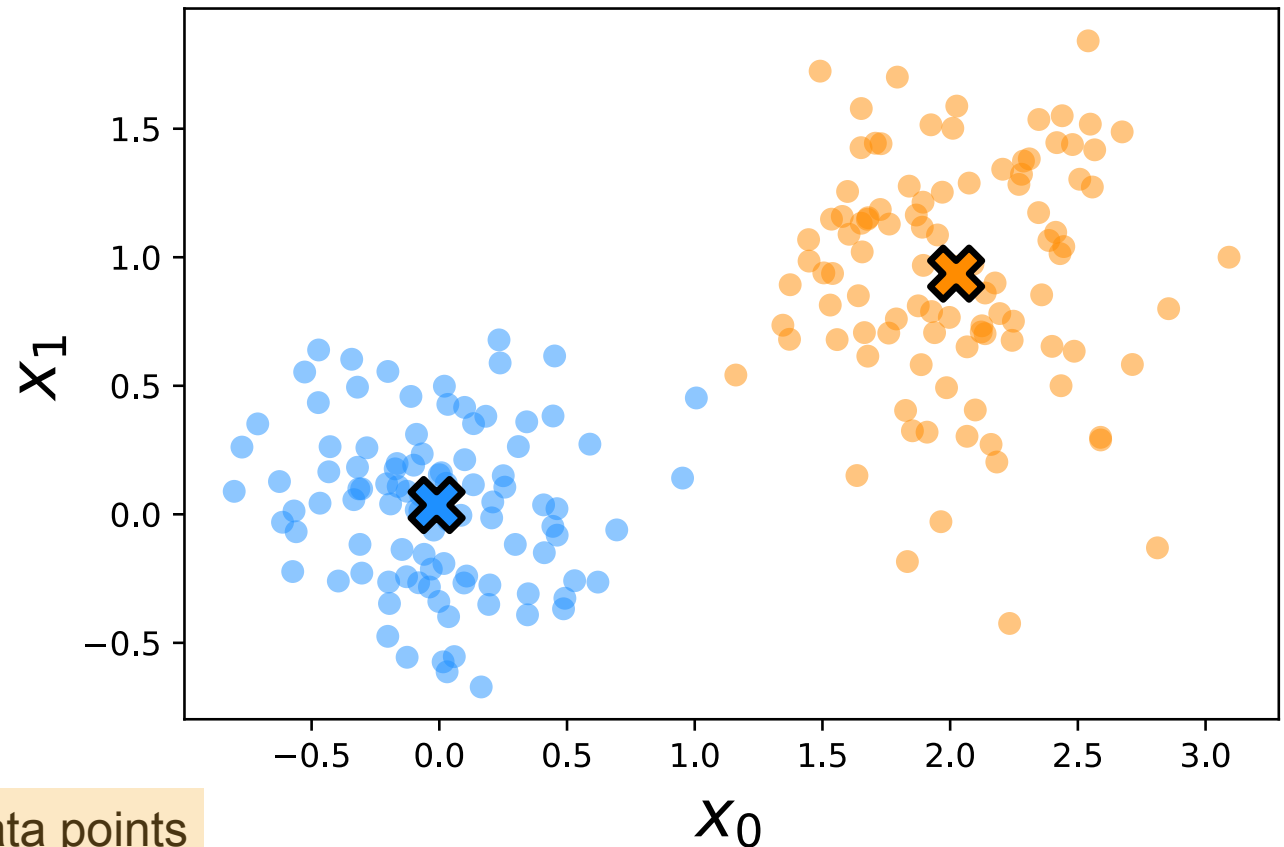
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Example: K-means

iter 3

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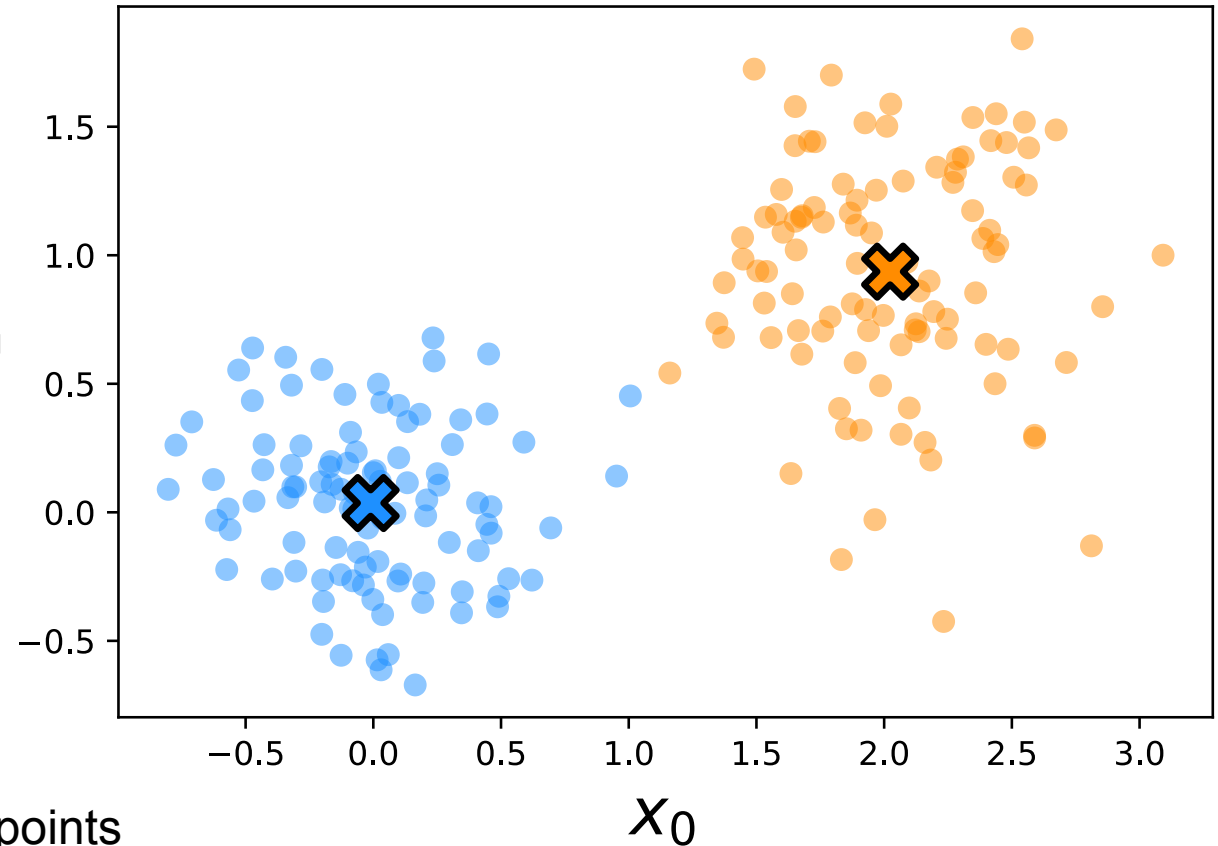
2. Repeat until convergence. ✓ x_1

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No change!

K-means algorithm

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2. Repeat until convergence:

Expectation:

Assign data points to closest centroids

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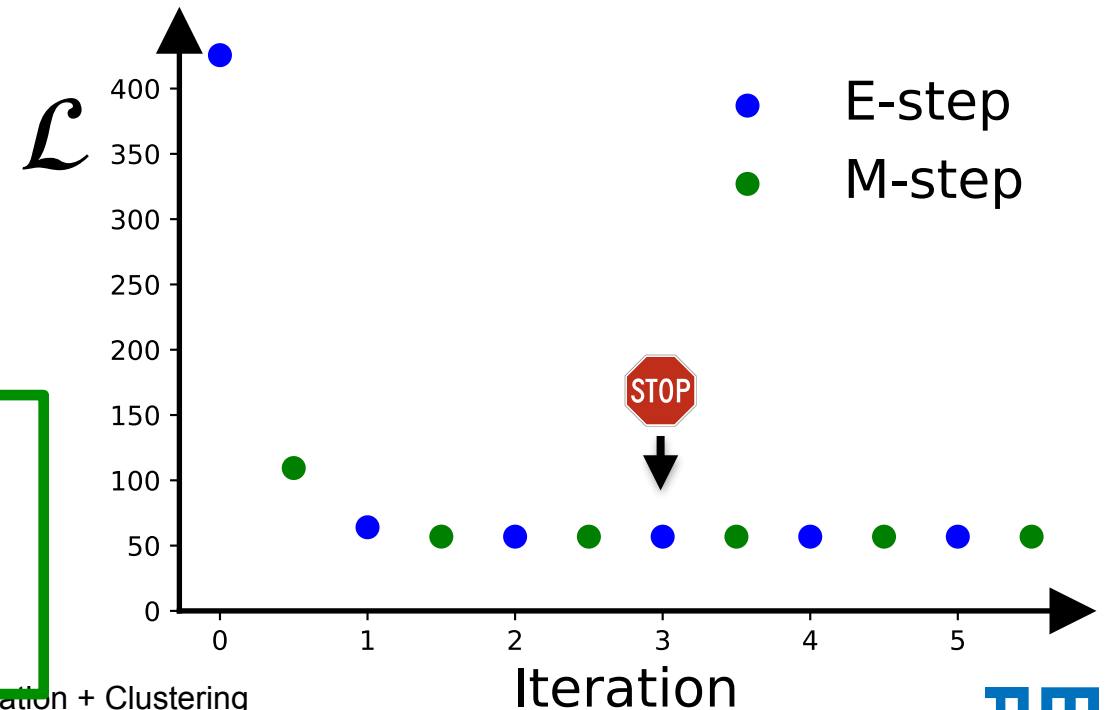
Maximization:

Reset centroids to mean of associated data points

$$\mu_j := \frac{\sum_{i=1}^n 1\{c^{(i)} = j\} x^{(i)}}{\sum_{i=1}^n 1\{c^{(i)} = j\}}$$

$$\mathcal{L}(c, \mu) = \sum_{i=1}^n ||x^{(i)} - \mu_{c^{(i)}}||^2$$

Cluster association
Cluster centers



Your turn!!



1+3 min

Q: Which step (the **E-step** or the **M-step**) uses gradients of $\mathcal{L}$?

Or, more explicitly:

Can you derive the centroid assignment from $\nabla_{c^{(i)}} \mathcal{L}$?

— or —

Can you derive the centroid update with $\nabla_{\mu_j} \mathcal{L}$?

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Derivation:

$$\nabla_{\mu_j} \mathcal{L} = -2 \sum_{i=1}^n (x^{(i)} - \mu_{c^{(i)}}) \cdot 1\{c^{(i)} = j\} = 0$$

$$\sum_{i=1}^n 1\{c^{(i)} = j\} x^{(i)} = \sum_{i=1}^n 1\{c^{(i)} = j\} \mu_j$$

$$\mu_j = \frac{\sum_{i=1}^n 1\{c^{(i)} = j\} x^{(i)}}{\sum_{i=1}^n 1\{c^{(i)} = j\}}$$

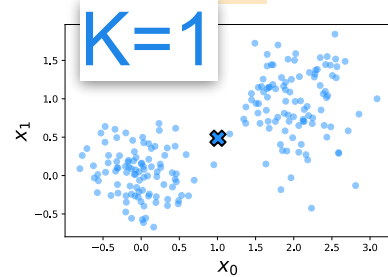
How to choose the complexity of the model?

INPUT: # of clusters, K

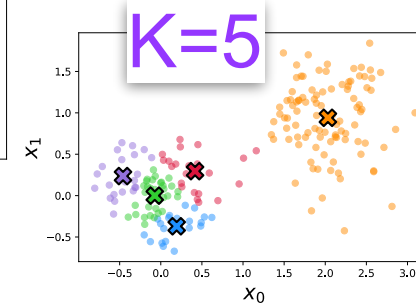
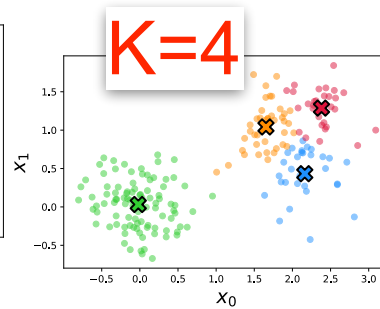
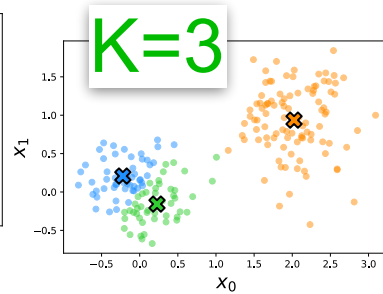
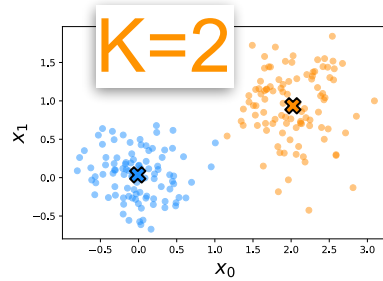
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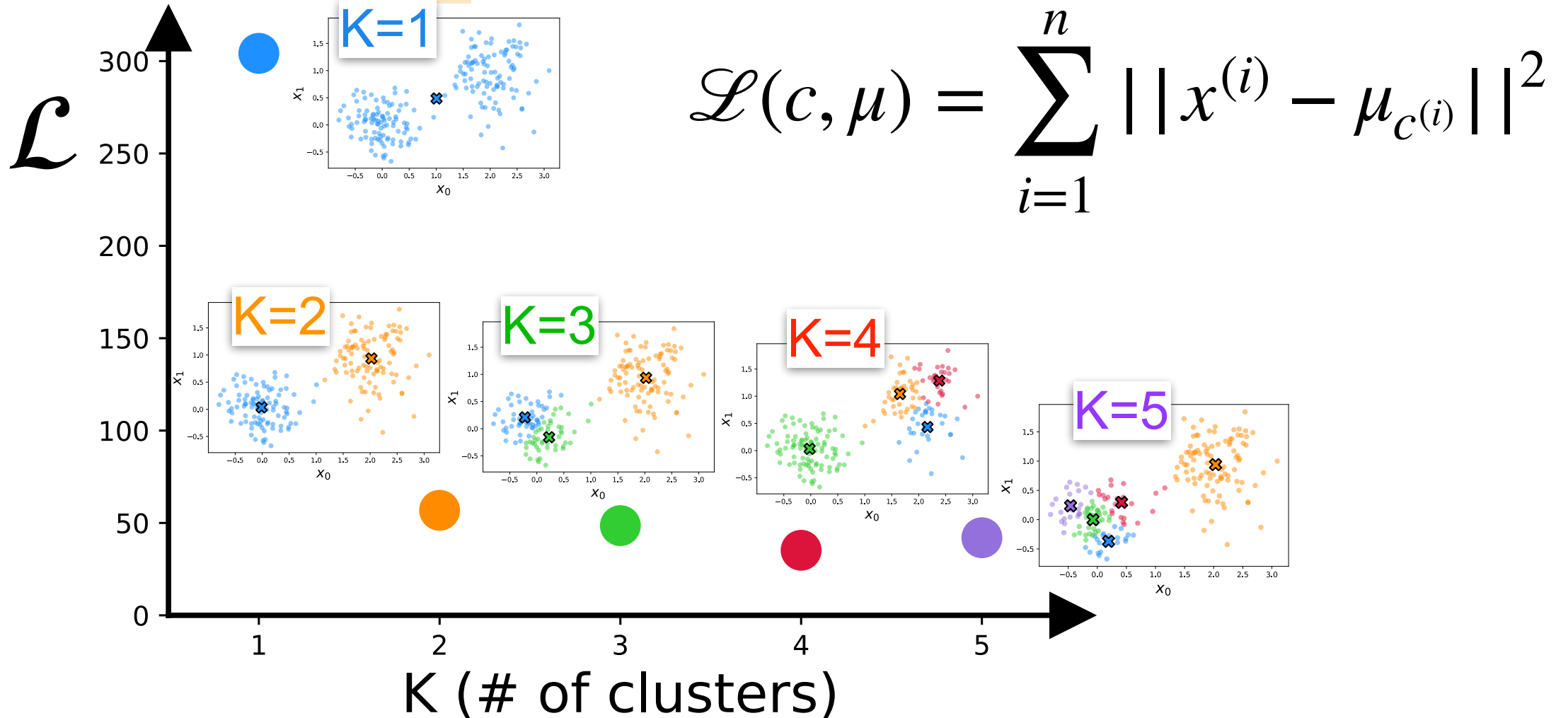


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Lecture 6: Classification + Clustering

How to choose the complexity of the model?

INPUT: # of clusters, K

Large $\Delta \mathcal{L}_2$
choose K=2
clusters

$$\mathcal{L}(c, \mu) = \sum_{i=1}^n ||x^{(i)} - \mu_{c^{(i)}}||^2$$

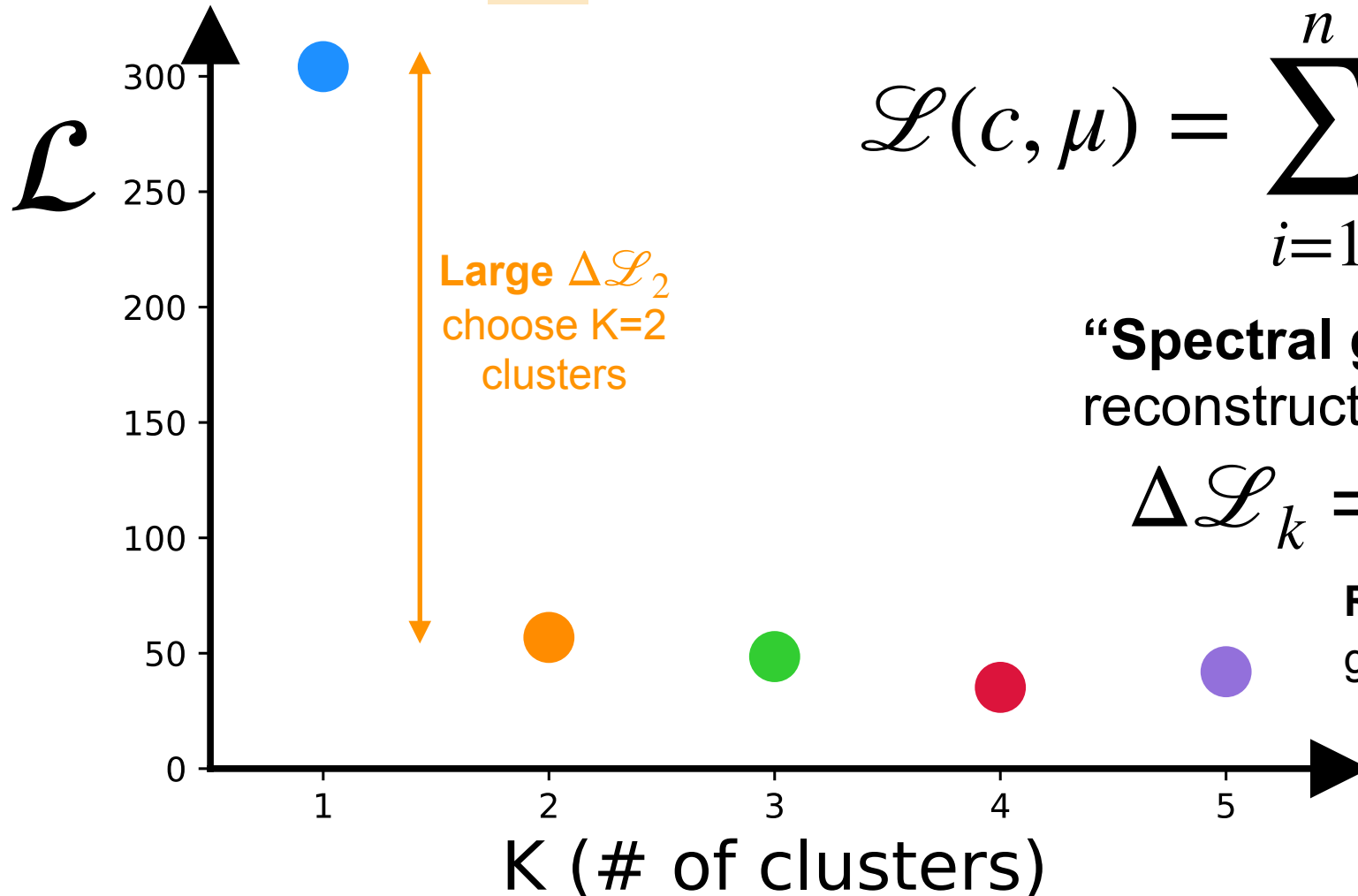
“Spectral gap”: Difference b/w the reconstruction errors for two K

$$\Delta \mathcal{L}_k = \mathcal{L}_k - \mathcal{L}_{k-1}$$

Rule of thumb: Large spectral gap $\implies$ stable clustering

How to choose the complexity of the model?

INPUT: # of clusters, K



$$\mathcal{L}(c, \mu) = \sum_{i=1}^n ||x^{(i)} - \mu_{c^{(i)}}||^2$$

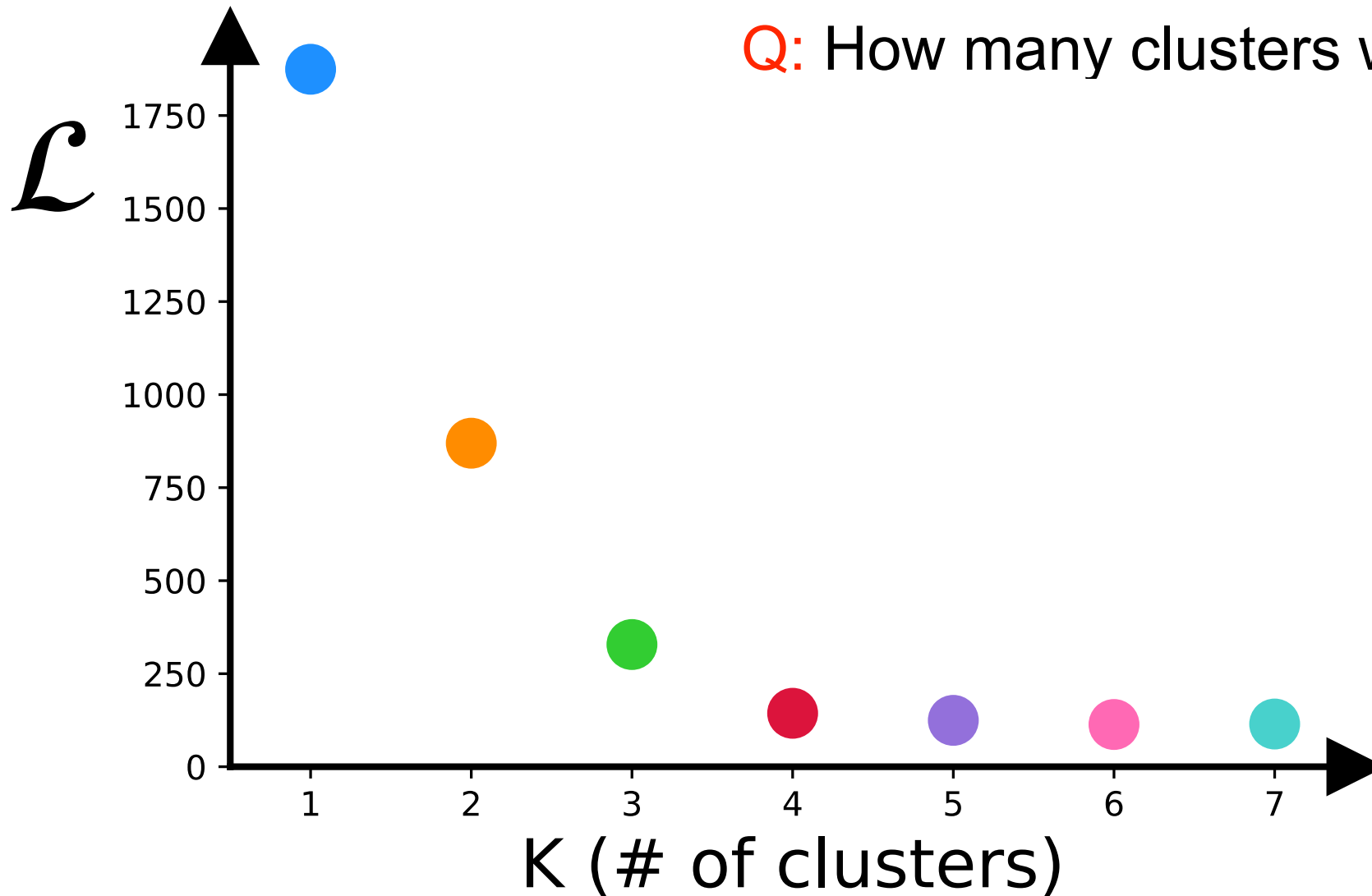
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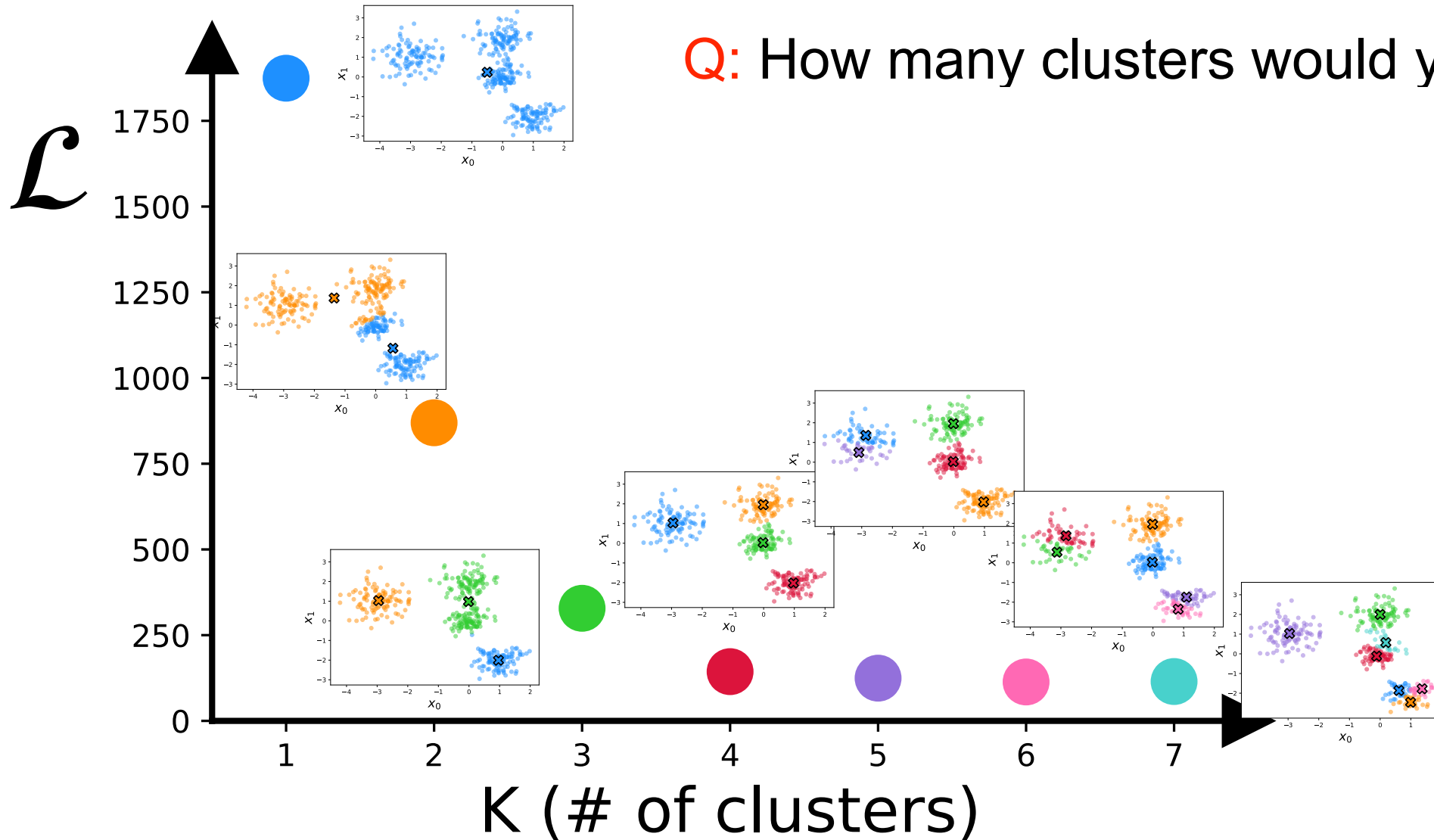
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Your turn!! Clustering a *new* data set with K-means....

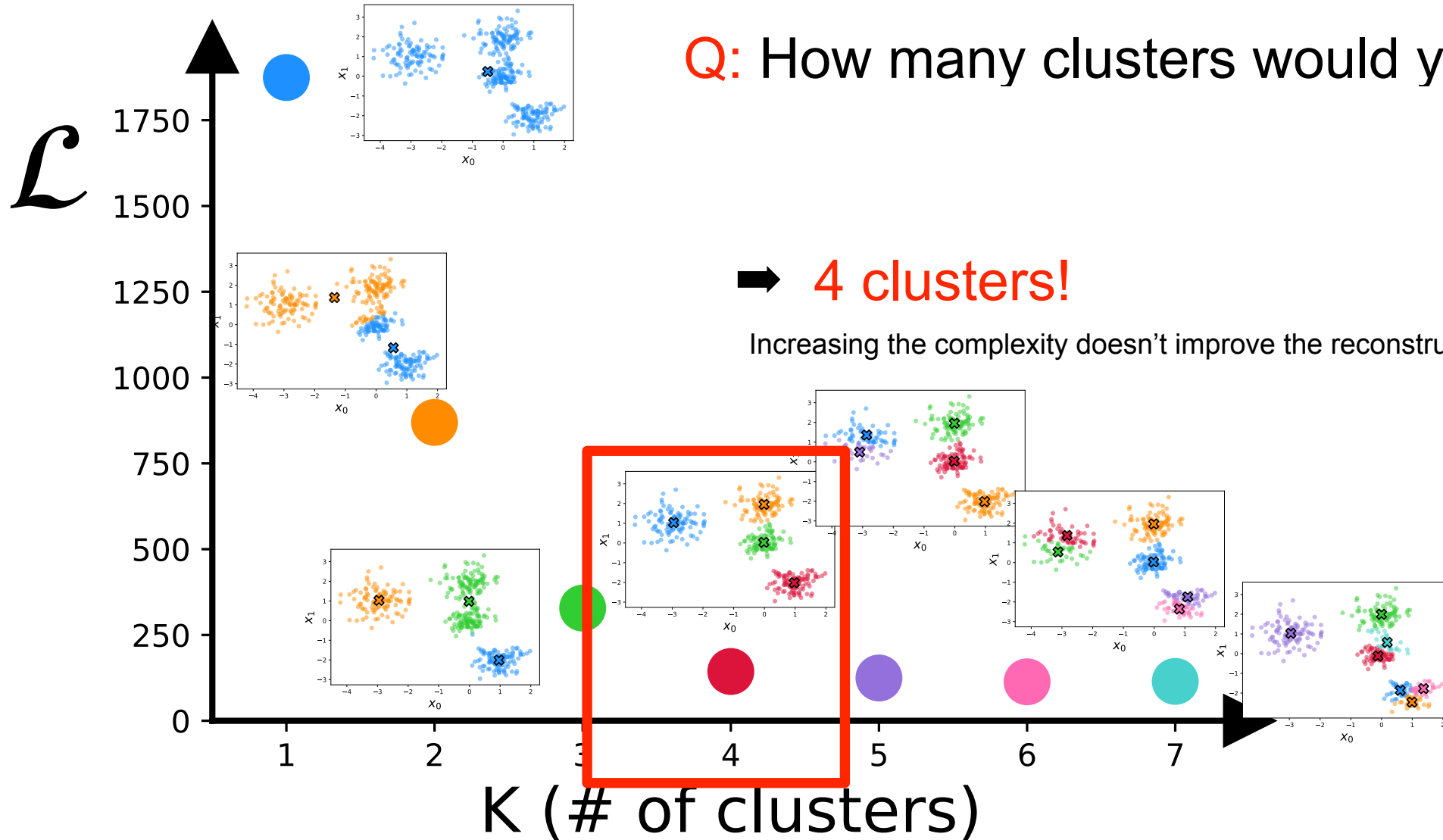
Q: How many clusters would you pick?



Your turn!! Clustering a *new* data set with K-means.... 1+1 min



Your turn!! Clustering a *new* data set with K-means.... 1+1 min



Outline

1

K-means

2

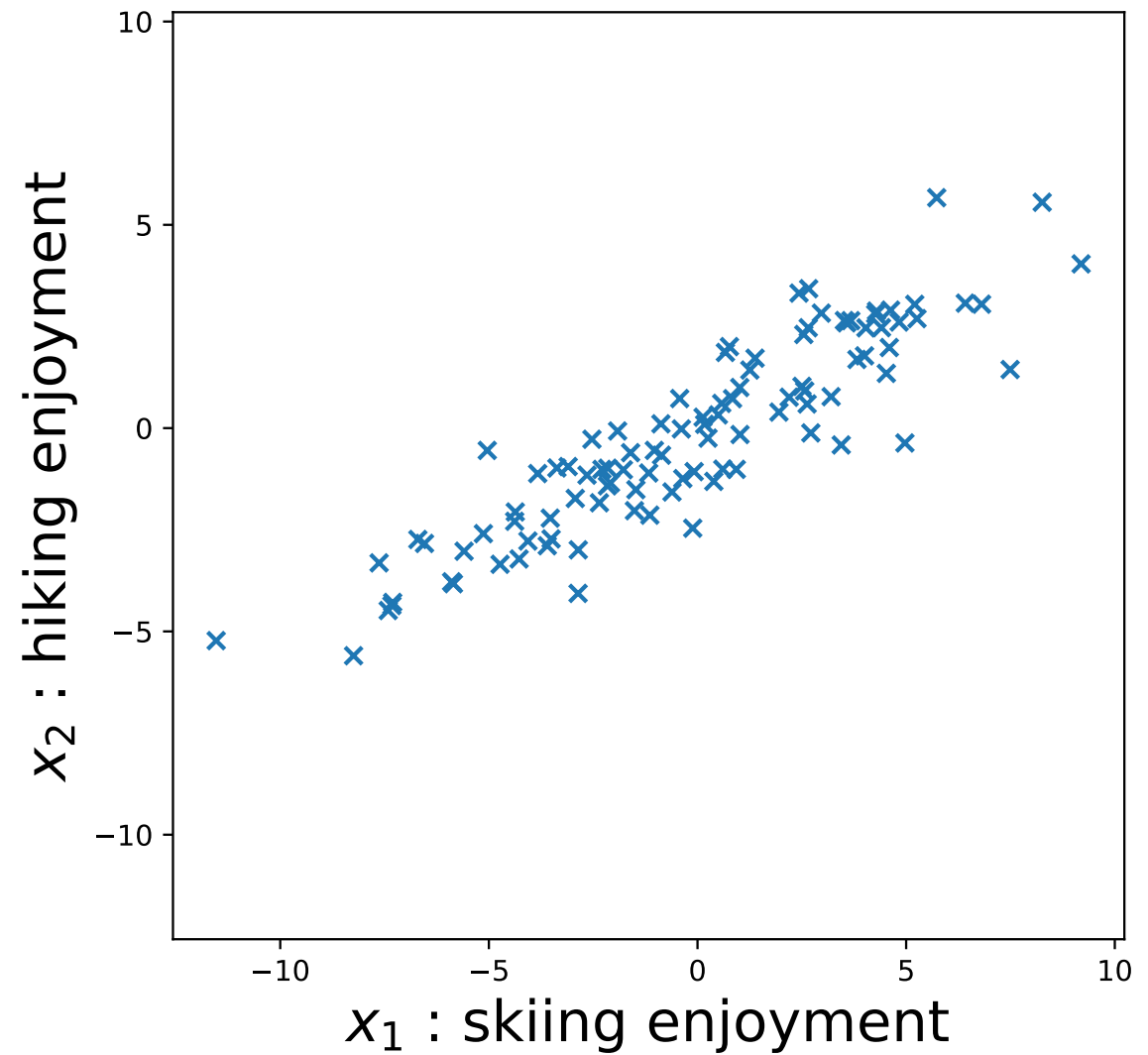
Principle Component Analysis

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Learning theory

Motivating example

Learn $p(X)$

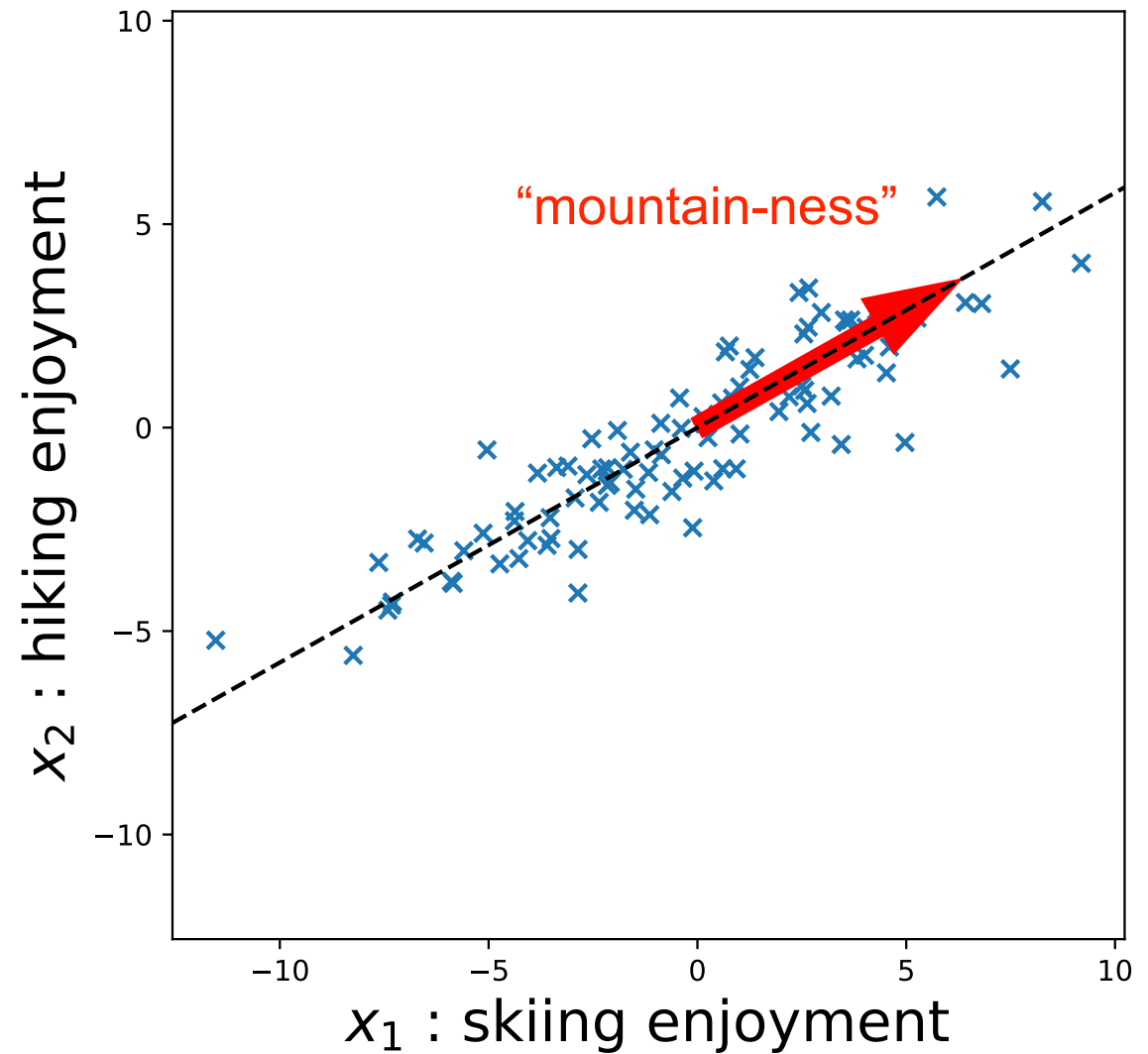


Motivating example

Learn $p(X)$



How to formalize this?



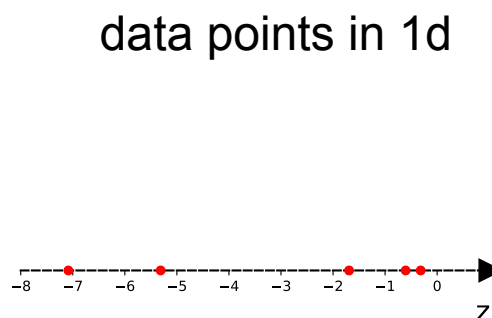
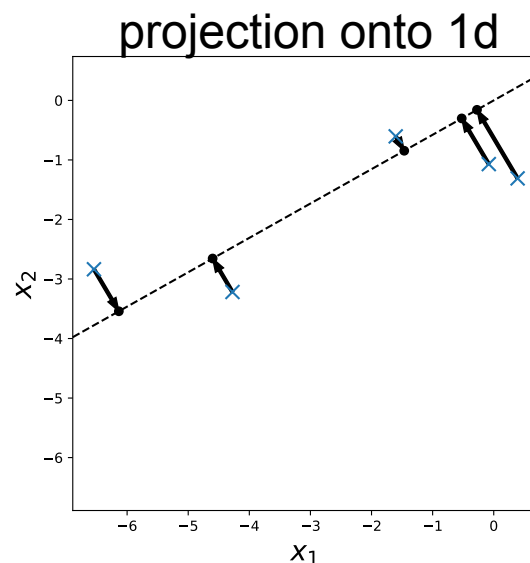
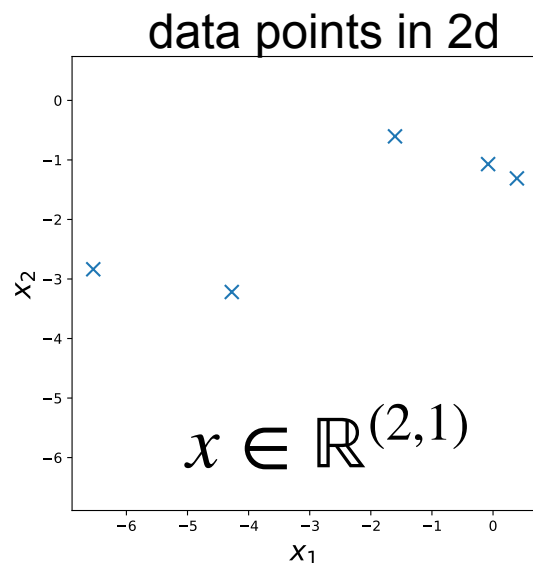
How do we do this?

Learn $p(x)$, minimize the reco error:

$$MSE = \frac{1}{N} \sum_{i=1}^N \left(x^{(i)} - \boxed{x_{||}^{(i)}} \right)^2$$

“reconstructed x”

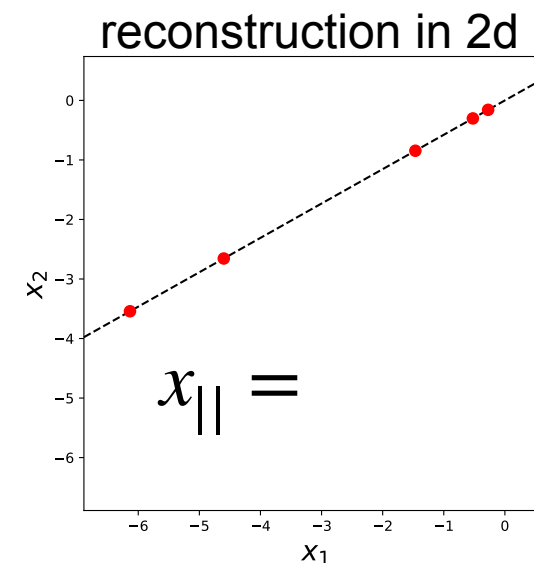
Let $v \in \mathbb{R}^{(2,1)}$ be the projection to the lower dim space



$$z = v^T x$$

1 x 2 2 x 1

(1d subspace)



How do we represent the 1d projection back in 2d again?
A.k.a, what is the formula for $x_{||}$?

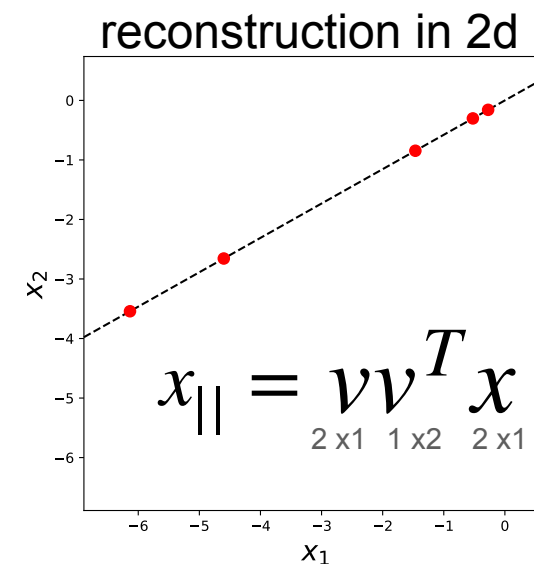
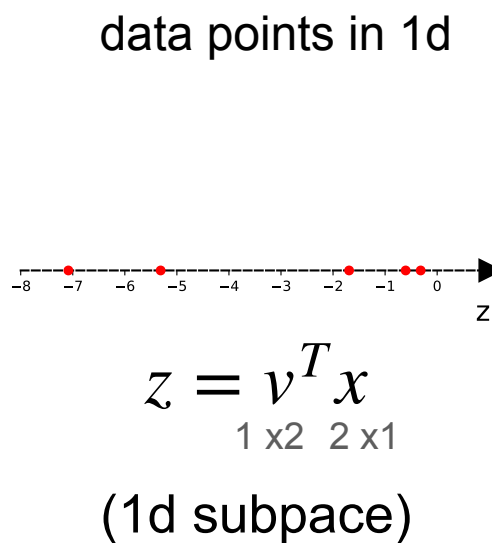
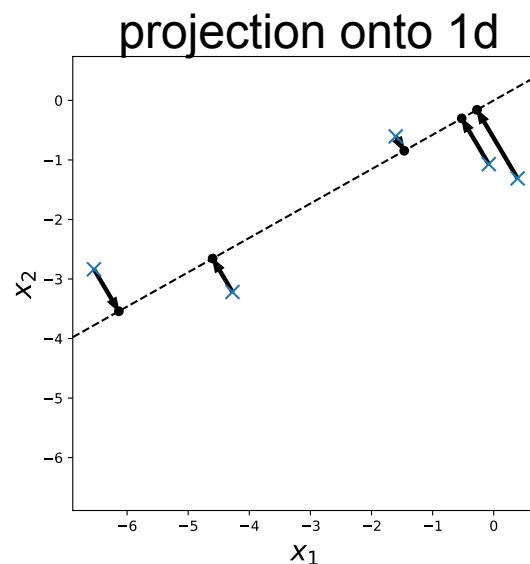
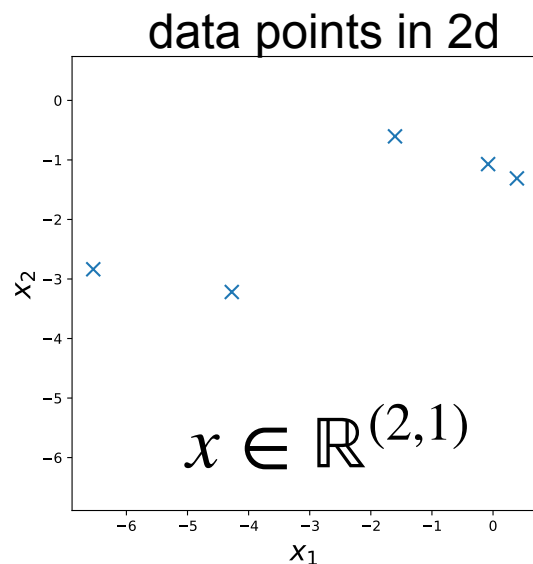
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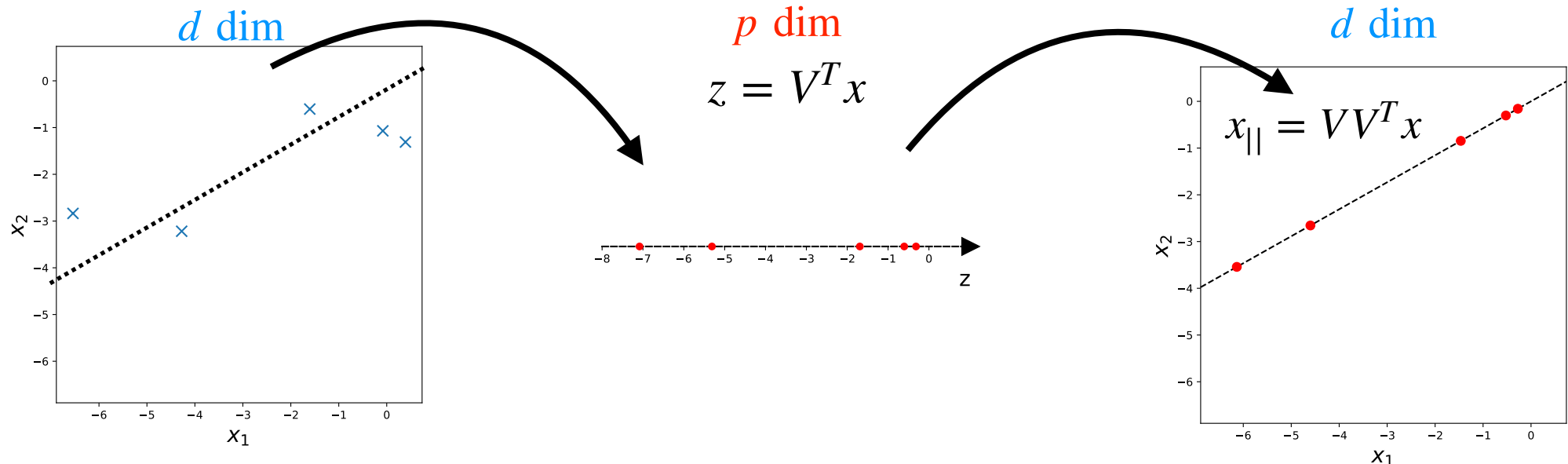
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Generalization to higher dimensions!!

- The points live in a high dim space, $x \in \mathbb{R}^{d \times 1}$
 - Assume the inputs have $\langle x \rangle = 0$ (already mean subtracted)
- Let $V \in \mathbb{R}^{d \times p}$ be the matrix such that V^T projects down to the p-dim subspace.
- Then $x_{||} = VV^T x$ is the projection back to the high dim space



Variance and Reconstruction Error

- Zero mean, the variance is the 2nd moment
 - Assume the inputs have $\langle x \rangle = 0$ (already mean subtracted)
- So the reconstruction error is:

$$E = \langle (x - x_{||})^T (x - x_{||}) \rangle = \langle (x - VV^T x)^T (x - VV^T x) \rangle$$

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 &= \langle x^T x \rangle - \langle z^T z \rangle
 \end{aligned}$$

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 &= \langle x^T x \rangle - \boxed{\langle z^T z \rangle}
 \end{aligned}$$

Maximizing the **variance of the reconstructed data** $\leftrightarrow$ minimizing the reconstruction error

How to solve for V ?

How do we know what this projection matrix is?

- Let $V = (v_1, v_2, \dots, v_p) \in \mathbb{R}^{d,p}$, with v_k the projection axes.
- Maximize variance means choosing these v_k to maximize:

$$\frac{1}{n} \sum_{i=1}^n (v_k^T x^{(i)})^2 = \frac{1}{n} \sum_{i=1}^n v_k^T x^{(i)} (x^{(i)})^T v_k = v_k^T \underbrace{\left(\frac{1}{n} \sum_{i=1}^n x^{(i)} (x^{(i)})^T \right)}_{\Sigma} v_k = v_k^T \Sigma v_k$$

Σ : empirical covariance matrix

See with the Method of Lagrange multipliers if you let $\mathcal{L} = v^T \Sigma v - \lambda v^T v$.

Then $\nabla_v \mathcal{L} =$

How to solve for V ?

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Σ : empirical covariance matrix

The v_k are the eigenvectors of Σ with the k largest eigenvalues!

See with the Method of Lagrange multipliers if you let $\mathcal{L} = v^T \Sigma v - \lambda v^T v$.

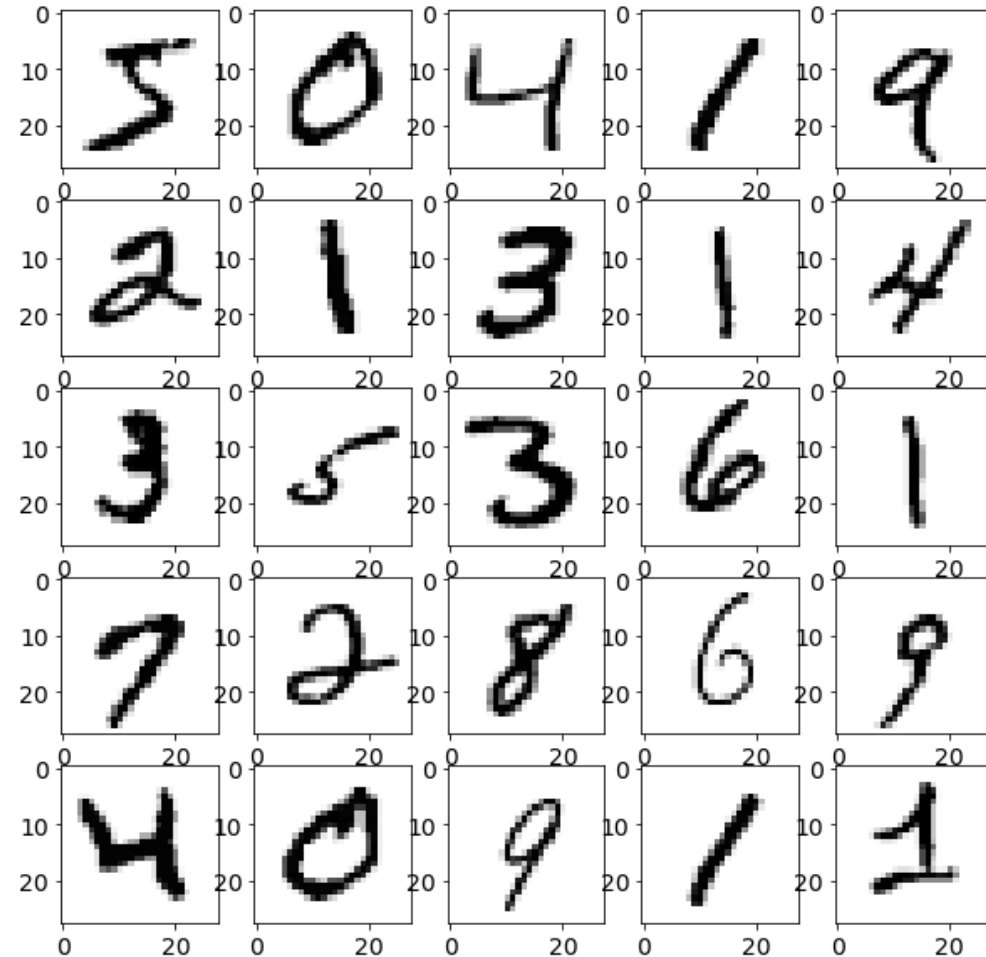
Then $\nabla_v \mathcal{L} = 2\Sigma v - 2\lambda v = 0 \implies \Sigma v = \lambda v$

Your turn: PCA

$$X = \mathbb{R}^{N \times 784}$$

Fun linear algebra Q: Will there be a more succinct way to write projections and covariances over all of these N training examples?

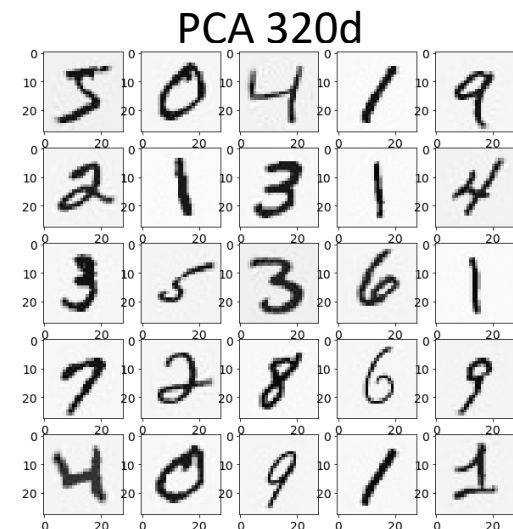
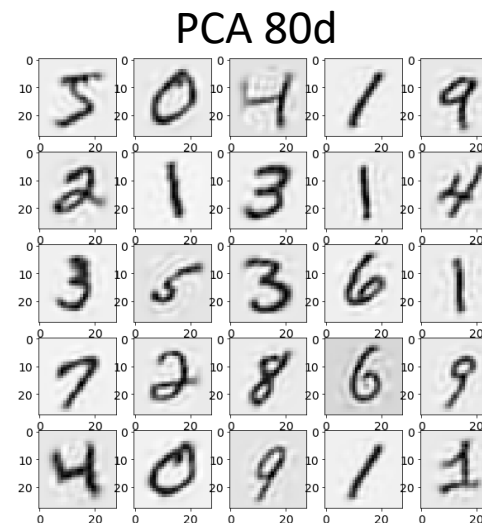
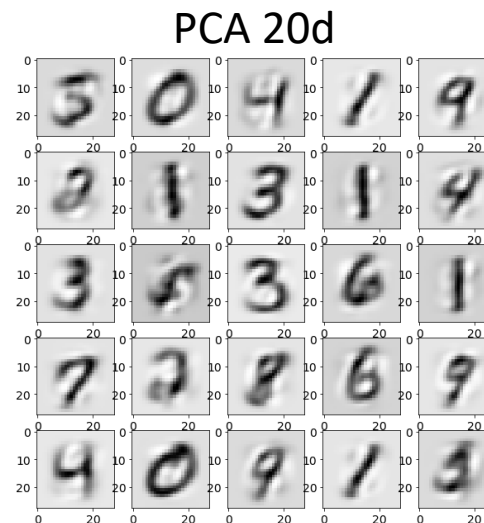
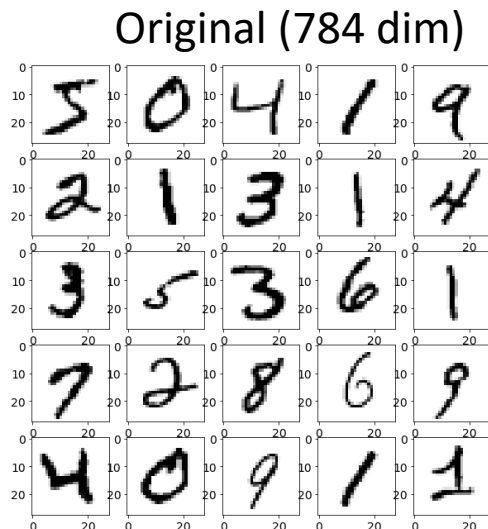
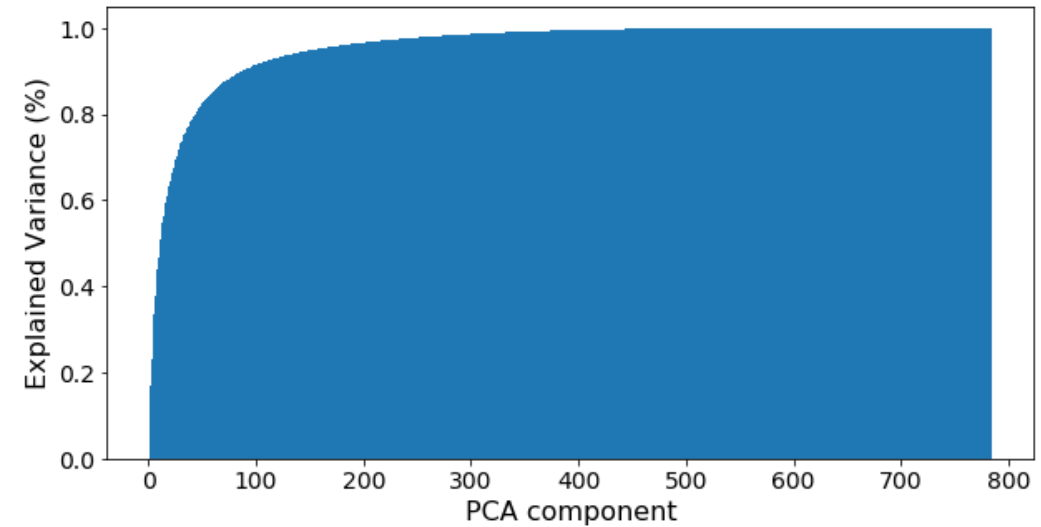
Original (784 dim)



Generalization to higher dimensions!!

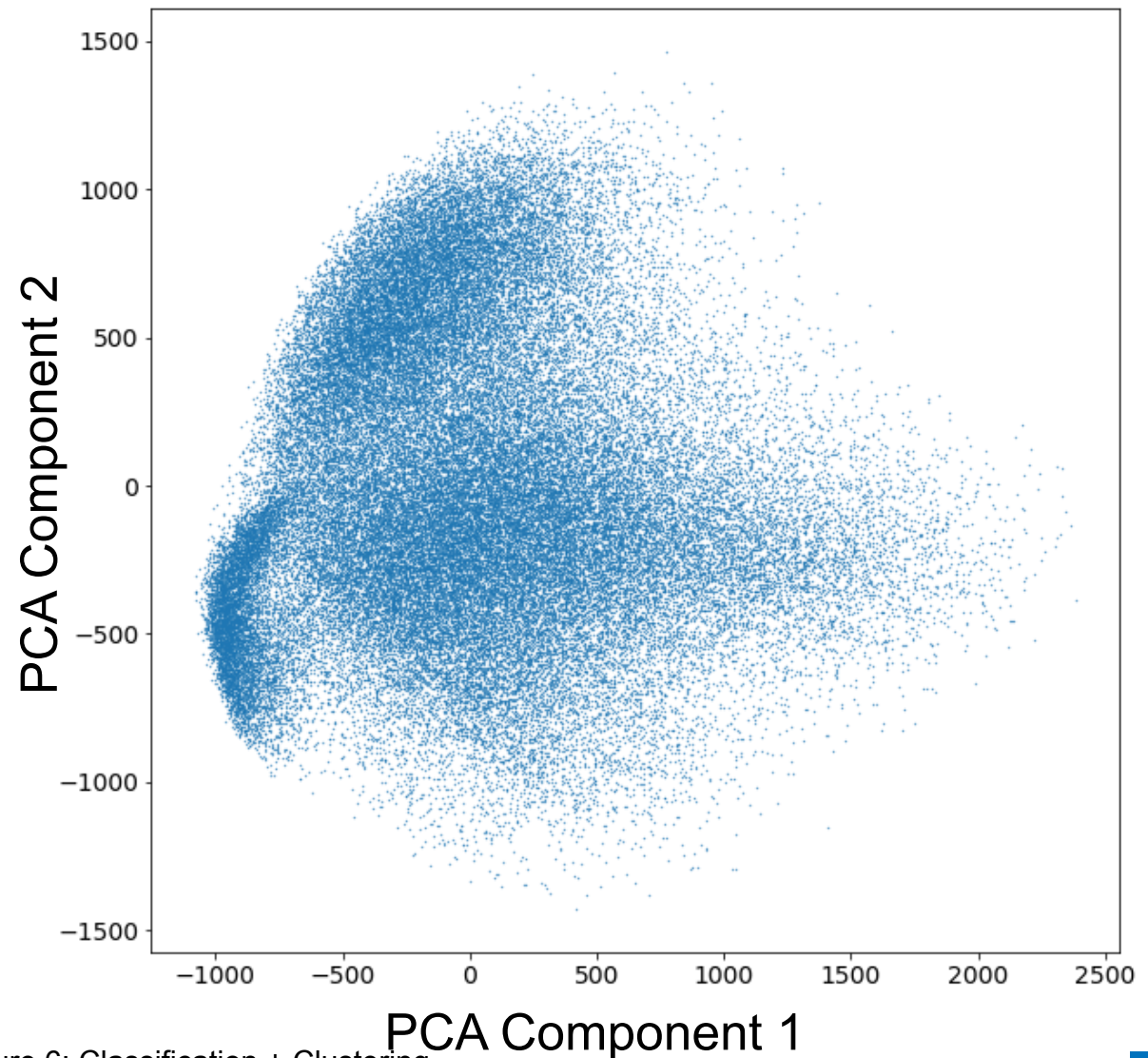
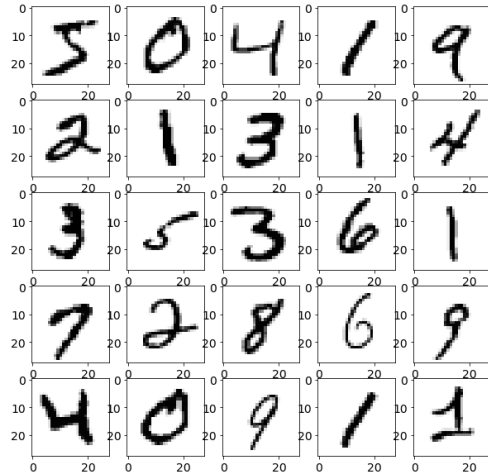
Goal: Minimize the error /
maximize the variation

Q: How well did we do??

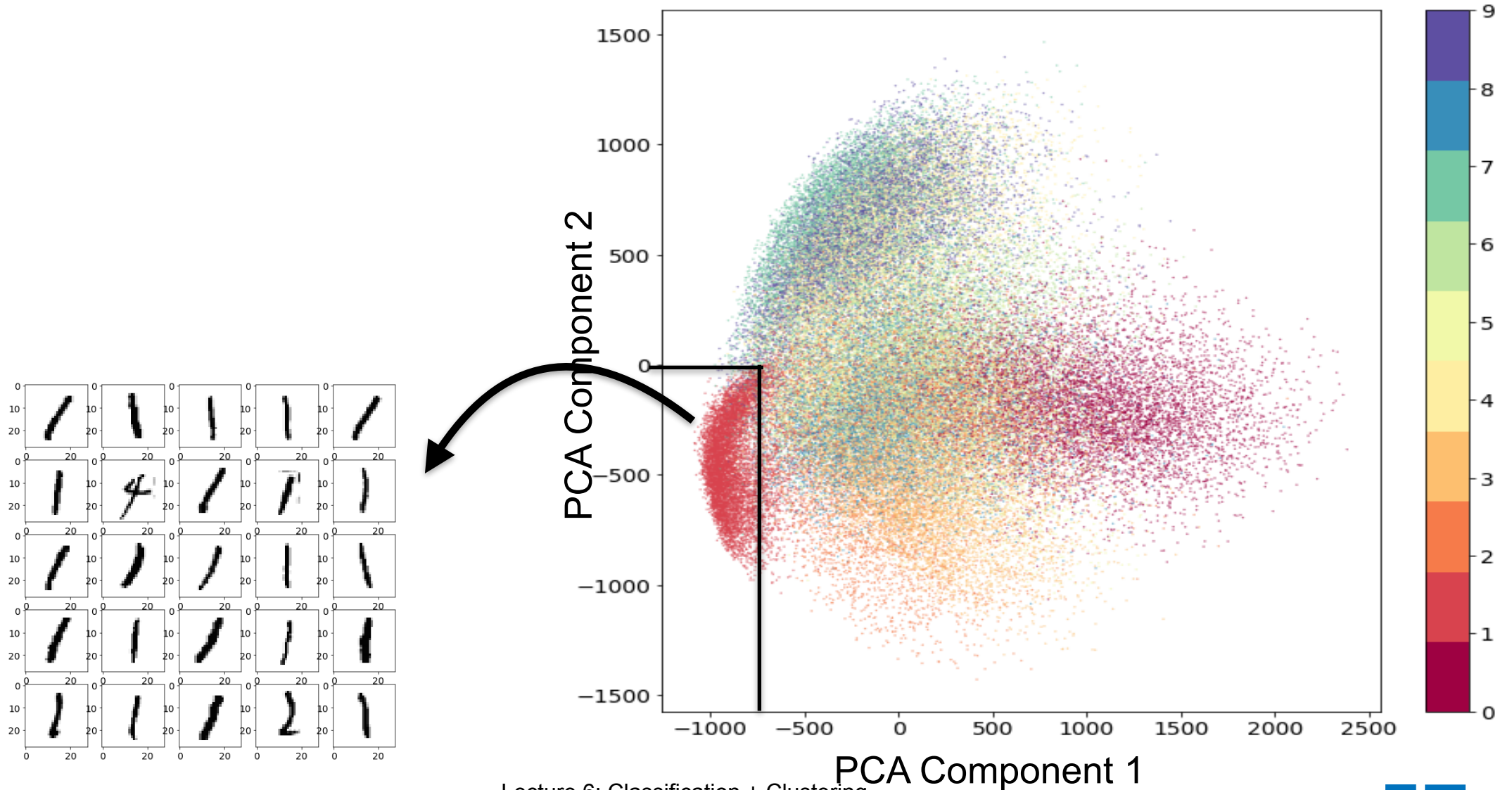


Core concept in ML: Data Visualization

Visualize the first two PCA components: are already revealing structure in the MNIST images!



Core concept in ML: Data Visualization



Outline

1

K-means

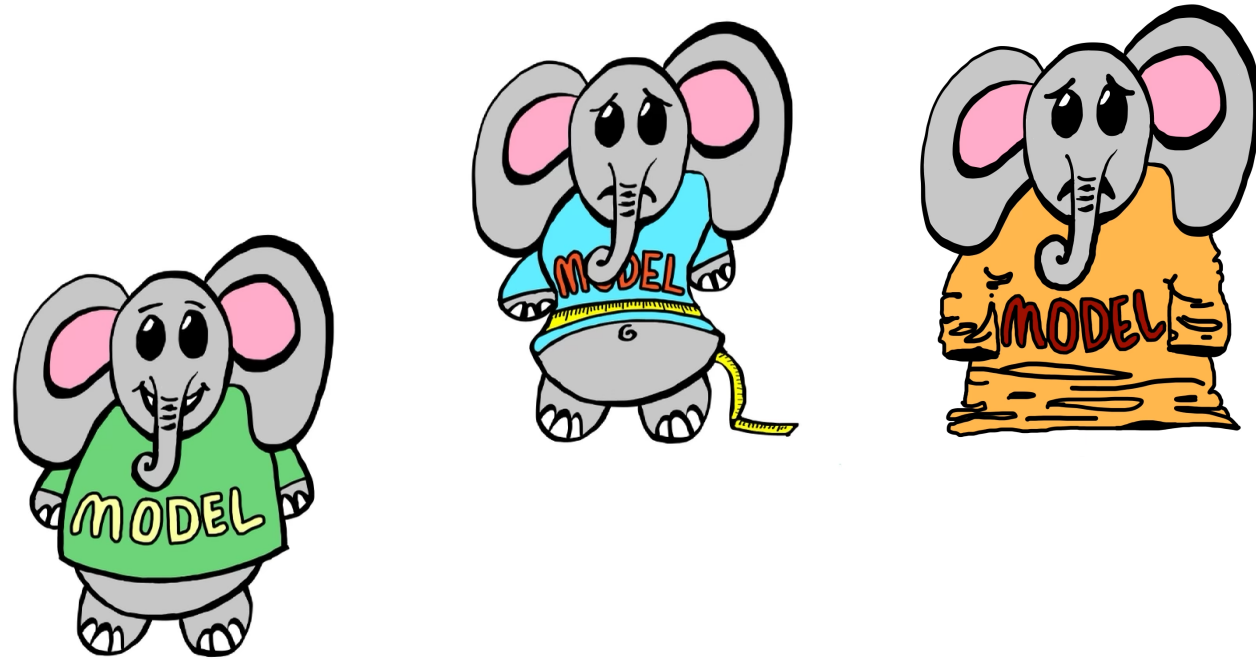
2

Principle Component Analysis

3

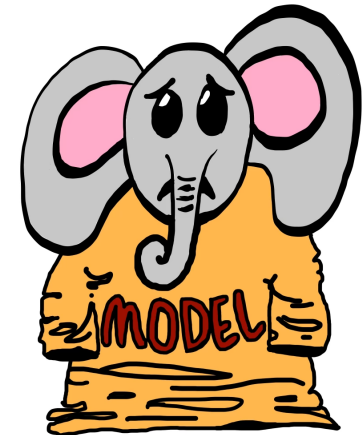
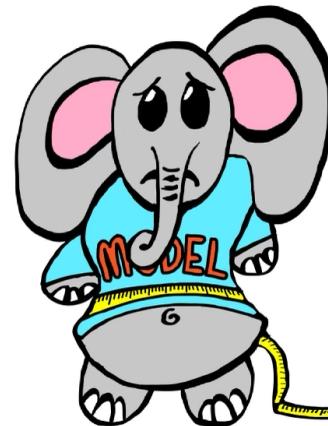
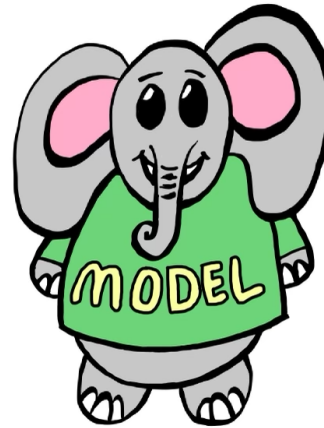
Learning theory

In the context of science, the well-known adage “a picture is worth a thousand words” might well be “a model is worth a thousand datasets”

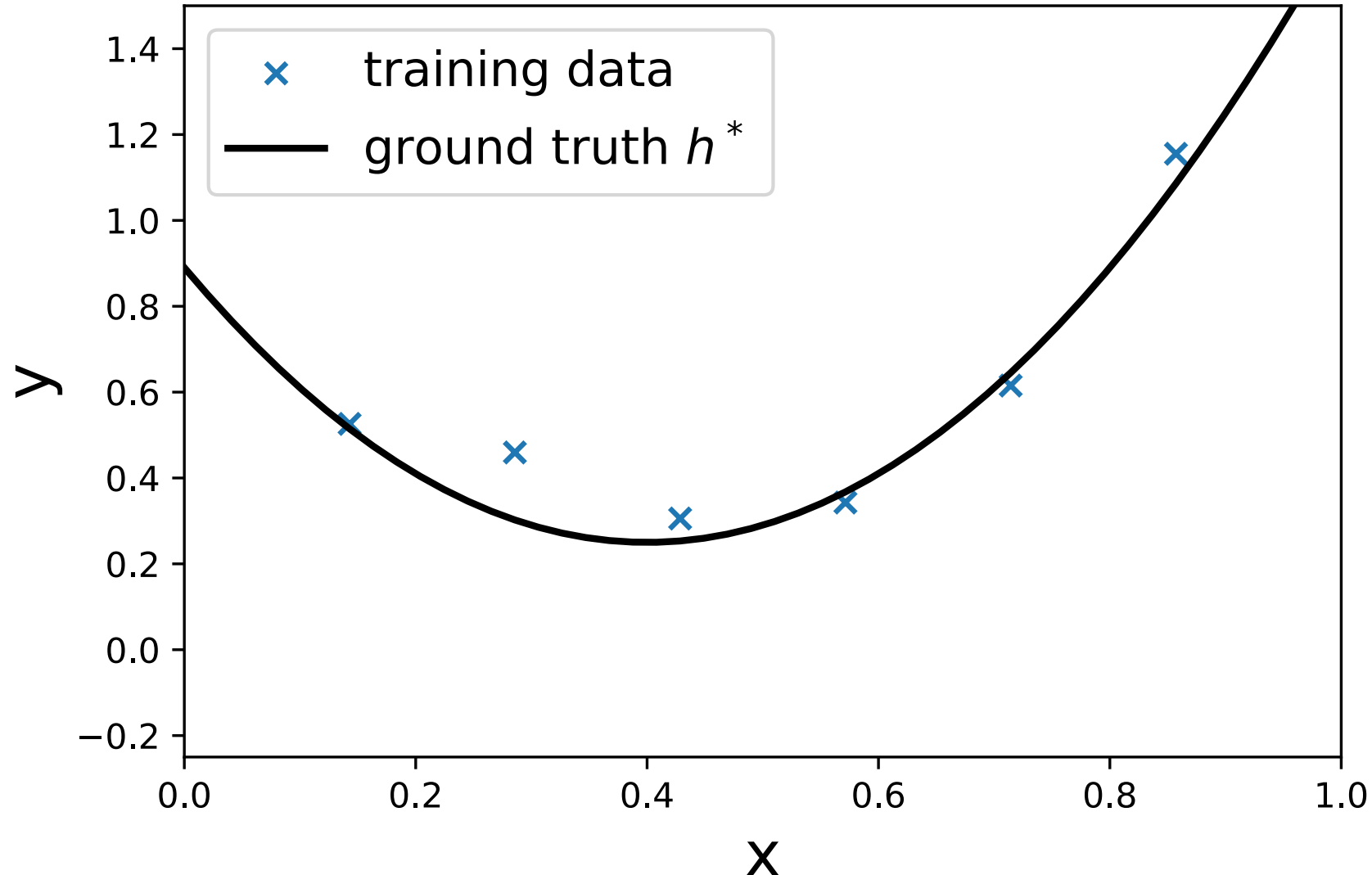


In the context of science, the well-known adage “a picture is worth a thousand words” might well be “a model is worth a thousand datasets”

How to choose
this model



Working example

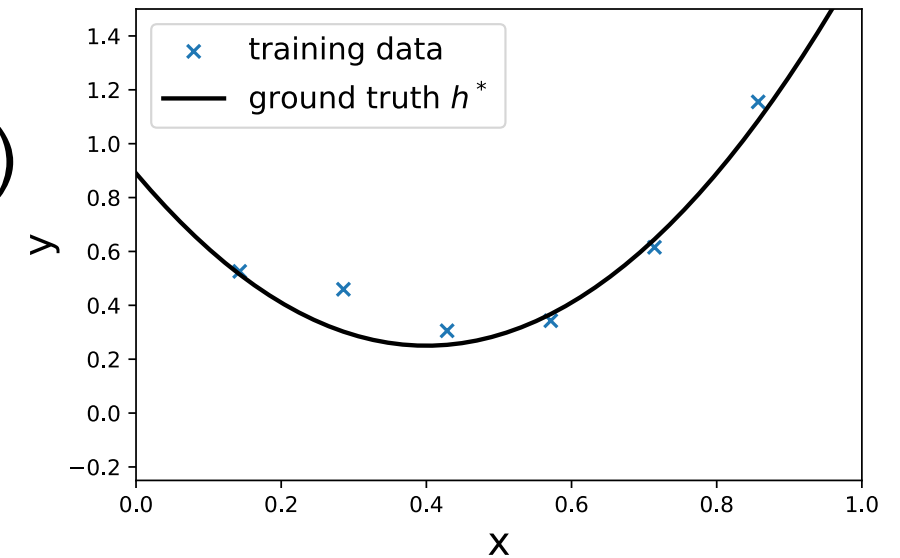
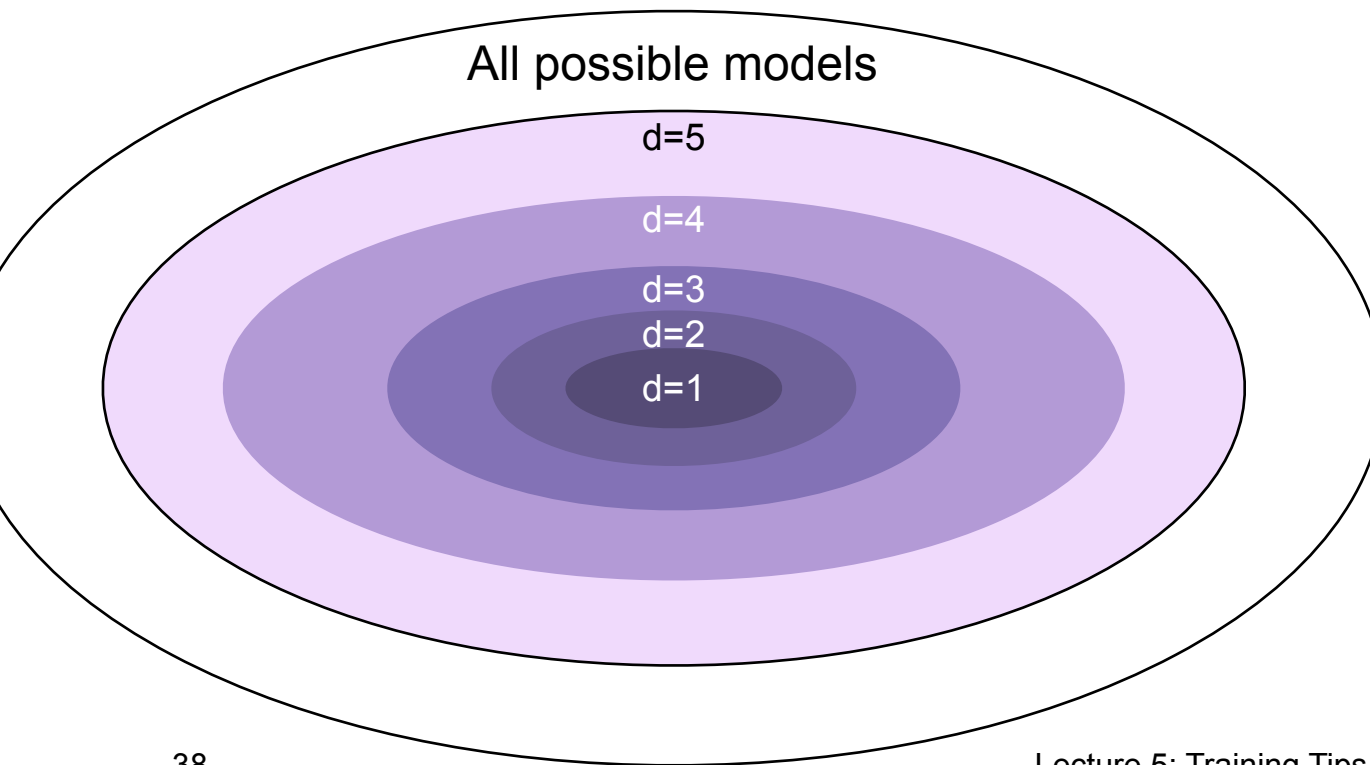


Hypothesis class

How to fit this dataset?

Hypothesis class: **polynomials** (degree d)

→ Back to linear models!!

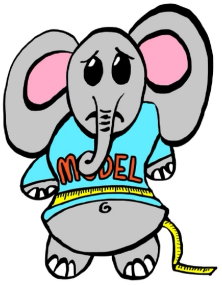


$$d=1: f(x) = w_0$$

$$d=2: f(x) = w_0 + w_1x$$

$$d=3: f(x) = w_0 + w_1x + w_2x^2$$

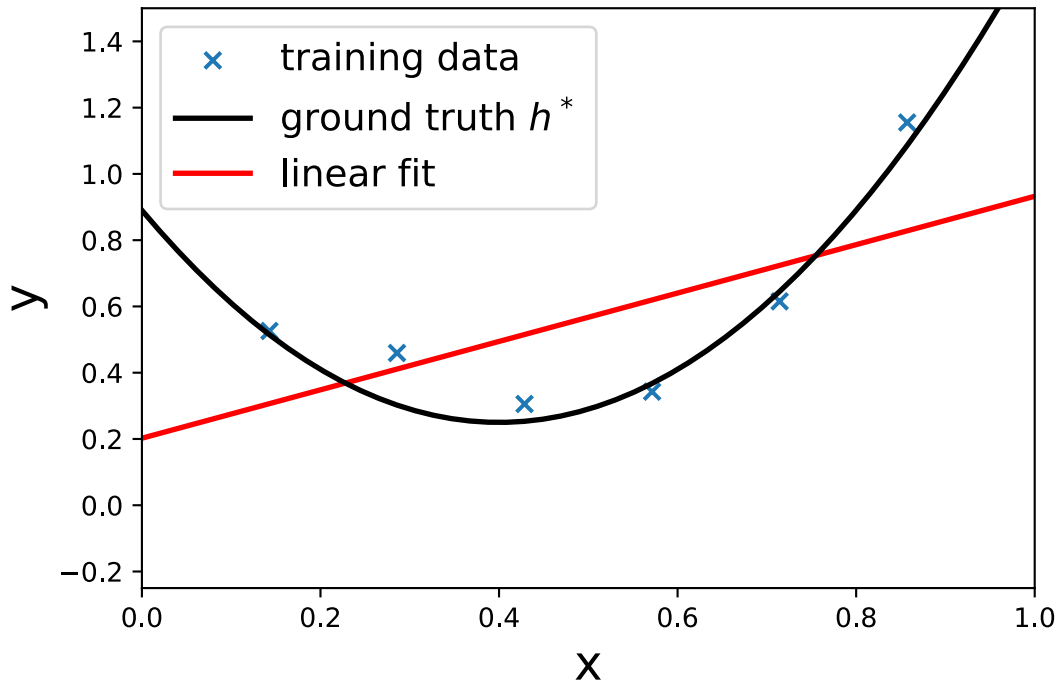
$$d=4: f(x) = w_0 + w_1x + w_2x^2 + w_3x^3$$



Underfitting

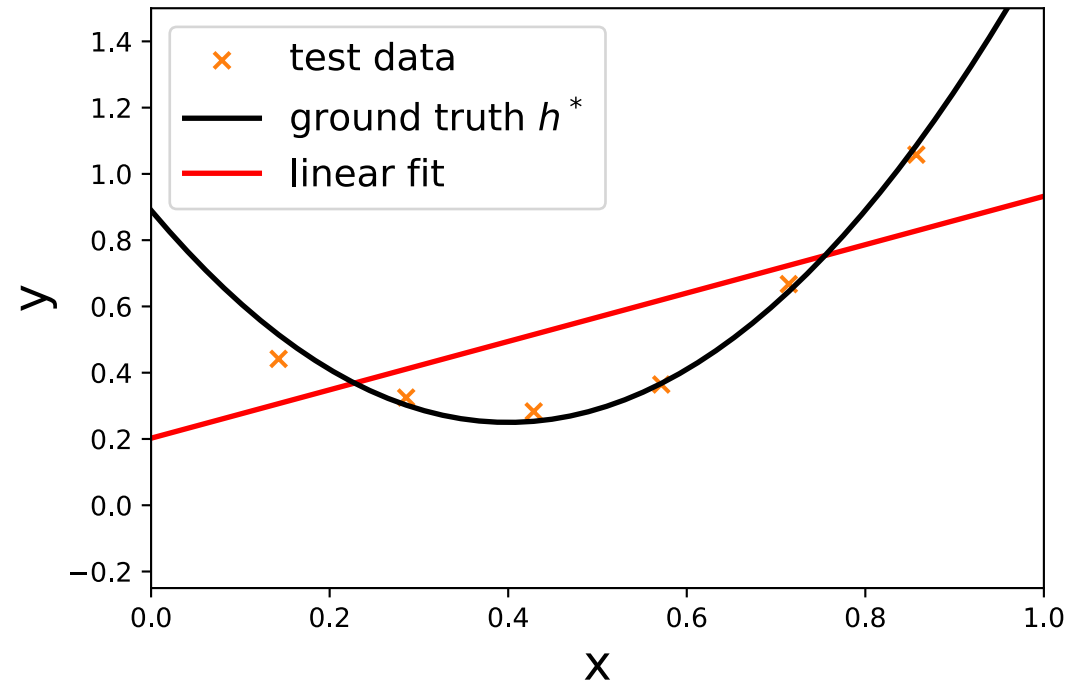
Model is not expressive enough

Training data

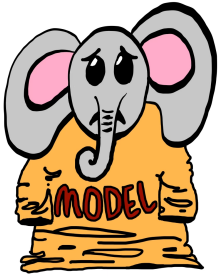


High training error

Test data



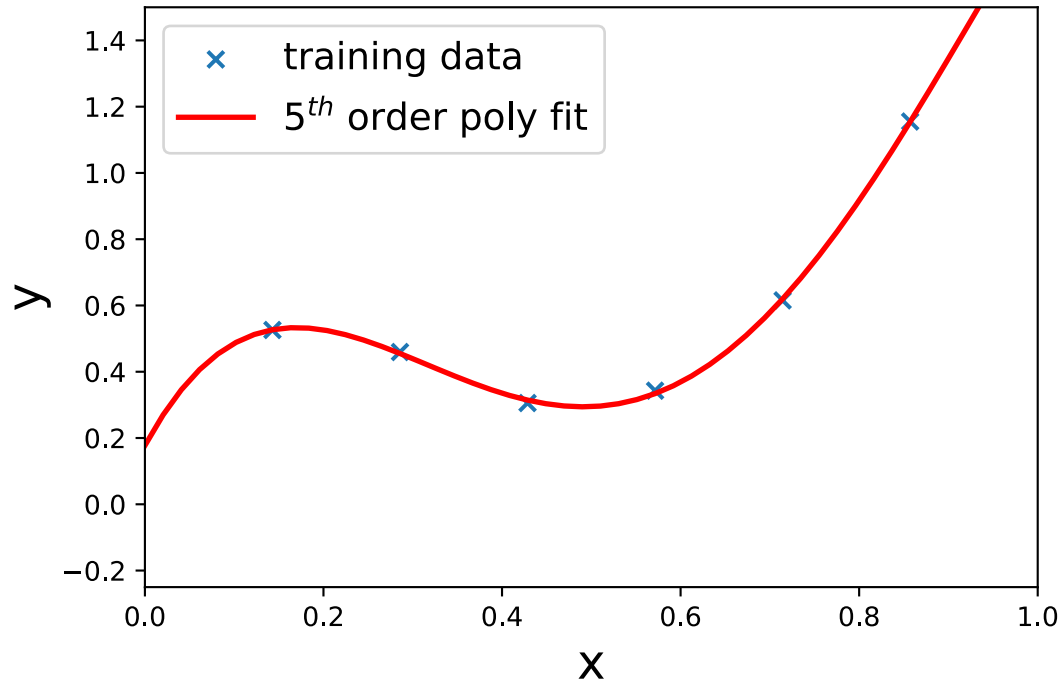
Also high test error



Overfitting

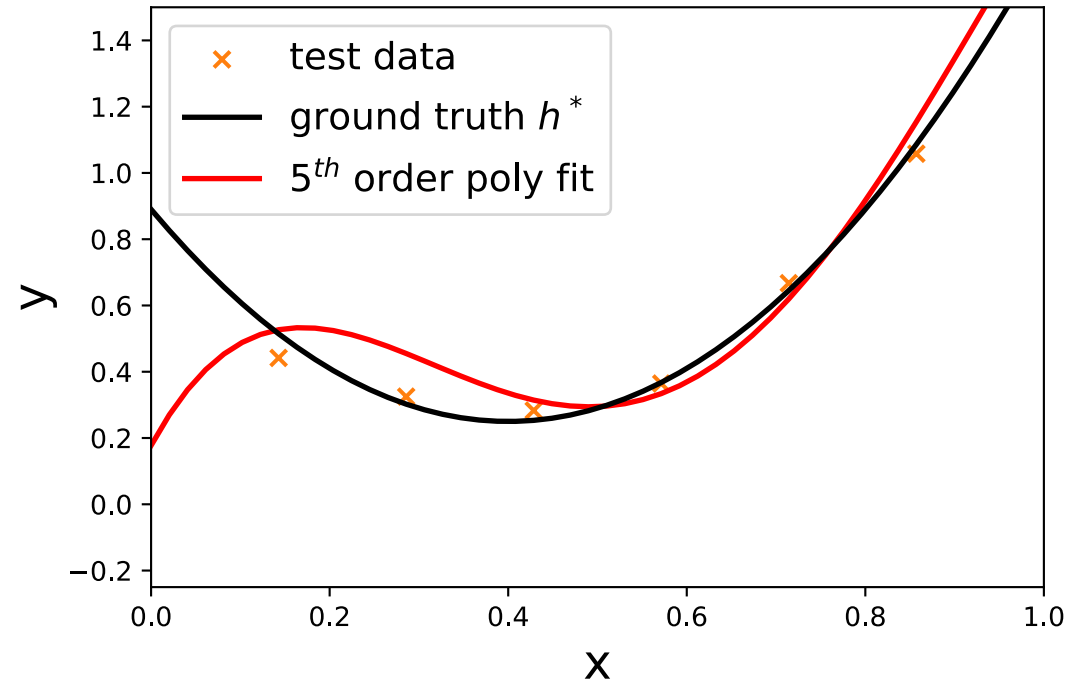
Model too expressive to generalize to unseen dataset

Training data

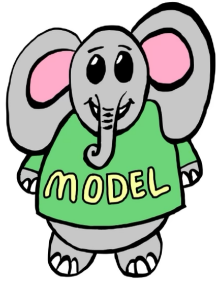


Small (zero) training error

Test data



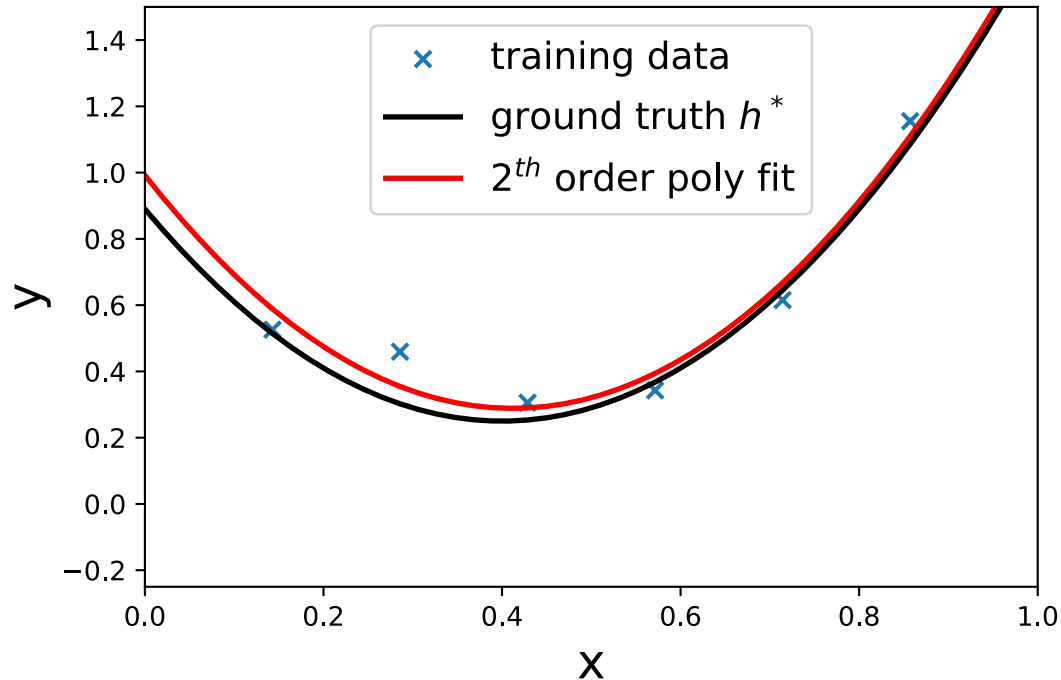
High test error



Optimal model complexity

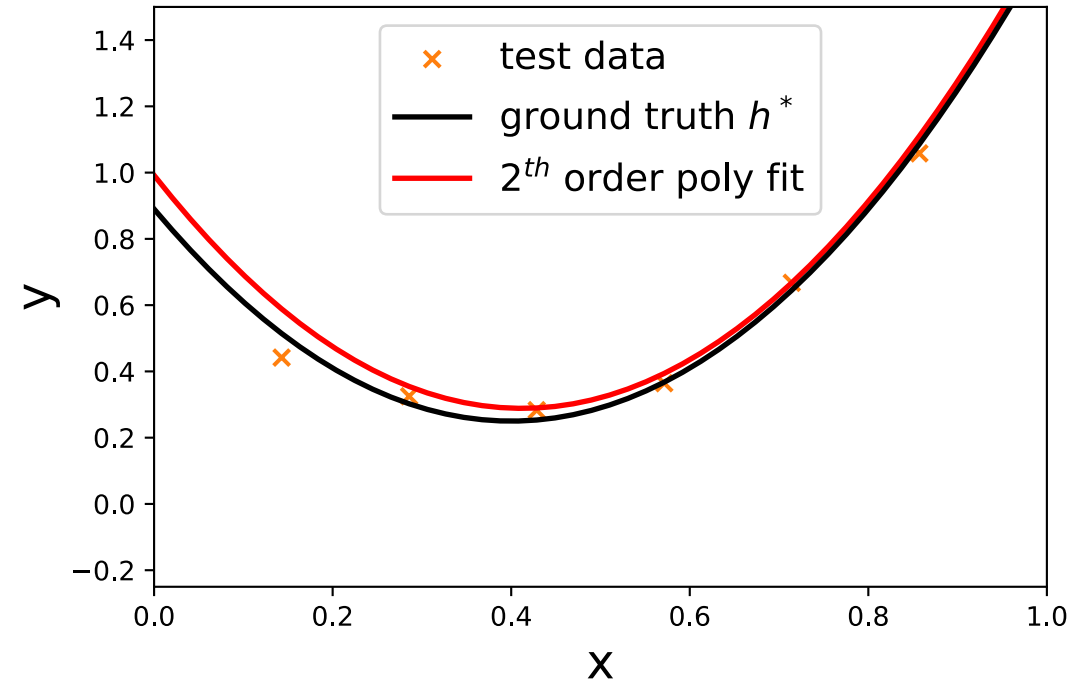
Fit 2nd order polynomial to quadratic distribution

Training data



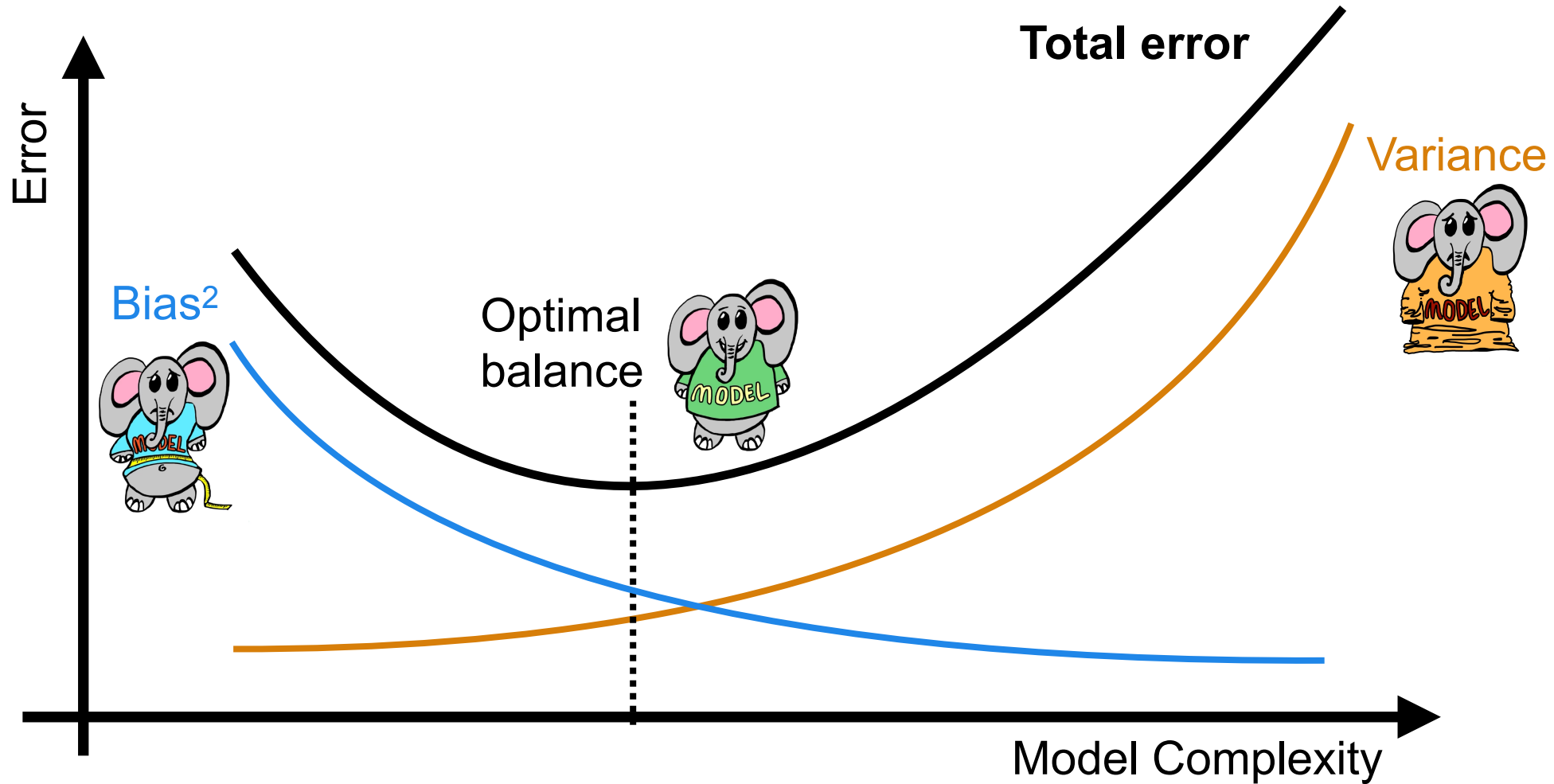
Small training error

Test data



Also small test error

Bias / variance tradeoff



Bias / variance tradeoff: maths 1

- Training dataset $\mathcal{D} = \{x^{(i)}, y^{(i)}\}_{i=1}^n$

- Truth labels $y = f^*(x) + \xi$

- f^* : ground truth function

- $\xi^{(i)} \sim \mathcal{N}(0, \sigma^2)$

- Train model f_D on dataset $\mathcal{D}$

- Consider test point (x, y) and quantify the *expected test error*:

$$\begin{aligned}
 MSE(x) &= \mathbb{E}_{\mathcal{D}, \xi} [(y - f_D(x))^2] \stackrel{\text{Defn of } y}{=} \mathbb{E}_{\mathcal{D}, \xi} [(f^*(x) + \xi - f_D(x))^2] \\
 &= \mathbb{E}_{\mathcal{D}, \xi} [\xi^2] + 2 \mathbb{E}_{\mathcal{D}, \xi} [\xi] \cdot \mathbb{E}_{\mathcal{D}, \xi} [f^*(x) - f_D(x)] + \mathbb{E}_{\mathcal{D}, \xi} [(f^*(x) - f_D(x))^2] \\
 &= \sigma^2 + \mathbb{E}_{\mathcal{D}} [(f^*(x) - f_D(x))^2]
 \end{aligned}$$

Bias / variance tradeoff: maths 2

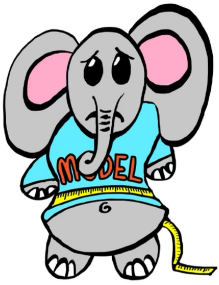
- Let $f_{avg}(x) = \mathbb{E}_D [f_D(x)]$ – the performance of the model trained on infinitely many datasets
- Substitute this back into the MSE expression

$$\begin{aligned}
 MSE(x) &= \sigma^2 + \mathbb{E}_{\mathcal{D}} [(f^*(x) - f_D(x))^2] \\
 &= \sigma^2 + \mathbb{E}_{\mathcal{D}} \left[\left(f^*(x) - f_{avg}(x) + f_{avg}(x) - f_D(x) \right)^2 \right] \\
 &= \sigma^2 + \underbrace{(f^*(x) - f_{avg}(x))^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}_{\mathcal{D}} [(f_{avg}(x) - f_D(x))^2]}_{\text{Variance}}
 \end{aligned}$$

No cross-term because
 $\mathbb{E} [f_{avg}(x) - f_D(x)] = 0$

Error on this class of models

How does this instantiation compare
with the other possible ones?



High bias: diagnostics

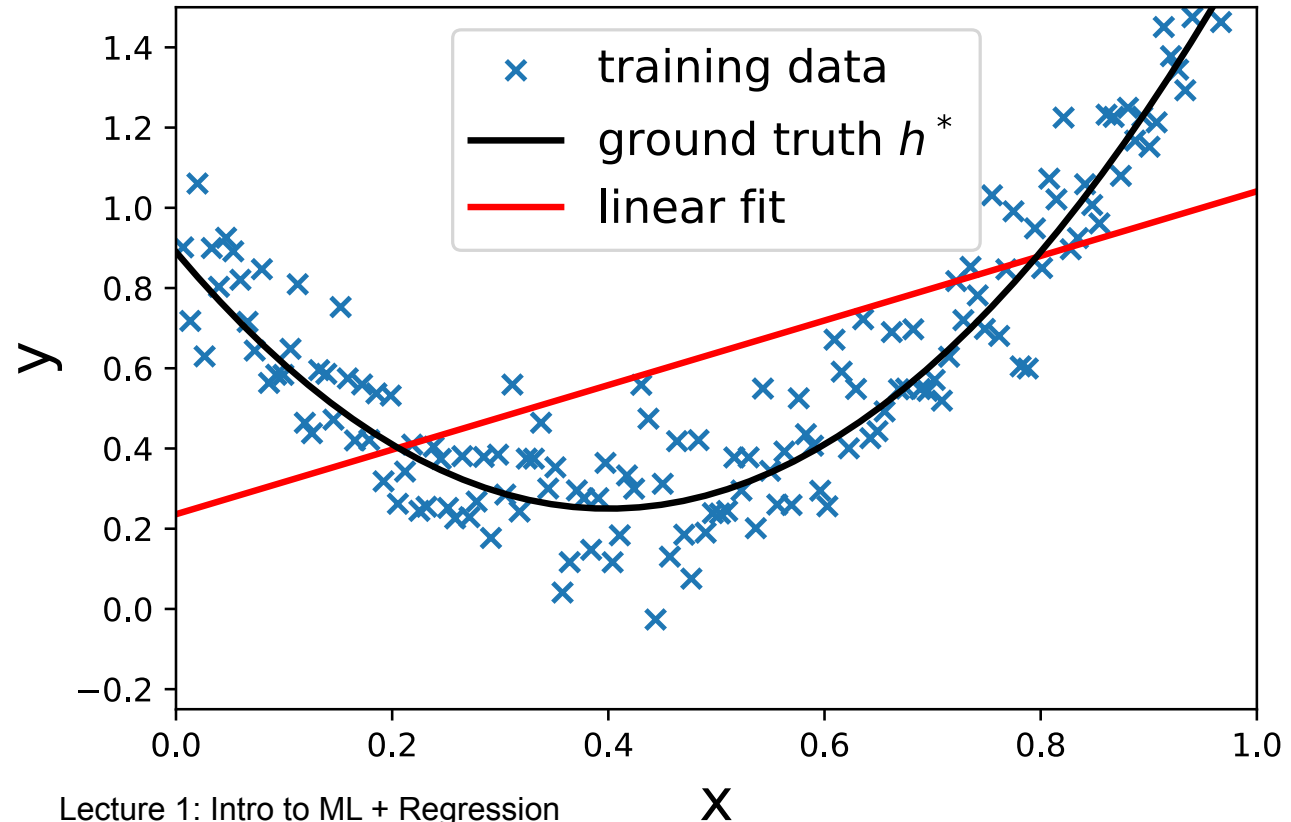
$$MSE(x) = \underbrace{(f^*(x) - f_{avg}(x))^2}_{\text{Bias}^2} + \mathbb{E}_{\mathcal{D}} \left[(f_{avg}(x) - f_S(x))^2 \right]$$

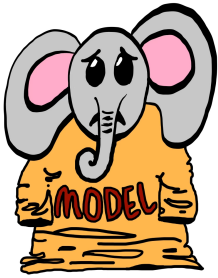
Linear fit

$$f_D(x) = w_0 + w_1 x$$

The training error high, even if we increase the training data.

Fit linear model on large dataset





High variance: diagnostics

$$MSE(x) = (f^*(x) - f_{avg}(x))^2 + \mathbb{E}_{\mathcal{D}} \left[(f_{avg}(x) - f_S(x))^2 \right]$$

Variance

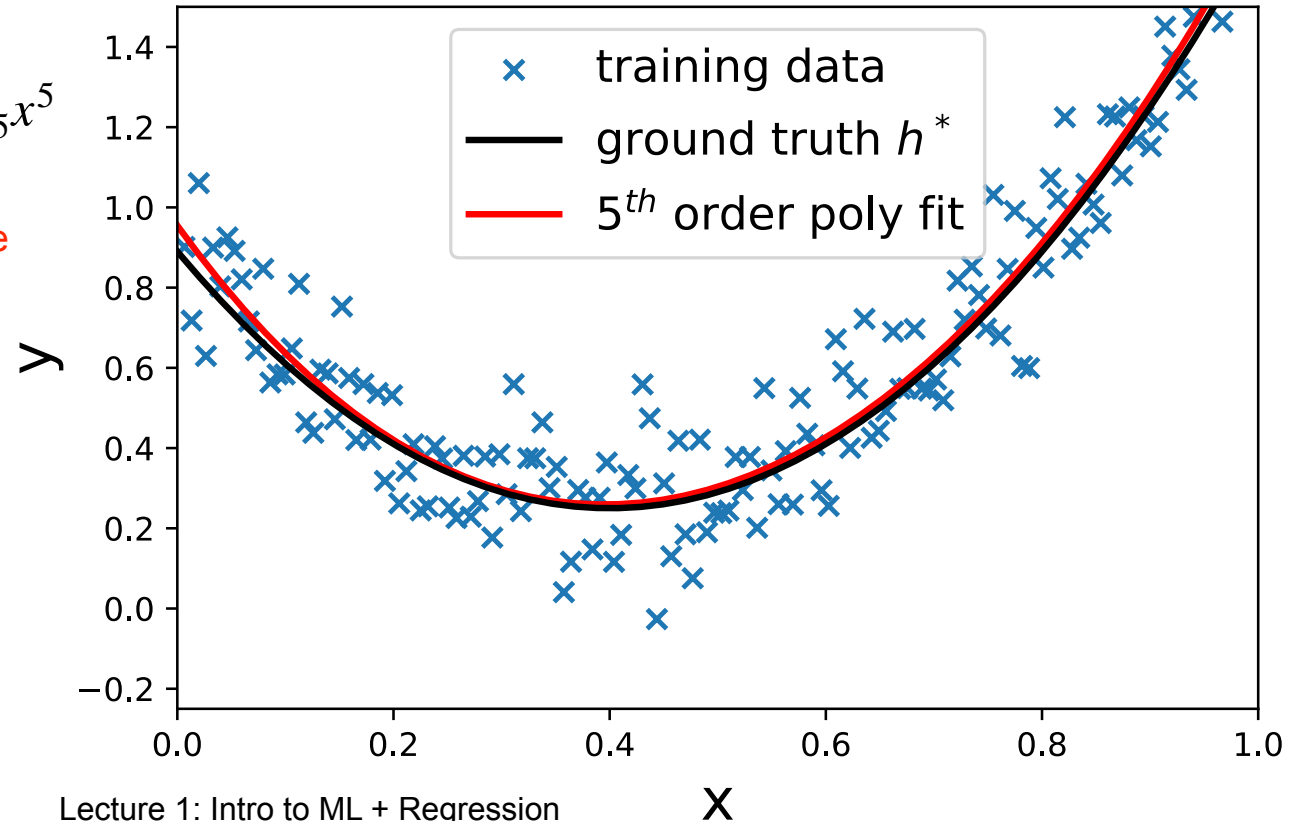
5th order polynomial fit

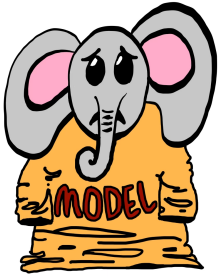
$$f_S(x) = w_0 + w_1x + w_2x^2 + w_3x^3 + w_4x^4 + w_5x^5$$

Can learn to set these
coefficients to 0

The training error decreases as we increase the training data.

Fit 5th order polynomial on large dataset



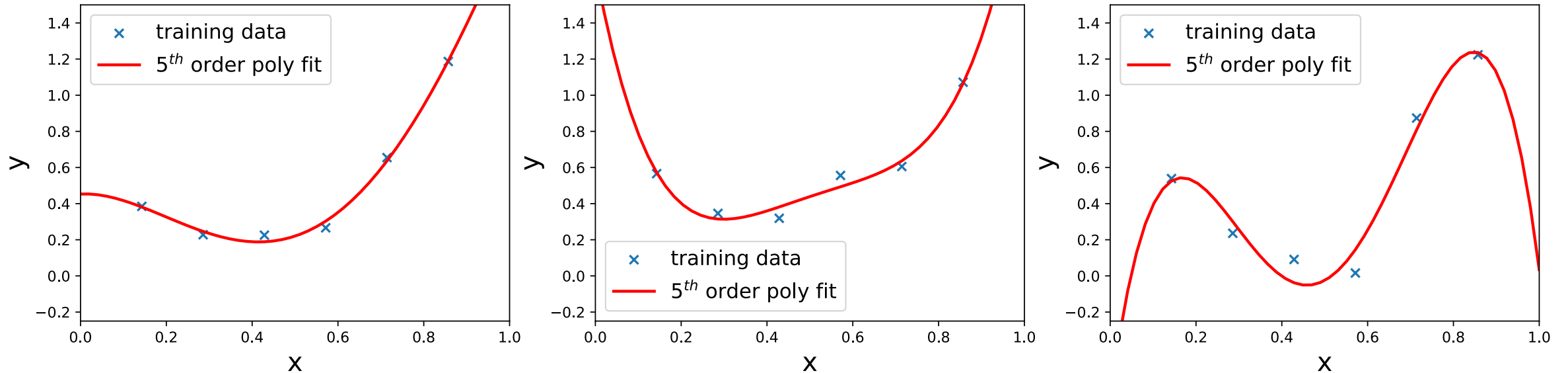


High variance: intuition

$$MSE(x) = (f^*(x) - f_{avg}(x))^2 + \mathbb{E} \left[(f_{avg}(x) - f_S(x))^2 \right]$$

Variance

fitting 5th degree polynomial on different datasets



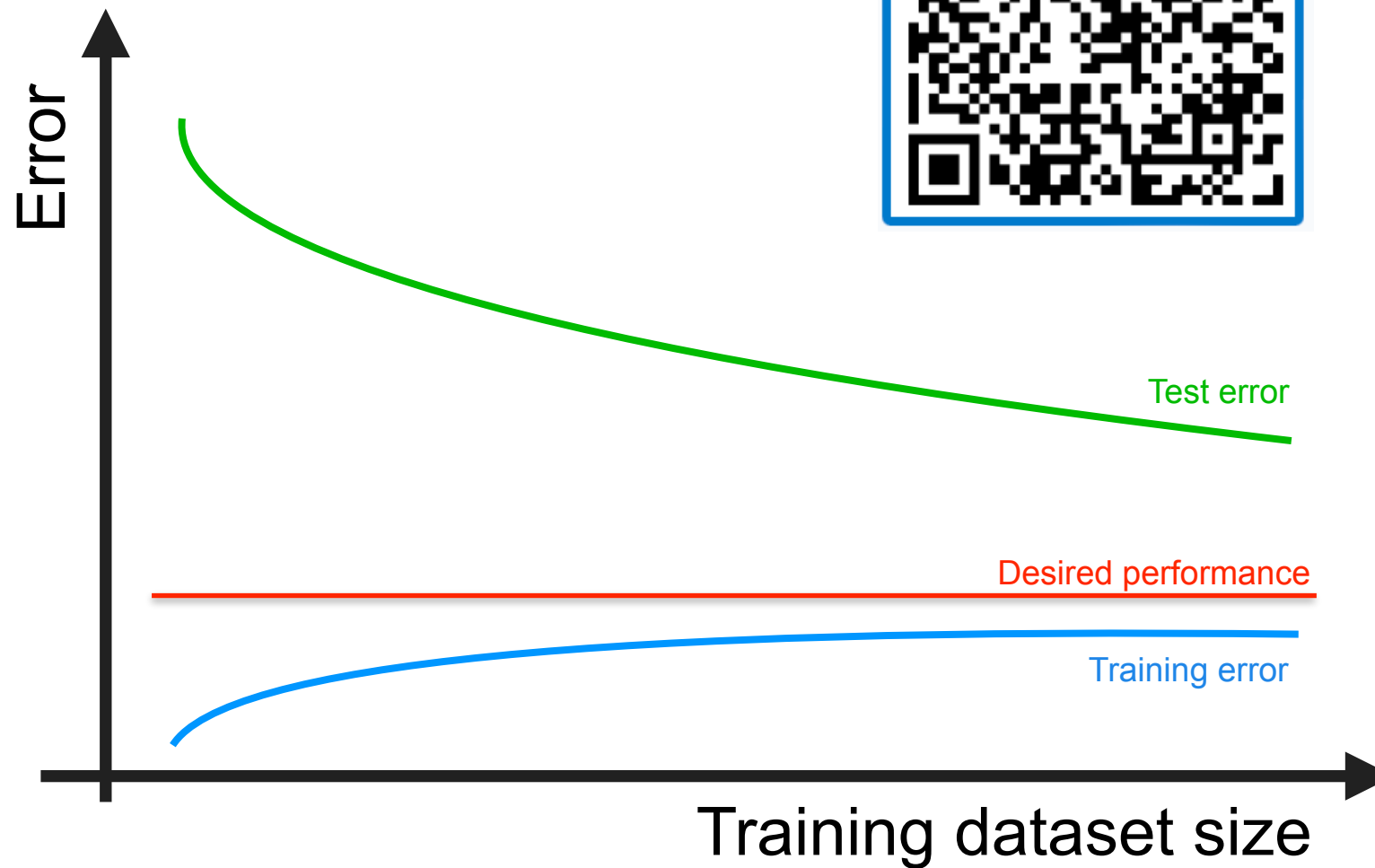
Lots of possibilities for the fitted function depending on the random realization of training data.

What's the culprit?



1+2 min

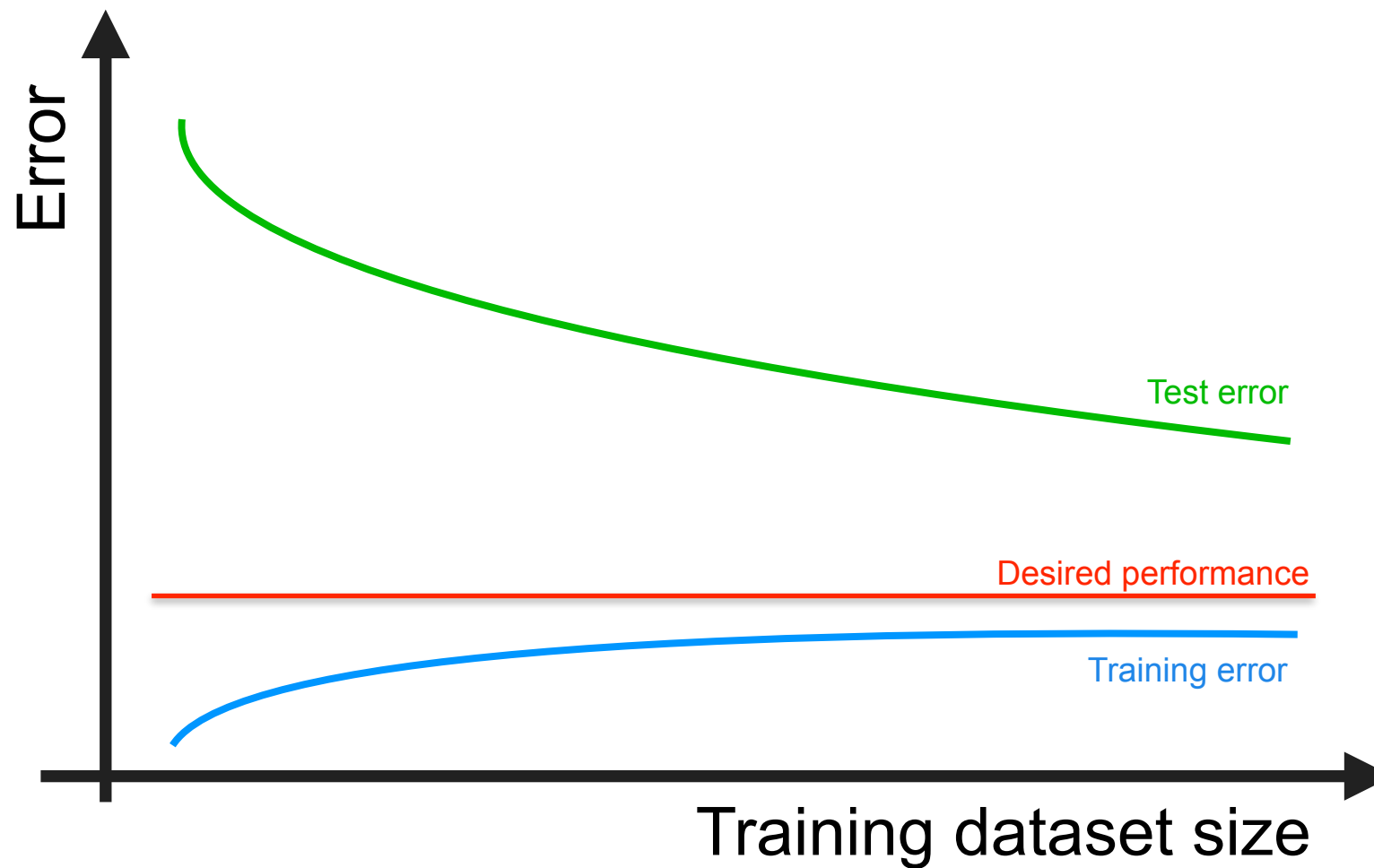
[PollEv.com/nicolehartman968](https://pollev.com/nicolehartman968)



High bias?

High Variance?

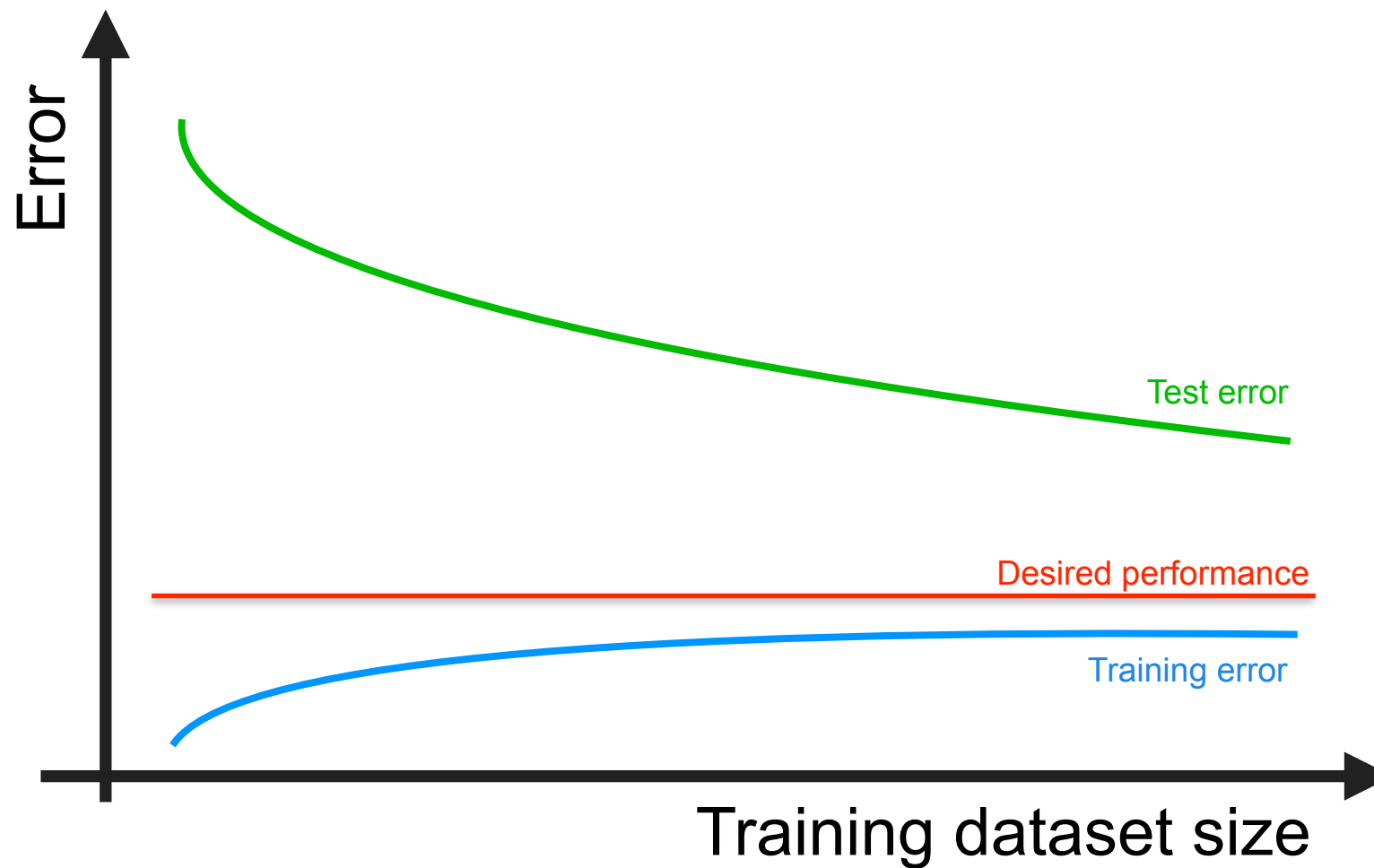
What's the culprit?



High bias?

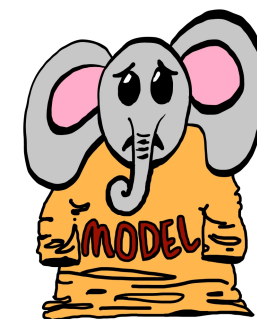
High Variance?

What's the culprit?

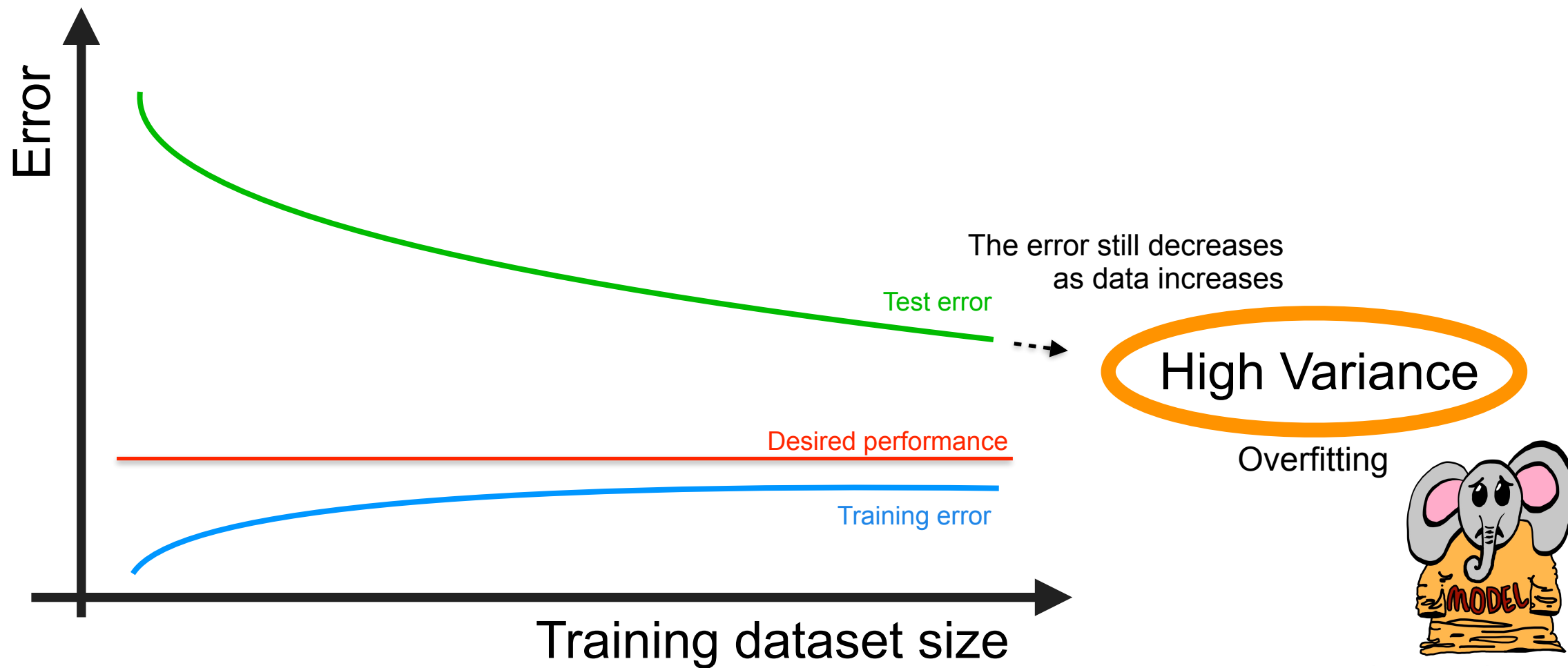


High Variance

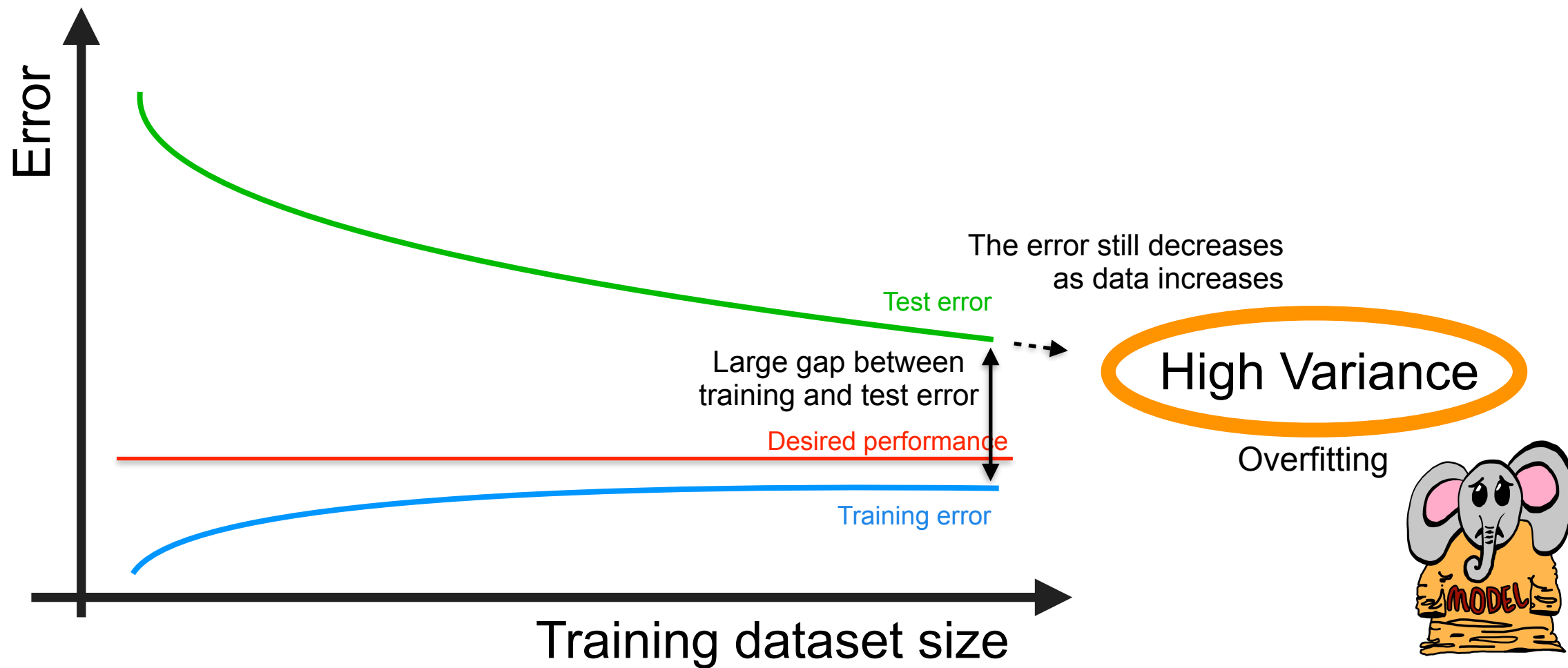
Overfitting



What's the culprit?



What's the culprit?



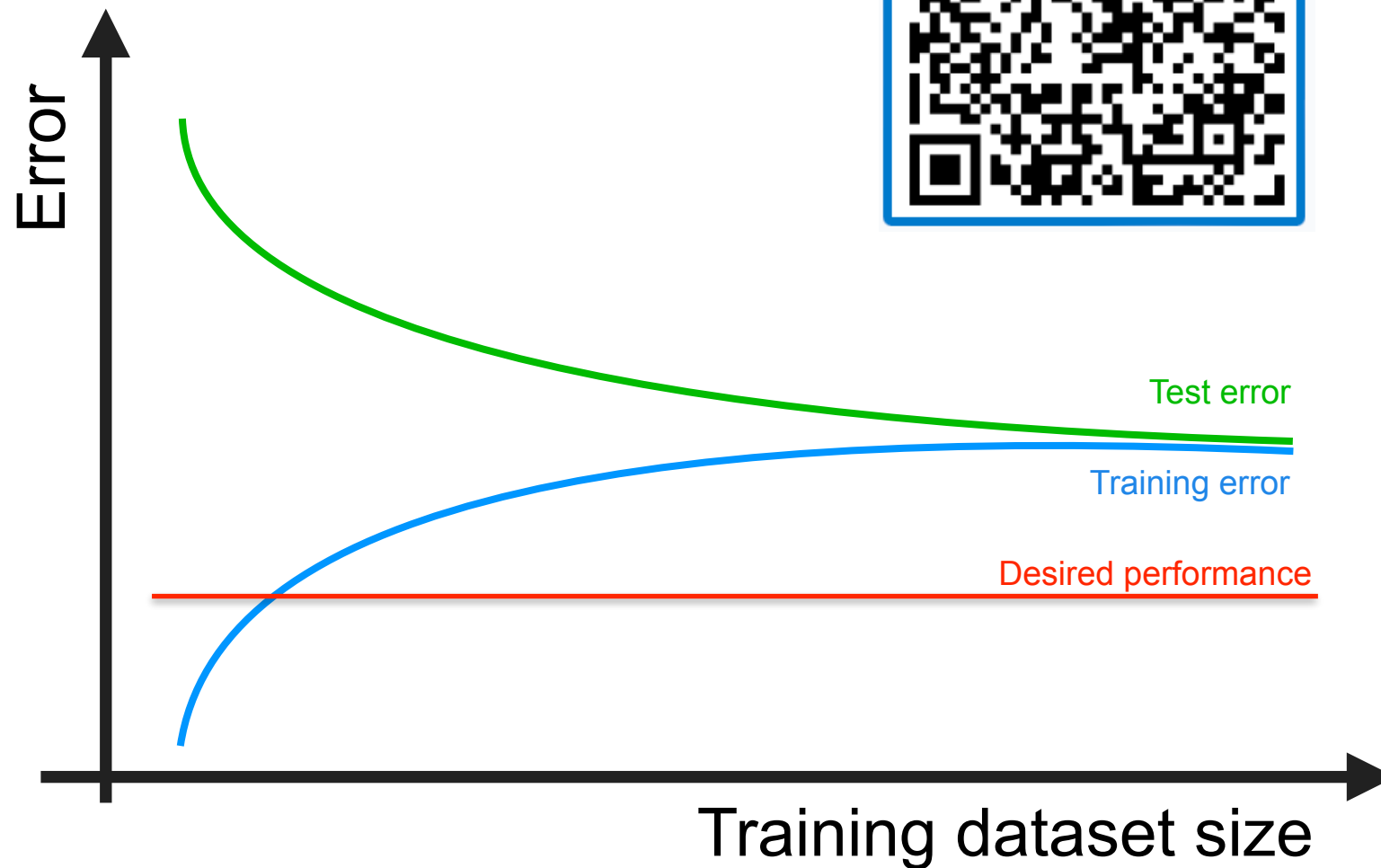
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1+2 min

CS 229 [Lecture](#)

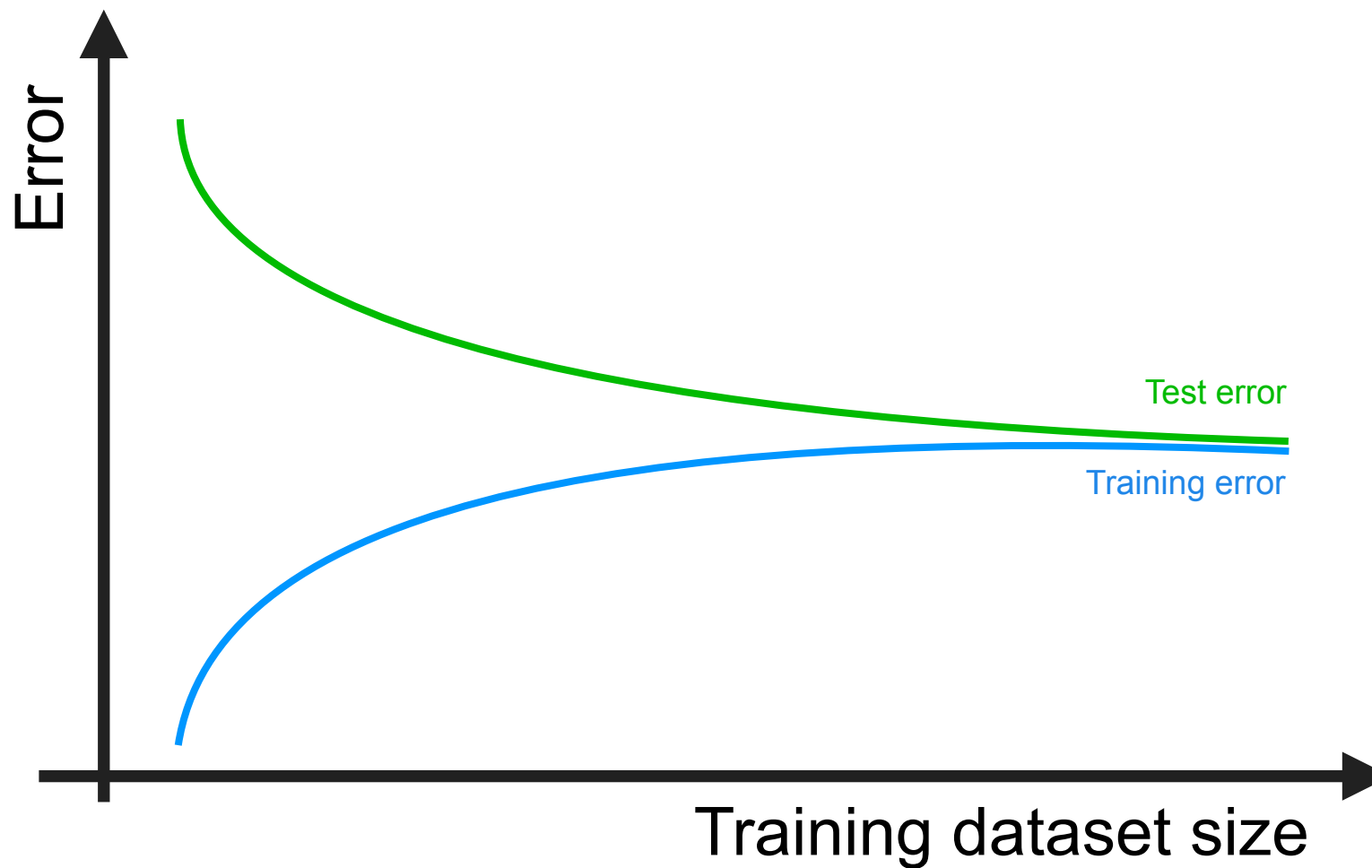
[PollEv.com/nicolehartman968](https://poll-ev.com/nicolehartman968)



High bias?

High Variance?

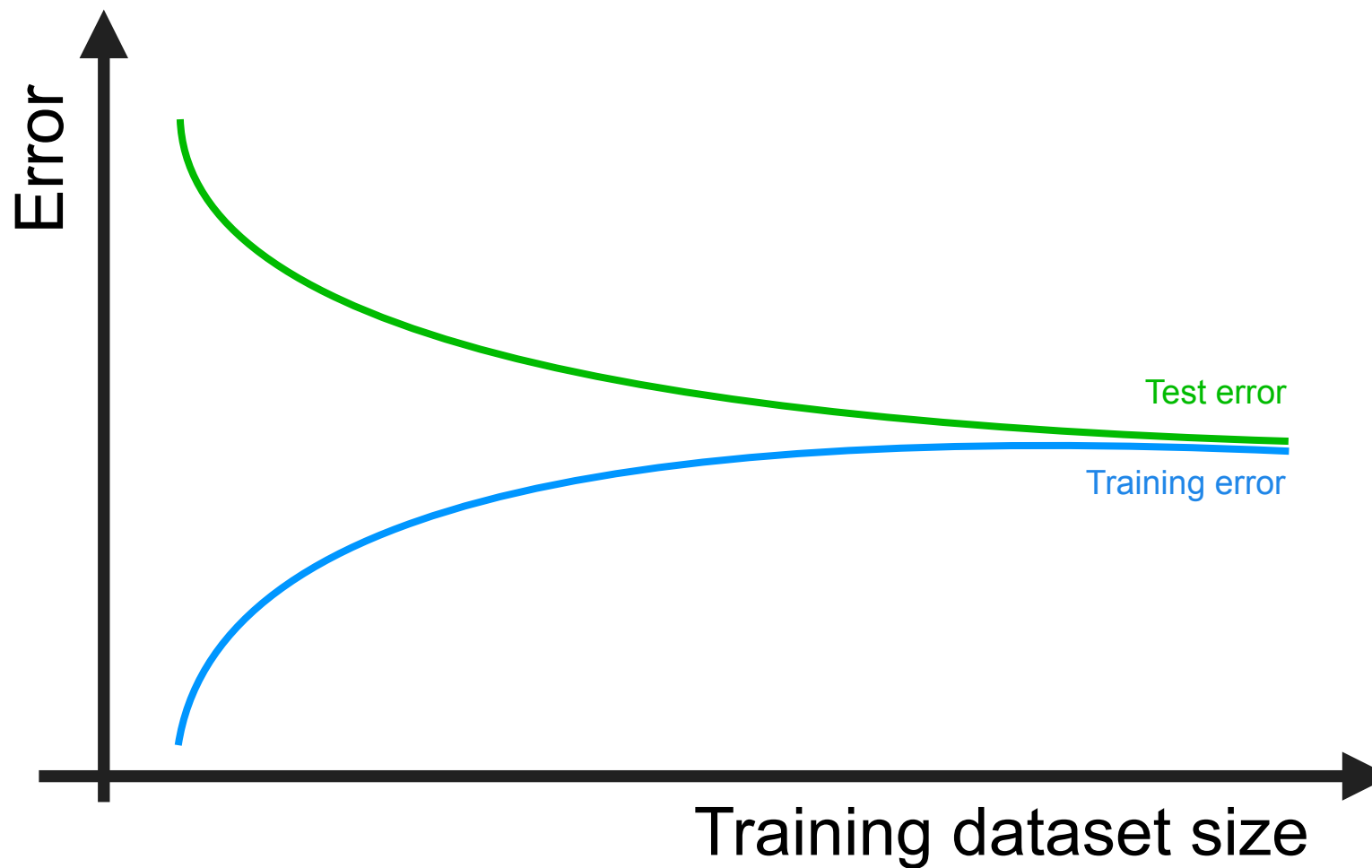
What's the culprit?



High bias?

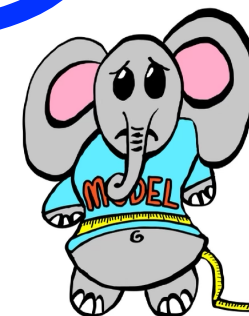
High Variance?

What's the culprit?

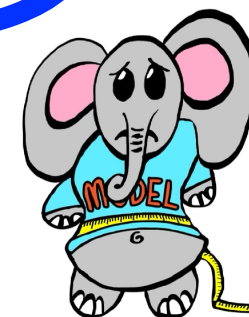
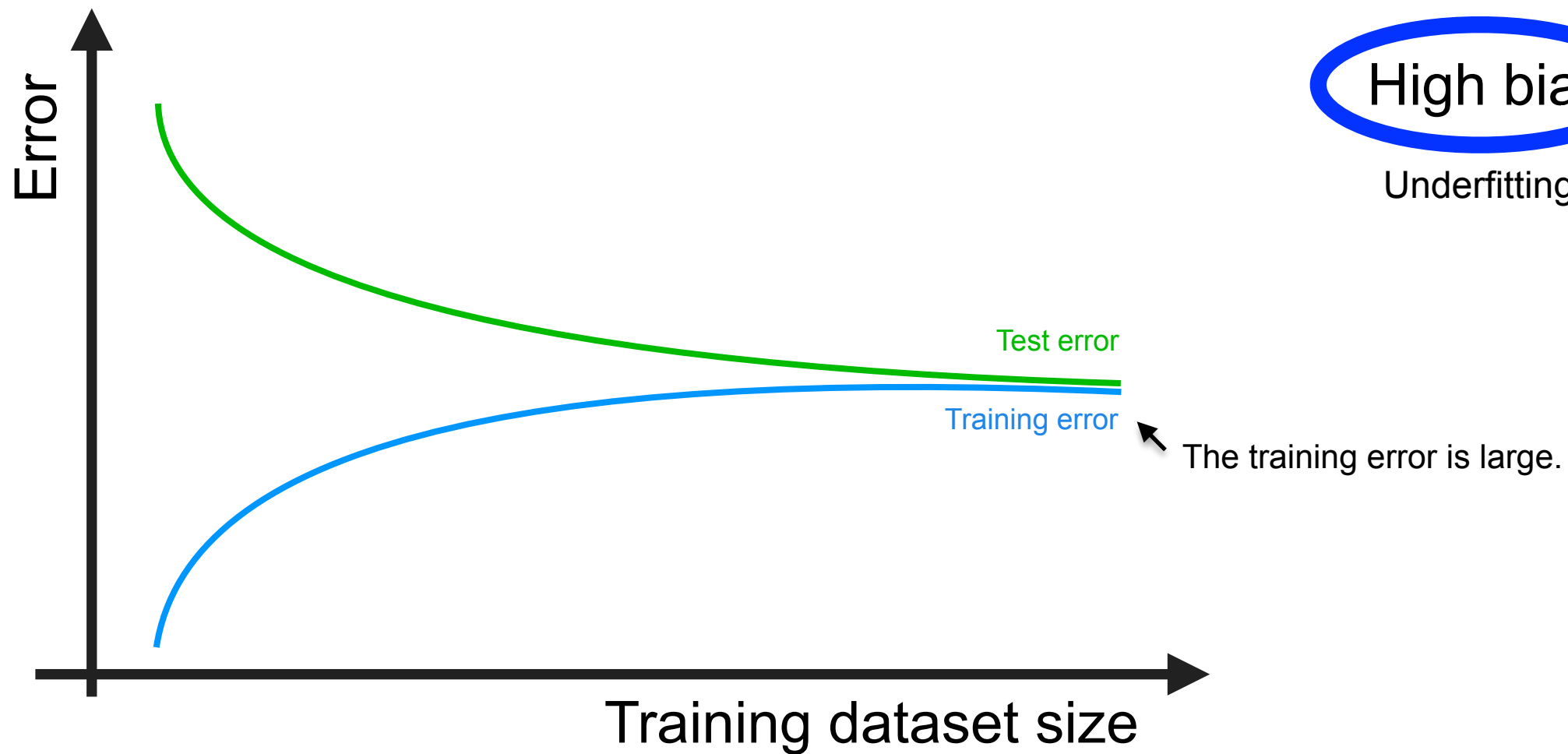


High bias

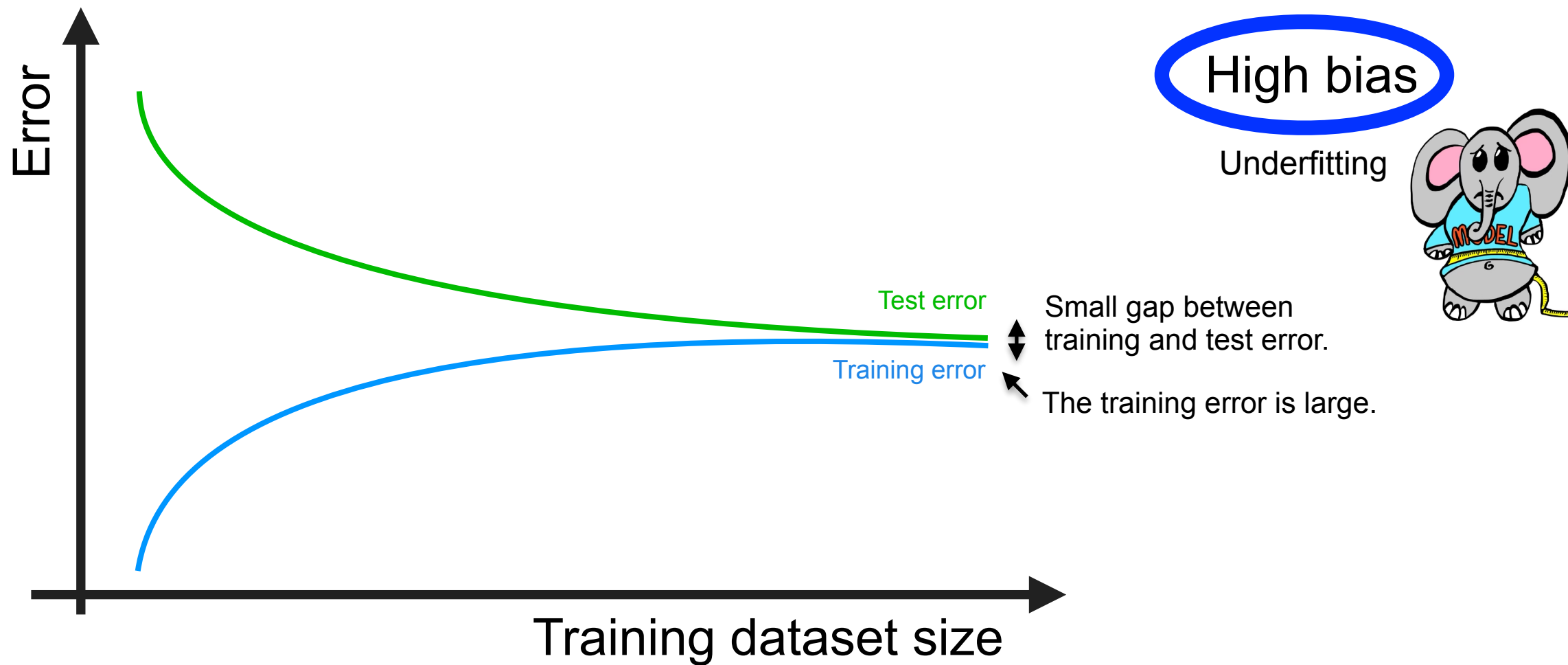
Underfitting



What's the culprit?



What's the culprit?

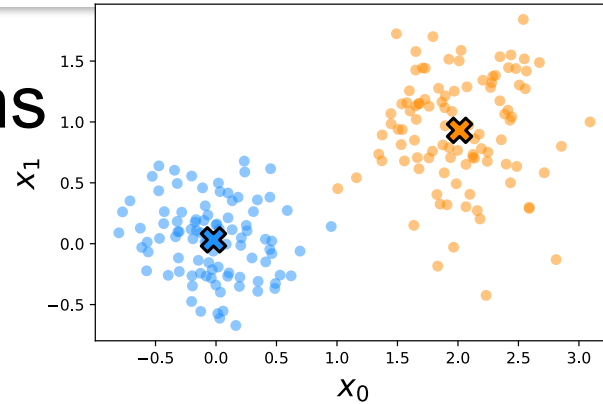


In summary

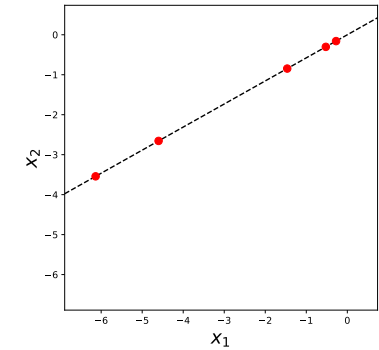
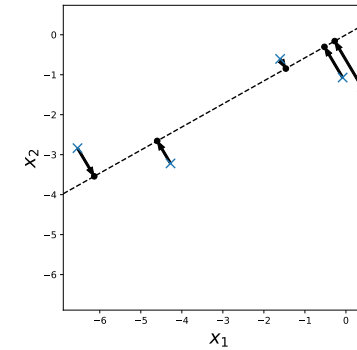
Unsupervised Learning

Train with reconstruction error

1 K-means



2 PCA

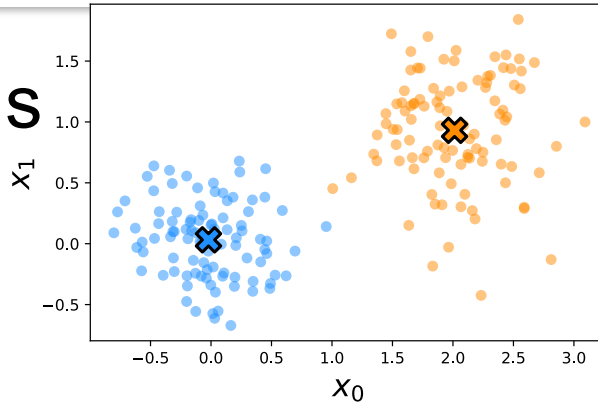


In summary

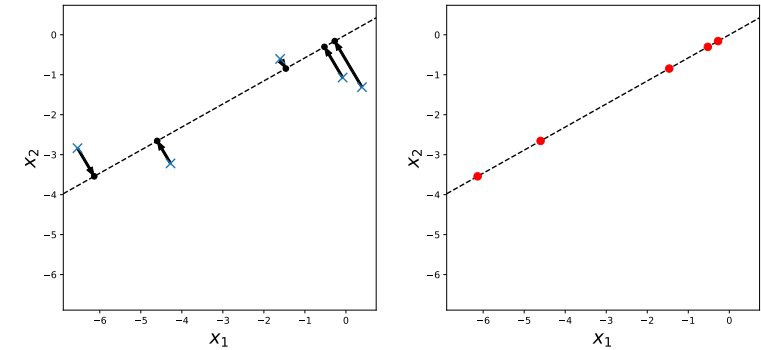
Unsupervised Learning

Train with reconstruction error

1 K-means

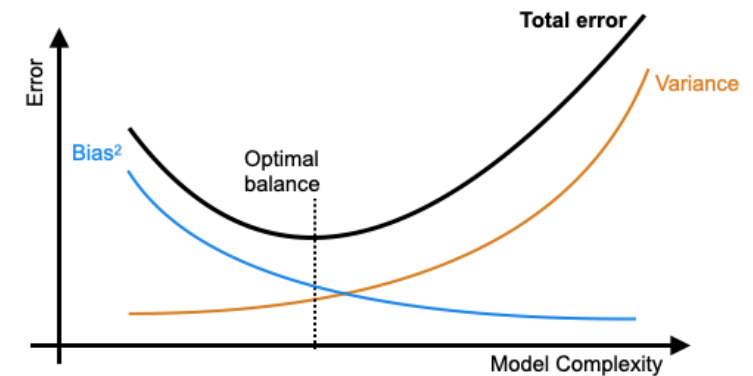
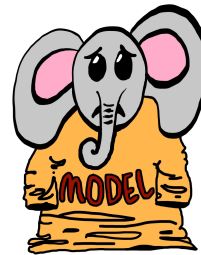
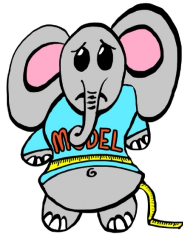


2 PCA



3 Total error = Bias² + Variance

Underfitting **Overfitting**



Bonus material

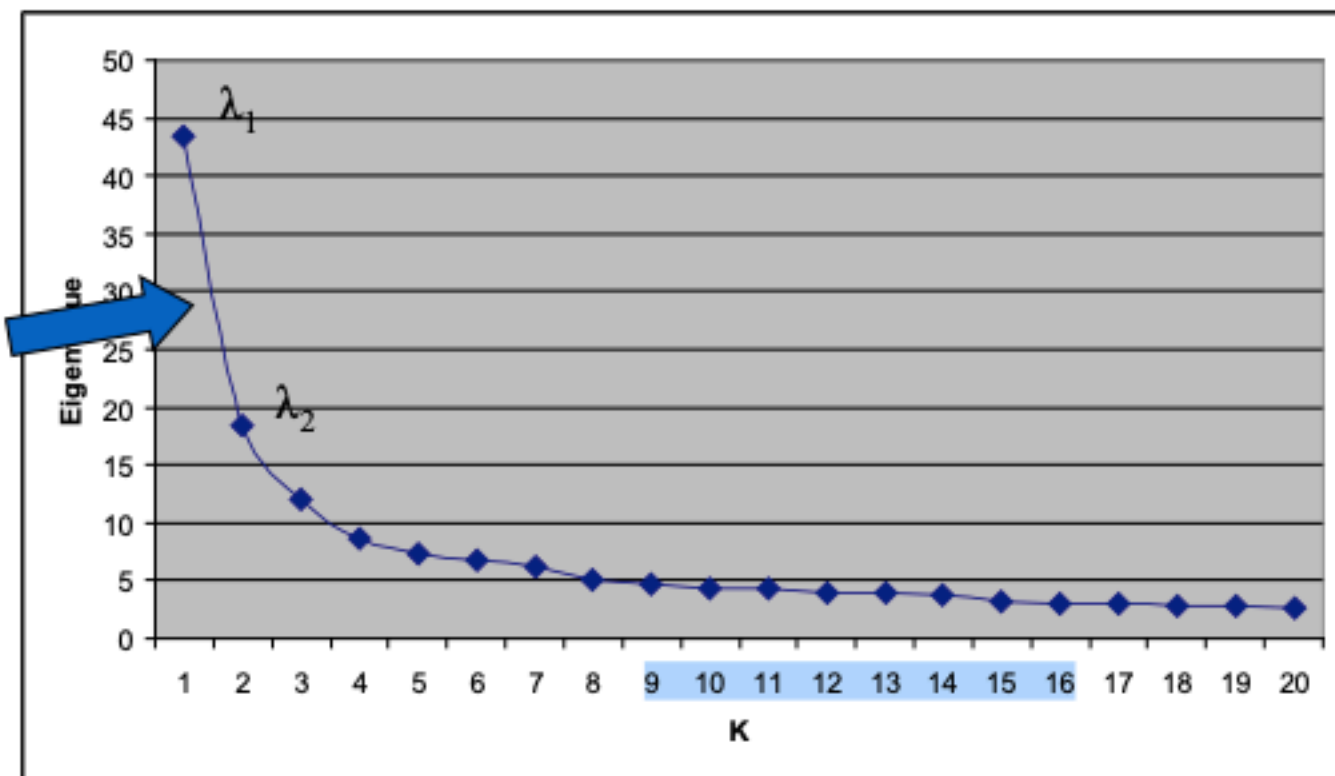
Spectral Gap: Selecting k

- ◆ *Eigengap*: the difference between two consecutive eigenvalues.
- ◆ Most stable clustering is generally given by the value k that maximizes the expression

$$\Delta_k = |\lambda_k - \lambda_{k-1}|$$

$$\max \Delta_k = |\lambda_2 - \lambda_1|$$

$\Rightarrow$ Choose $k=2$



GMMs

Mixtures of Gaussians

Combine *simple models* into complex models

$$p(X) = \sum_{k=1}^K \pi_k \mathcal{N}(X | \mu_k, \Sigma_k)$$

$$\forall k : \pi_k \geq 0, \quad \sum_{k=1}^K \pi_k = 1$$

Follow-up Q: What would be a limitation of this method?
-> Need to know the number of classes

Unobserved data

Marginalize over the unobserved “targets”

$$\arg \max_{\theta} \prod_i p(X_i) = \arg \max_{\theta} \prod_i \sum_{k=1}^K p(Y_i = k, X_i)$$


Max the log prob

$$\arg \max_{\theta} \log \prod_i p(X_i) = \arg \max_{\theta} \log \prod_i \sum_{k=1}^K p(Y_i = k, X_i)$$

$$= \arg \max_{\theta} \sum_i \log \sum_{k=1}^K p(Y_i = k, X_i)$$



Our fav opt trick
from today

EM algorithm

I think θ might be slightly confusing, might be better to just specify mean and variance?

Goal: $\arg \max_{\theta} \sum_i \log \sum_{k=1}^K p(Y_i = k, X_i)$



Marginal log likelihood

EM for GMMs: only learning means

Expectation

What is the associated probability for a given point to belong to each class?

Iterate: On the t 'th iteration let our estimates be

$$\lambda_t = \{ \mu_1^{(t)}, \mu_2^{(t)} \dots \mu_K^{(t)} \}$$

E-step

Compute “expected” classes of all datapoints

$$P(Y_j = k | x_j, \mu_1 \dots \mu_K) \propto \exp\left(-\frac{1}{2\sigma^2} \|x_j - \mu_k\|^2\right) P(Y_j = k)$$

Maximization

Basically like a “fuzzy” mean
Or update the class means based on how likely you are to belong to each class

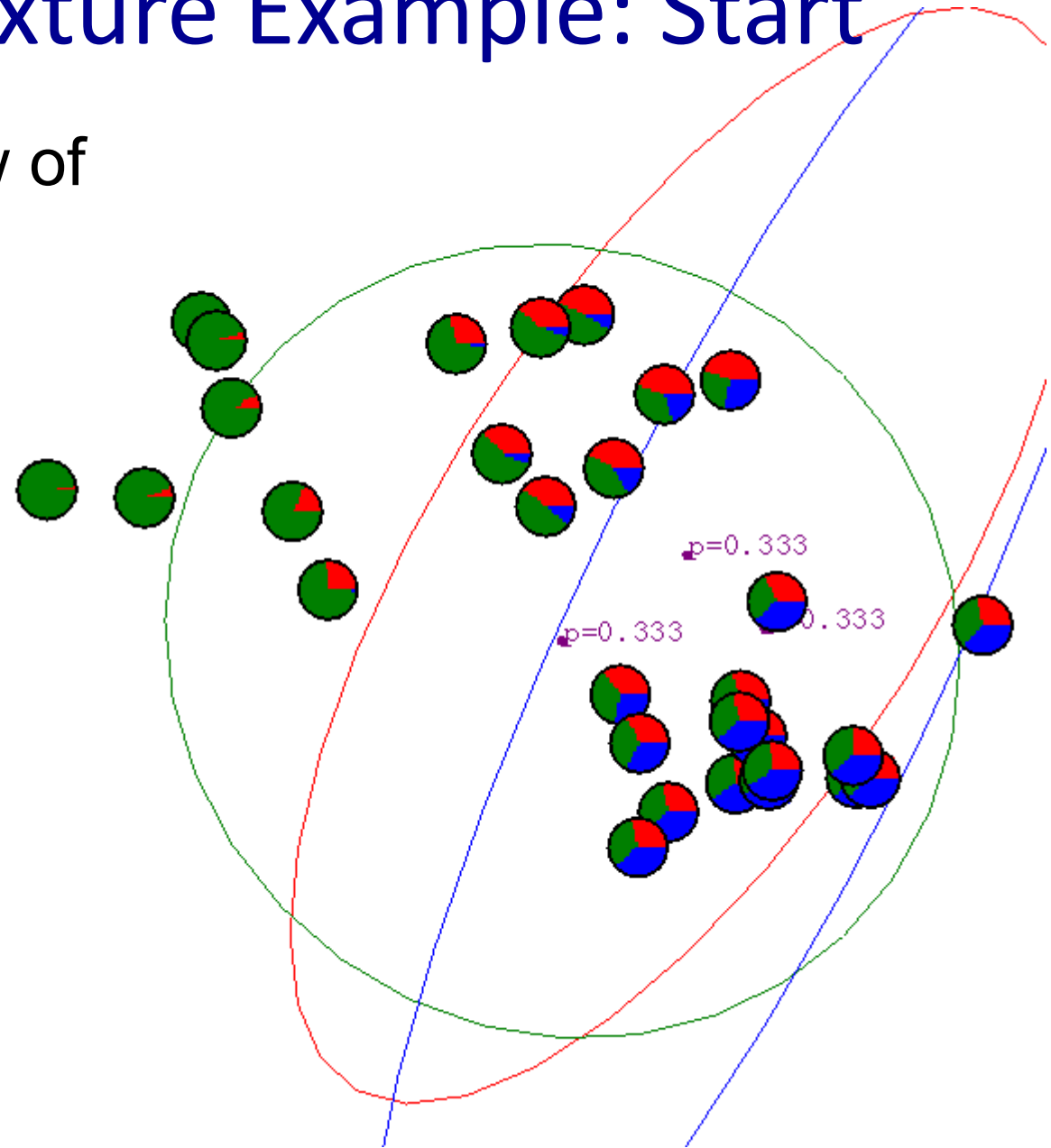
M-step

Compute most likely new μ s given class expectations

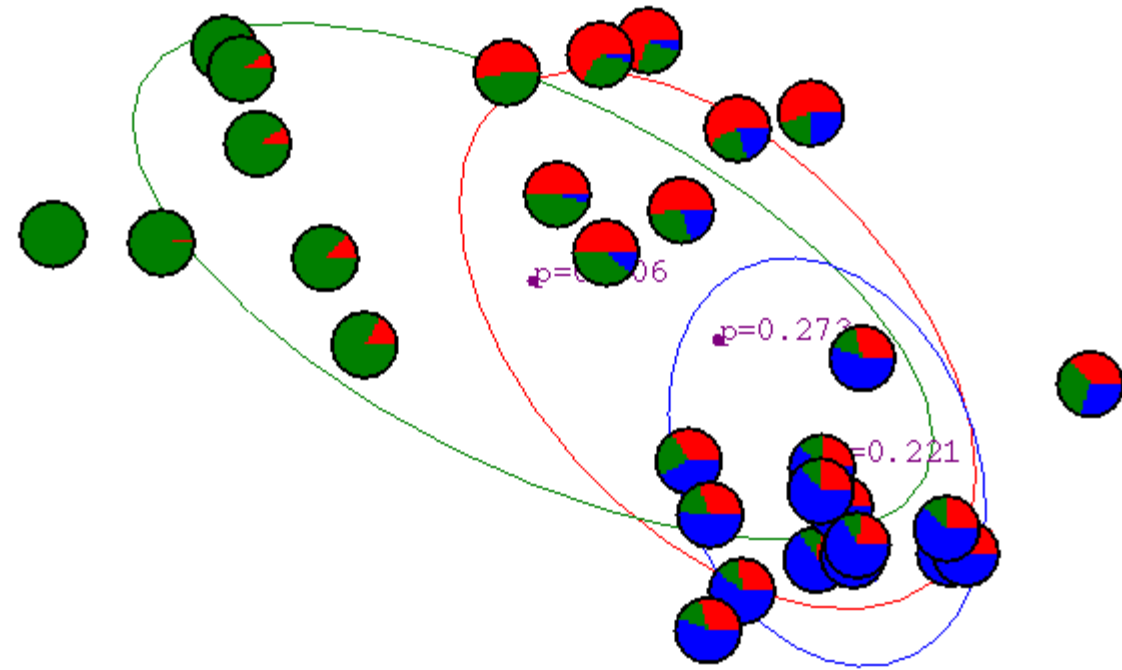
$$\mu_k = \frac{\sum_{j=1}^m P(Y_j = k | x_j) x_j}{\sum_{j=1}^m P(Y_j = k | x_j)}$$

Gaussian Mixture Example: Start

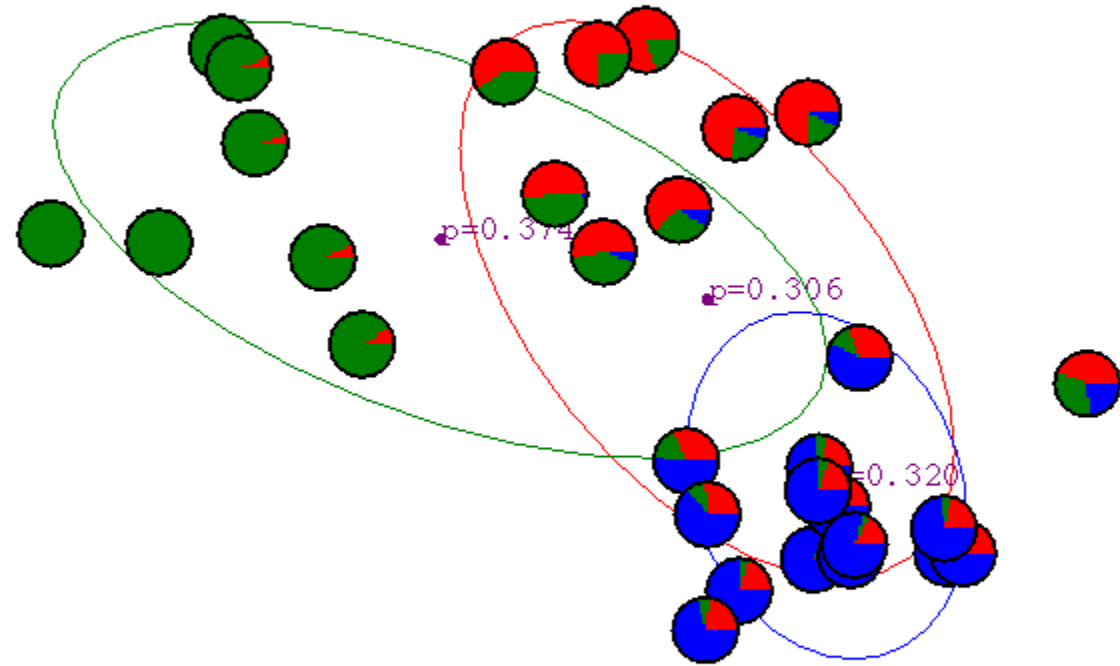
If I decide this fits in the flow of my story... I'll prob end up remaking them.



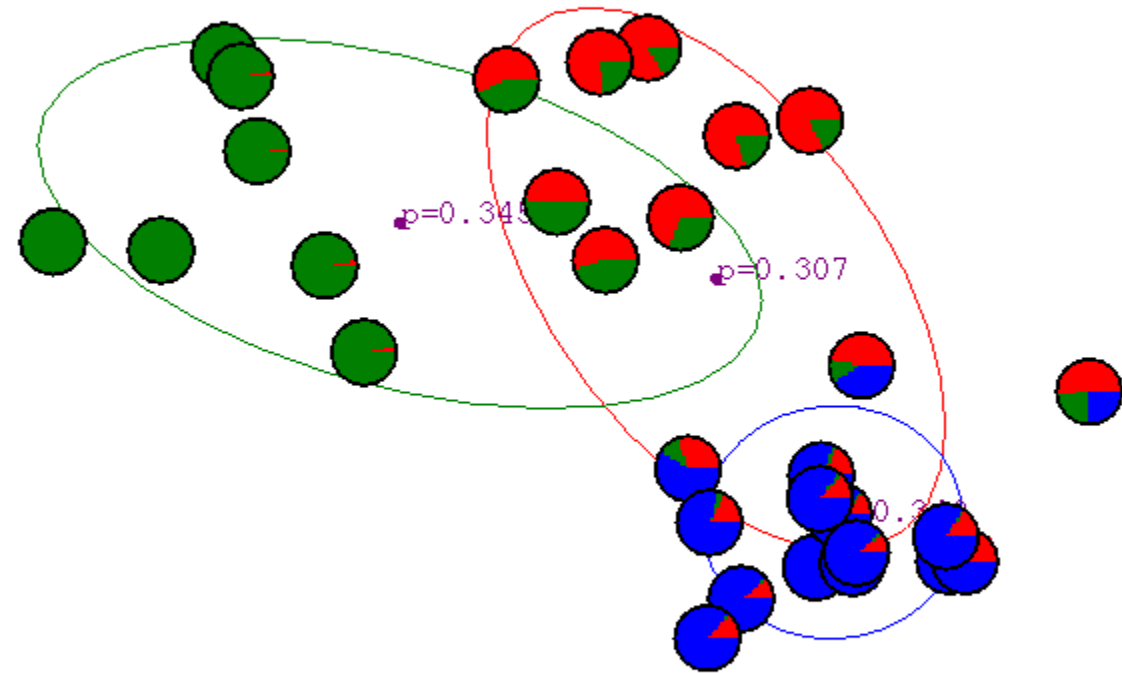
After first iteration



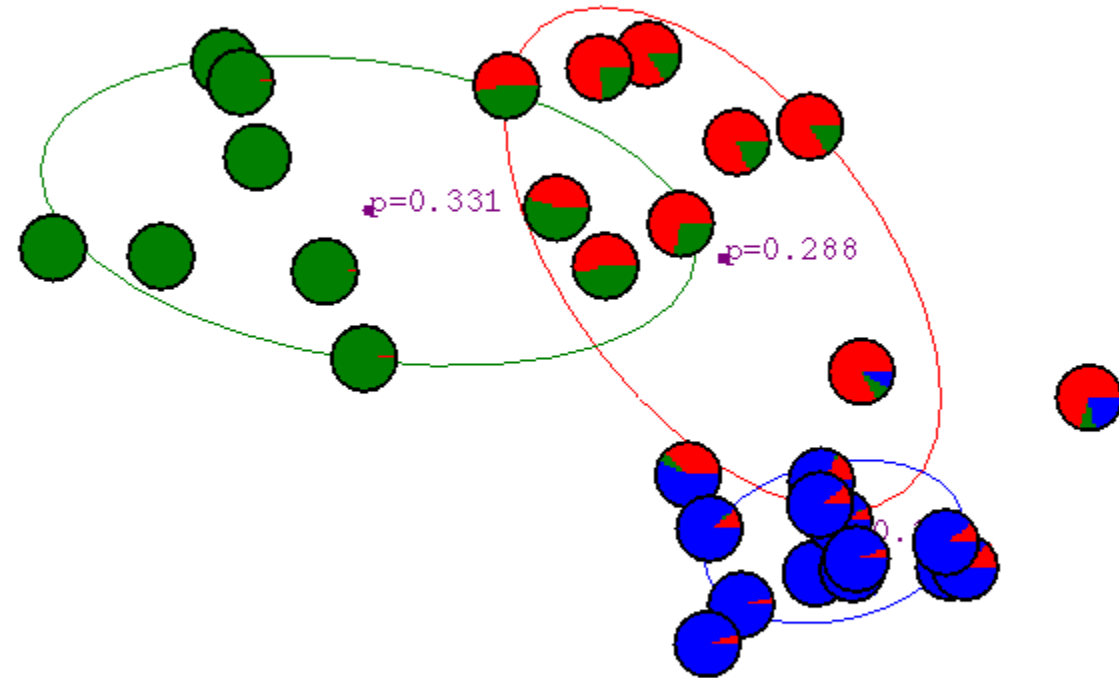
After 2nd iteration



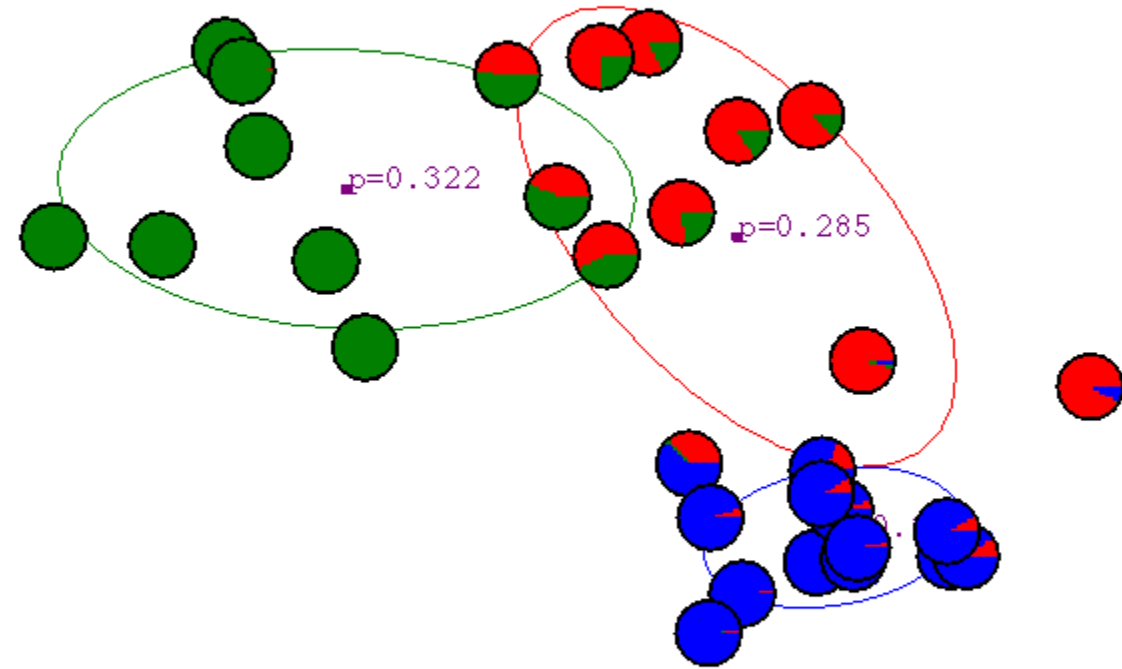
After 3rd iteration



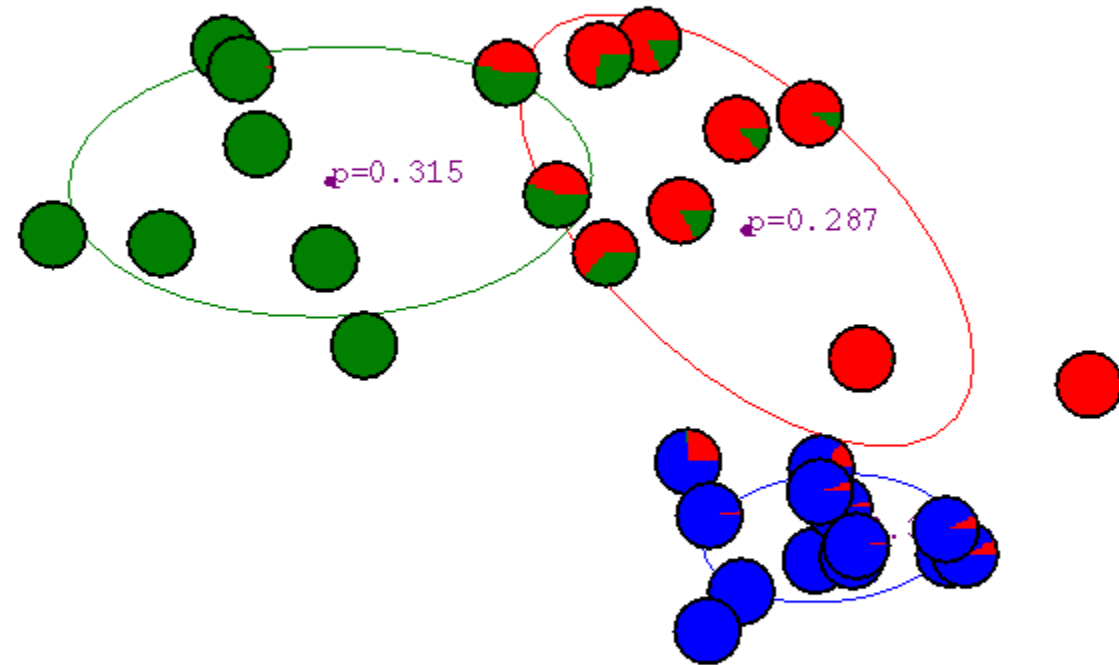
After 4th iteration



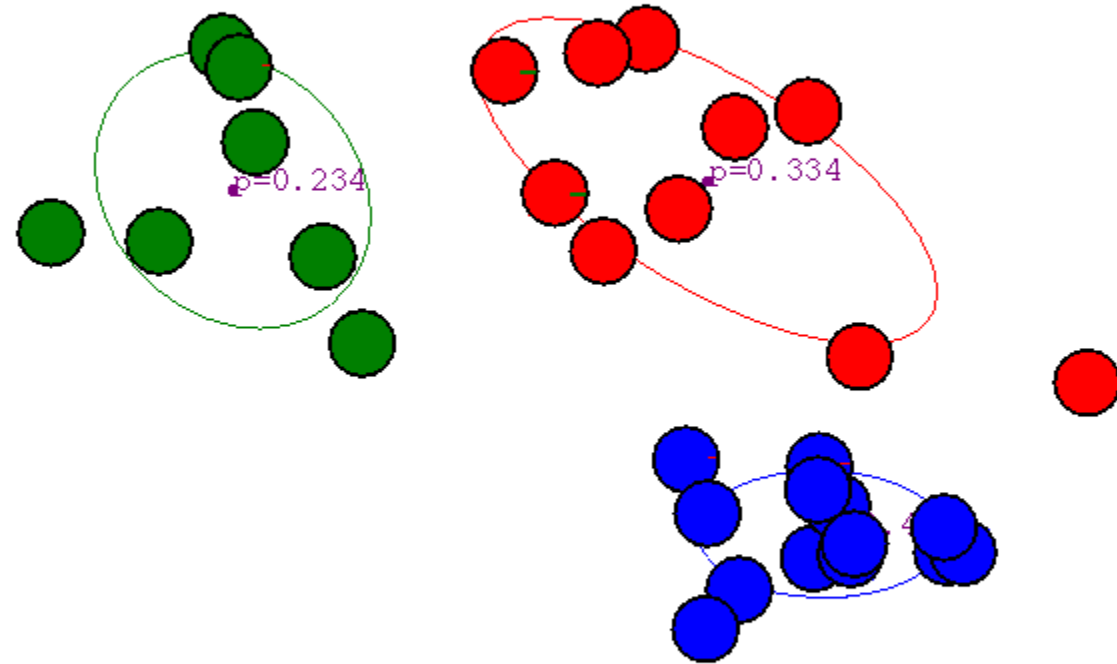
After 5th iteration



After 6th iteration



After 20th iteration



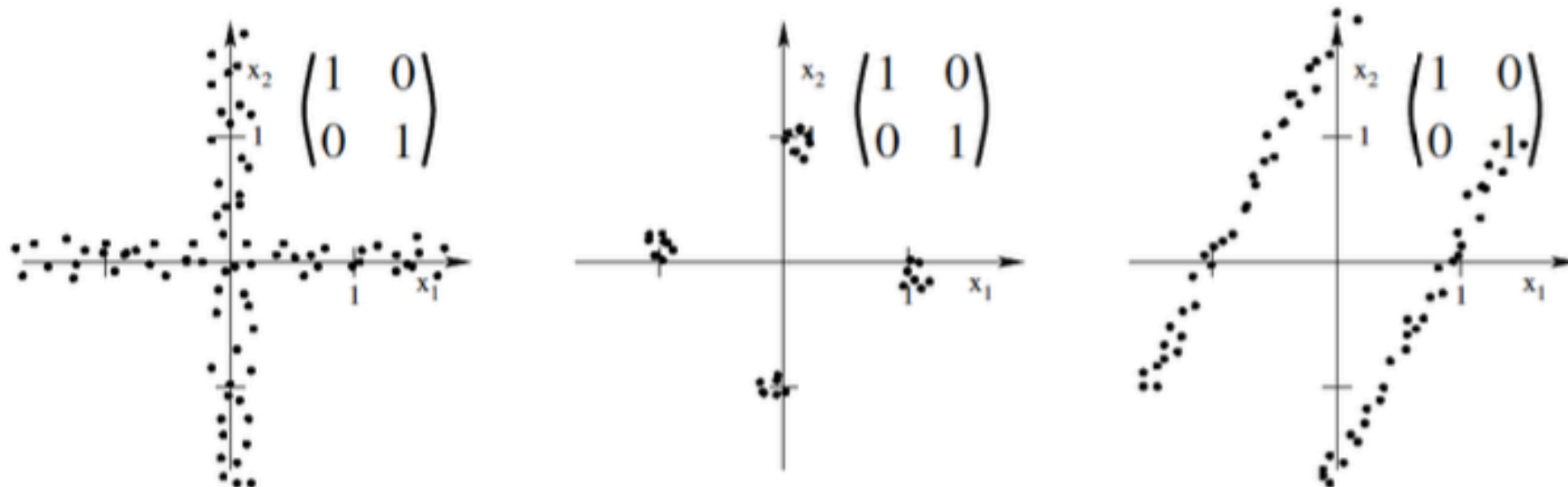
Learning rule for BCE

$$\mathcal{L}(x, y) = - (1 - y) \log(1 - f(x)) - y \log f(x)$$

$$\begin{aligned} \frac{\partial}{\partial \theta_j} \ell(\theta) &= \left(y \frac{1}{g(\theta^T x)} - (1 - y) \frac{1}{1 - g(\theta^T x)} \right) \frac{\partial}{\partial \theta_j} g(\theta^T x) \\ &= \left(y \frac{1}{g(\theta^T x)} - (1 - y) \frac{1}{1 - g(\theta^T x)} \right) g(\theta^T x)(1 - g(\theta^T x)) \frac{\partial}{\partial \theta_j} \theta^T x \\ &= (y(1 - g(\theta^T x)) - (1 - y)g(\theta^T x)) x_j \\ &= (y - h_\theta(x)) x_j \end{aligned}$$

Covariance is not the full story

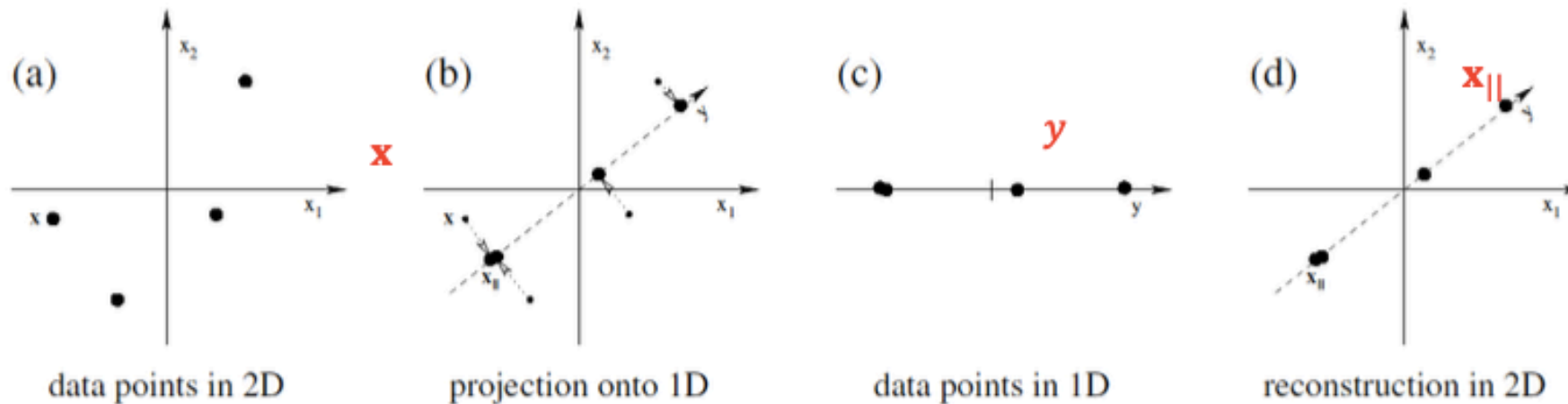
- The covariance matrix tells us about the 2nd order moments, but not about the higher-order structure of the data.



- When it comes to 1st and 2nd order moments, might as well assume a normal distribution, if all we know is the covariance matrix.

Projection and reconstruction error

$\mathbf{x} = (x_1, x_2)$ a point; $\mathbf{v}$ a unit vector normal to projection space
 $\mathbf{x}_{||} := \mathbf{v}\mathbf{v}^T \mathbf{x}$; $y := \mathbf{v}^T \mathbf{x}$; therefore $\mathbf{x}_{||} := \mathbf{v} y$



$$\text{Error } E := \left\langle \left\| \mathbf{x}^\mu - \mathbf{x}_{||}^\mu \right\|^2 \right\rangle_\mu = \frac{1}{M} \sum_{\mu=1}^M \sum_{i=1}^I (x_i^\mu - x_{||i}^\mu)^2$$

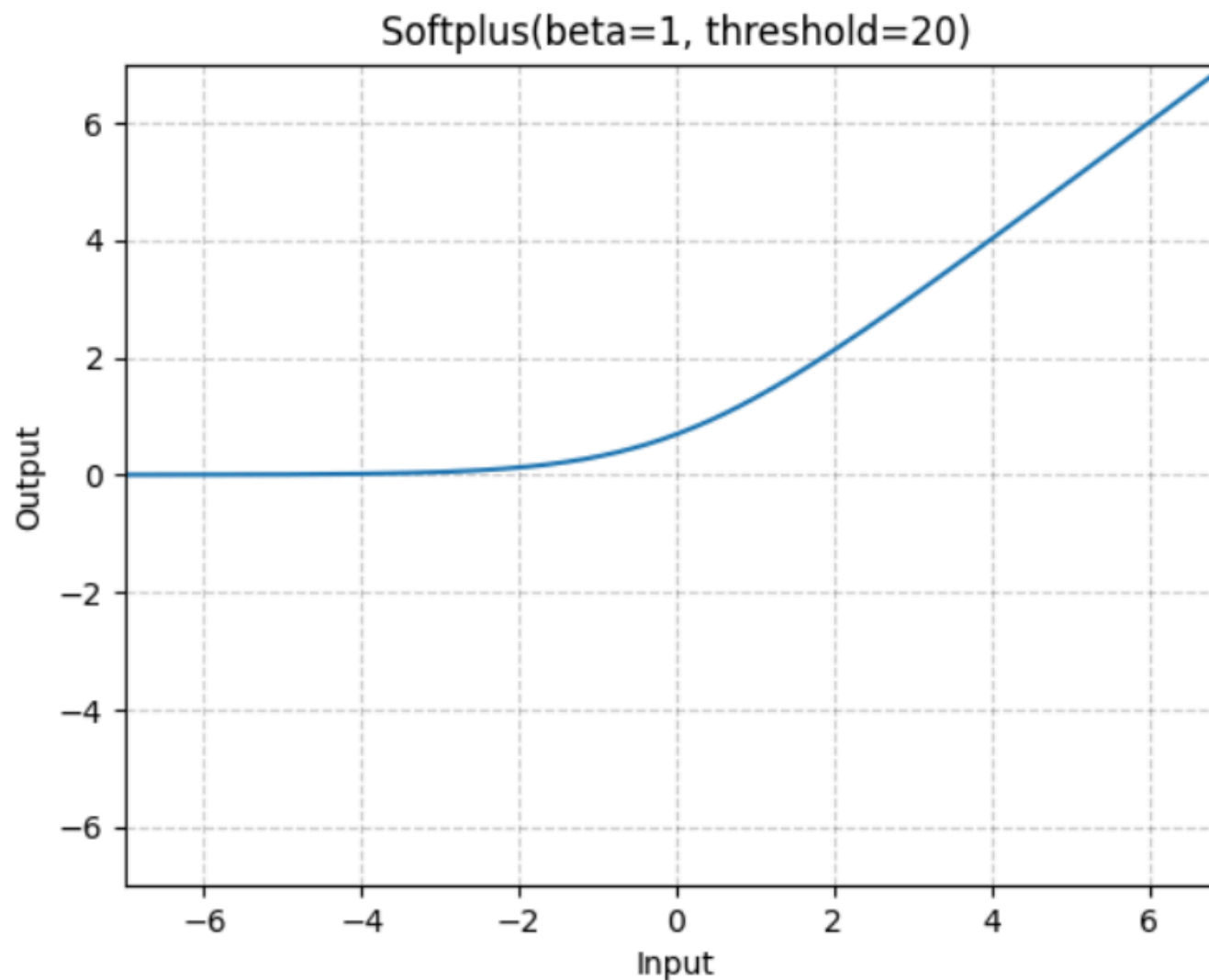
M points, I dimensions

Lecture 6: Classification + Clustering

Softplus

Smooth approx to ReLU

$$\text{Softplus}(x) = \frac{1}{\beta} * \log(1 + \exp(\beta * x))$$



Connection with KL divergence

KL divergence

$$D_{KL}(p || q) = \sum_i p_i \log \frac{p_i}{q_i}$$

Entropy

$$H(p) = - \sum_i p_i \log p_i$$

Cross-entropy

$$H(p, q) = H(p) + D_{KL}(p || q)$$